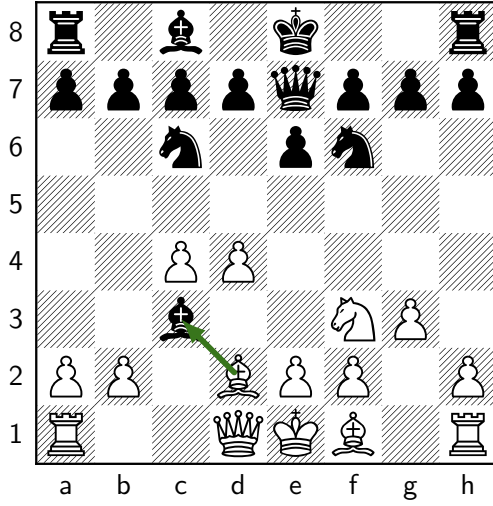


# Melanie-BOTS (CCRL engines) — Bogo-Indian

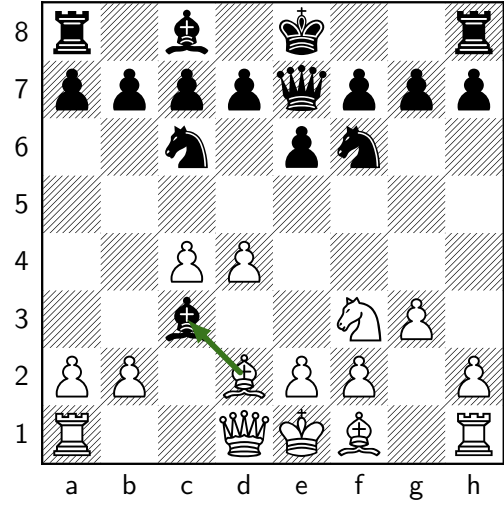
Melanie (Deep\_Vision)

## Abstract

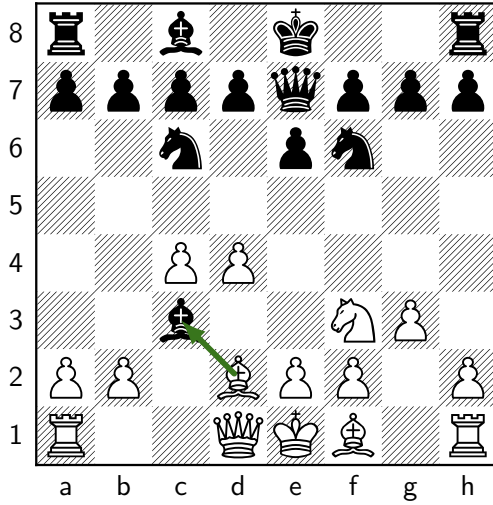
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



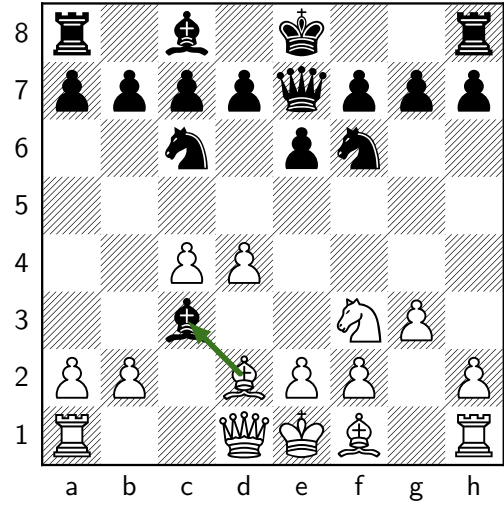
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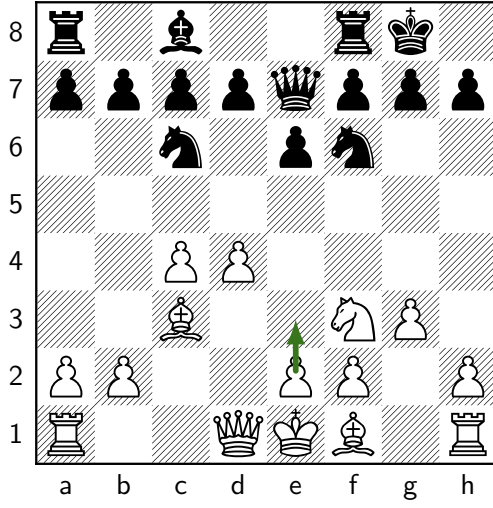


(c)

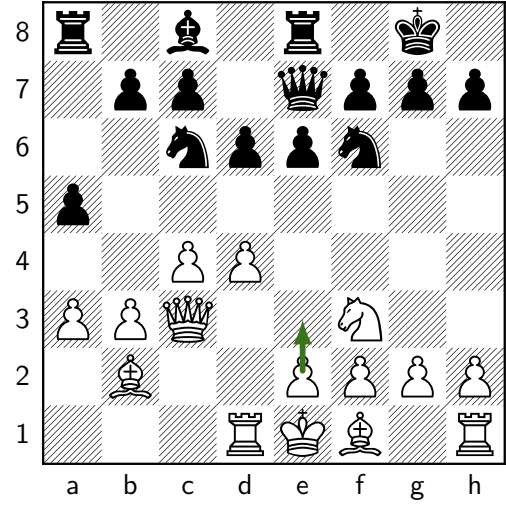


(d)

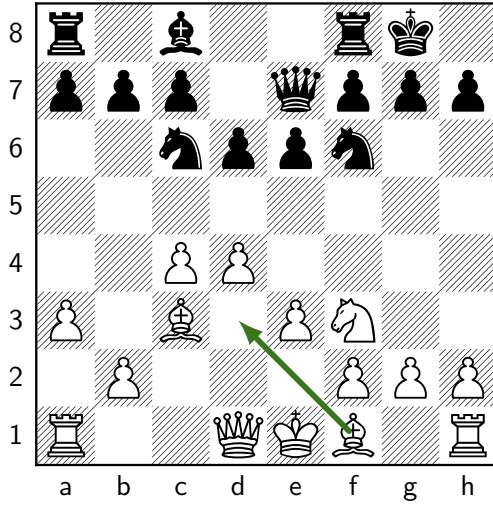
Figure 1: **Move 7, White: Bxc3.** (a) Our position; Melanie plays Bxc3 (d2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#3642 (ply 12) [structural distance  $d=0$ , QAP obj=193]: the master played Bxc3; under  $\mu$  our Bd2 plays the role of that game’s Bd2; cross-type roles: our Pe2 as their Bf1, our ra8 as their Pe2; transferred move d2c3 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#3643 (ply 12) [structural distance  $d=0$ , QAP obj=193]: the master played Bxc3; under  $\mu$  our Bd2 plays the role of that game’s Bd2; cross-type roles: our Pe2 as their Bf1, our ra8 as their Pe2; transferred move d2c3 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#1512 (ply 12) [structural distance  $d=0$ , QAP obj=193]: the master played Bxc3; under  $\mu$  our Bd2 plays the role of that game’s Bd2; cross-type roles: our Pe2 as their Bf1, our ra8 as their Pe2; transferred move d2c3 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Bxc3 (summed QAP objective 2863, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



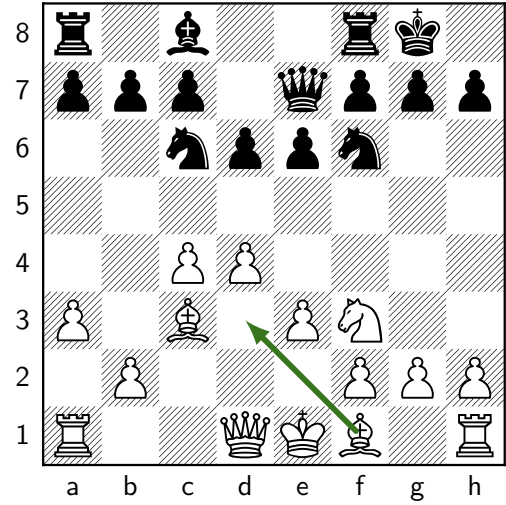
(a)



(b)

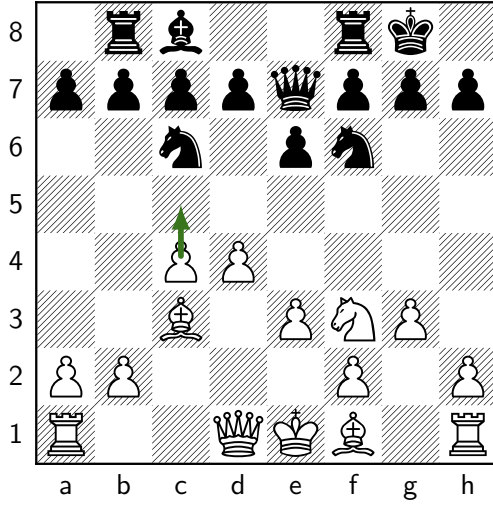


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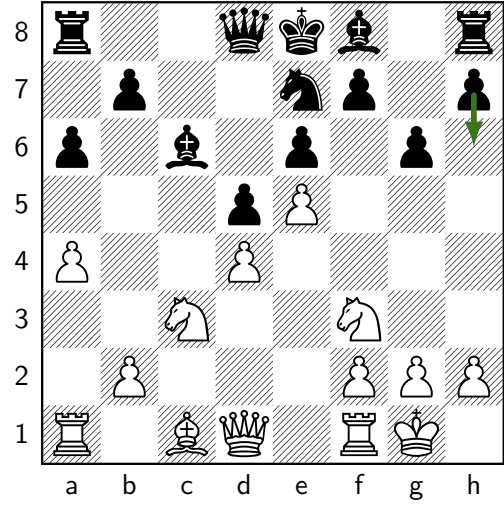


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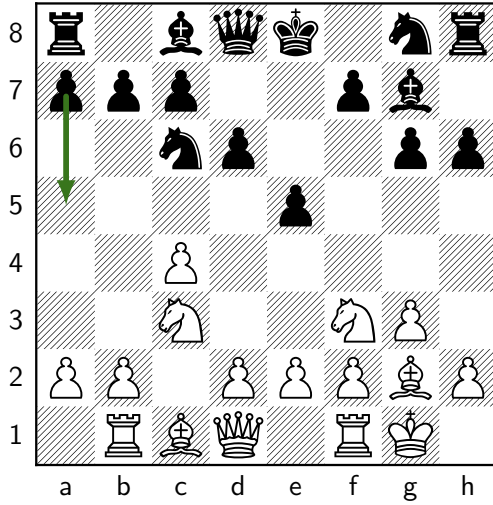
Figure 2: **Move 8, White: e3**. (a) Our position; Melanie plays e3 (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#856 (ply 20) [structural distance  $d=50$ , QAP obj=41]: the master played e3; under  $\mu$  our *Pe2* plays the role of that game’s *Pe2*; cross-type roles: our *Qd1* as their *Rd1*, our *pb7* as their *Bb2*; transferred move e2e3 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#3433 (ply 16) [structural distance  $d=21$ , QAP obj=34]: the master played Bd3; under  $\mu$  our *Pe2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *qe7* as their *Ph2*, our *Nf3* as their *Bc3*; transferred move e2e3 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#3434 (ply 16) [structural distance  $d=21$ , QAP obj=34]: the master played Bd3; under  $\mu$  our *Pe2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *qe7* as their *Ph2*, our *Nf3* as their *Bc3*; transferred move e2e3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to e3 (summed QAP objective 185, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



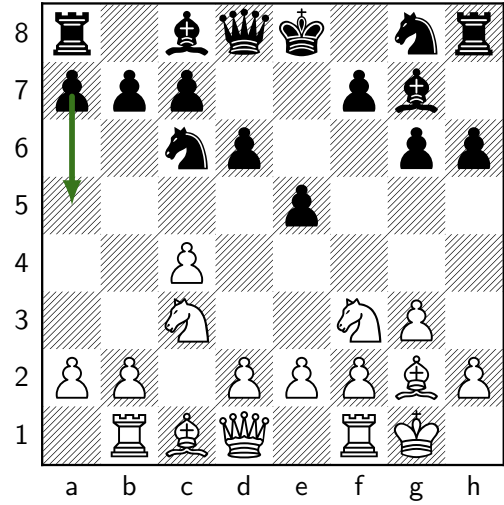
(a)



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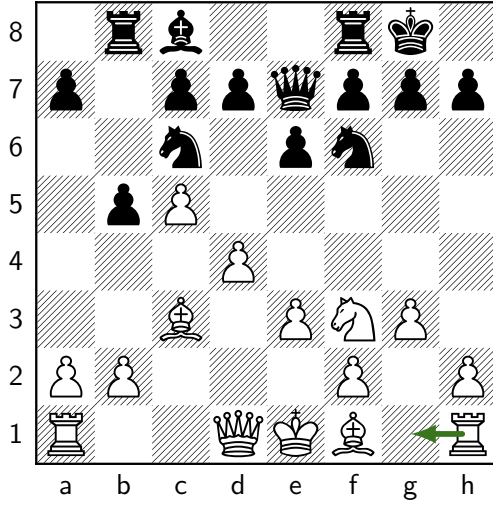


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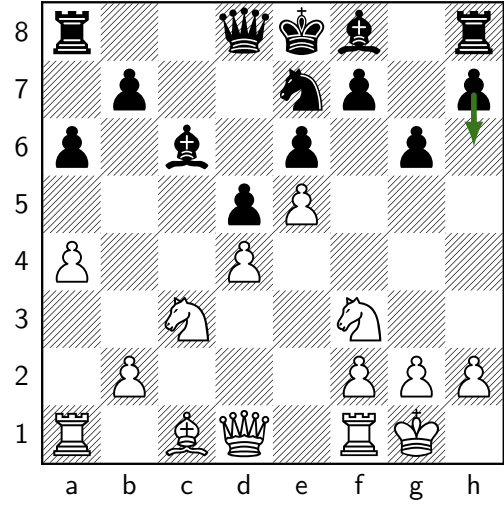


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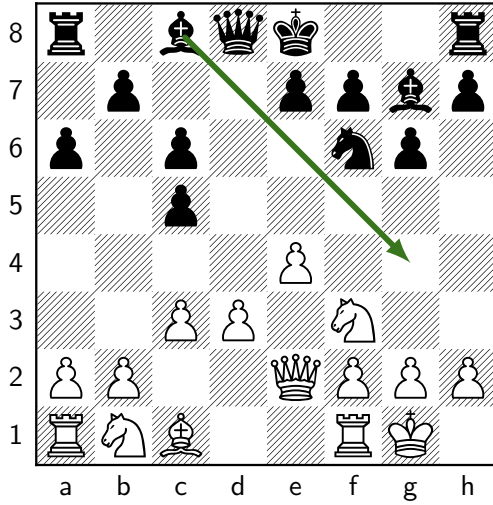
Figure 3: **Move 9, White: c5.** (a) Our position; Melanie plays c5 (c4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3757` (ply 25) [structural distance  $d=55$ , QAP obj=44]: the master played h6; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Ph2$ ; cross-type roles: our  $pe6$  as their  $Ra8$ , our  $Bc3$  as their  $Qd8$ ; transferred move c4c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3839` (ply 13) [structural distance  $d=59$ , QAP obj=25]: the master played a5; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Pa2$ ; cross-type roles: our  $pe6$  as their  $Bc8$ , our  $Bc3$  as their  $Qd8$ ; transferred move c4c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3840` (ply 13) [structural distance  $d=59$ , QAP obj=25]: the master played a5; under  $\mu$  our  $Pc4$  plays the role of that game’s  $Pa2$ ; cross-type roles: our  $pe6$  as their  $Bc8$ , our  $Bc3$  as their  $Qd8$ ; transferred move c4c5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to c5 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



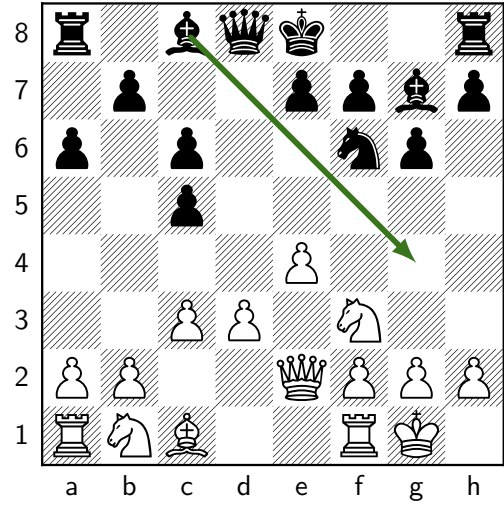
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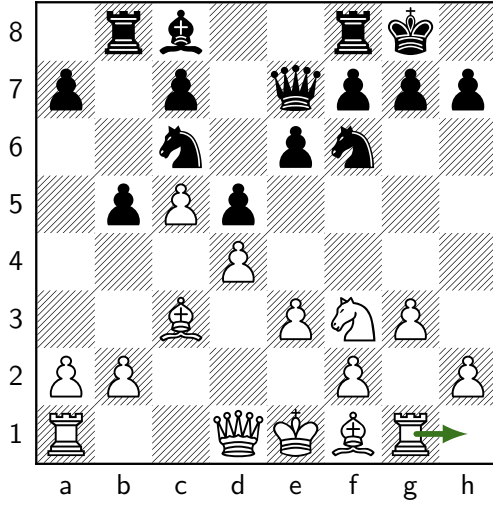


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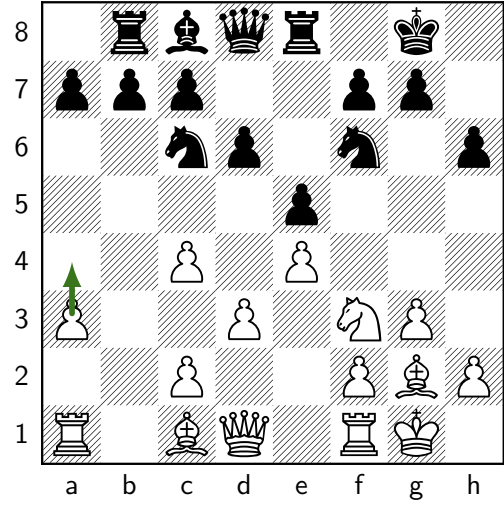


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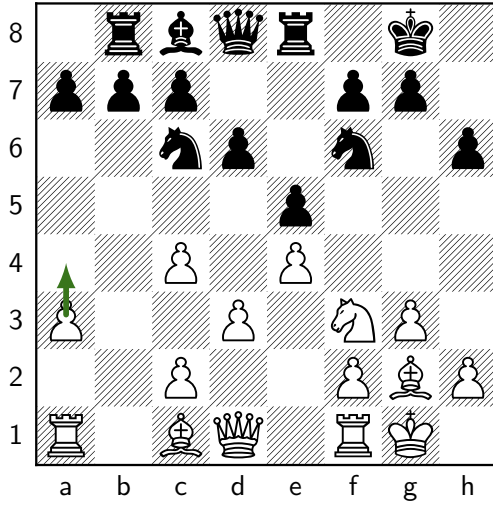
Figure 4: **Move 10, White: Rg1.** (a) Our position; Melanie plays Rg1 (h1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3757` (ply 25) [structural distance  $d=34$ , QAP obj=51]: the master played h6; under  $\mu$  our *Rh1* plays the role of that game's *Ph2* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *Bf8*, our *Bc3* as their *Rh8*; transferred move h1g1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1084` (ply 15) [structural distance  $d=65$ , QAP obj=33]: the master played Bg4; under  $\mu$  our *Rh1* plays the role of that game's *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pg3* as their *Ra8*, our *pe6* as their *Bg7*; transferred move h1g1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1085` (ply 15) [structural distance  $d=65$ , QAP obj=33]: the master played Bg4; under  $\mu$  our *Rh1* plays the role of that game's *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pg3* as their *Ra8*, our *pe6* as their *Bg7*; transferred move h1g1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rg1 (summed QAP objective 118, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



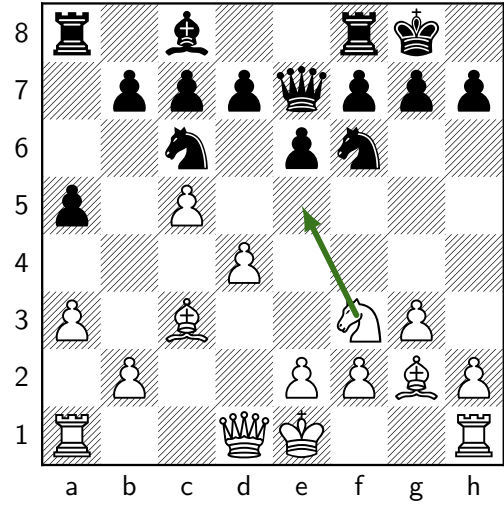
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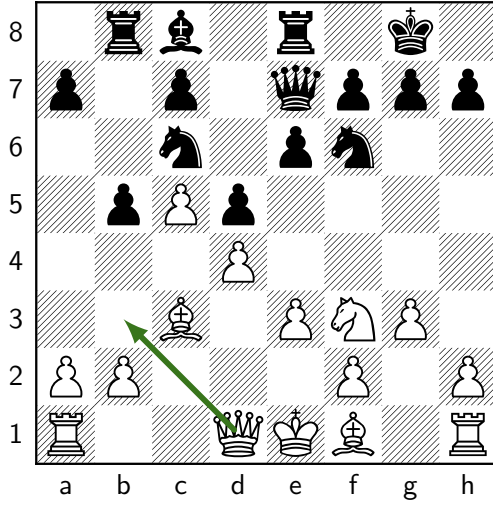


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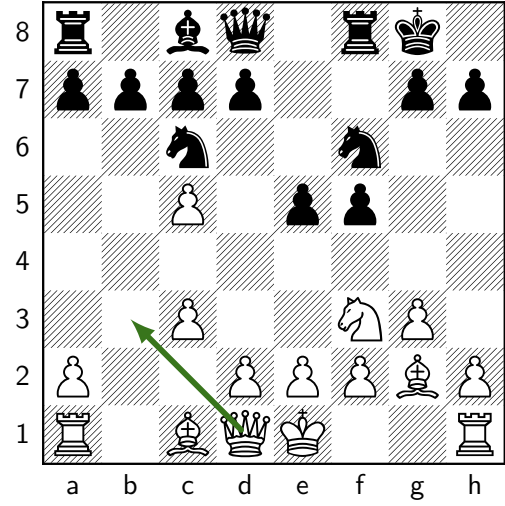


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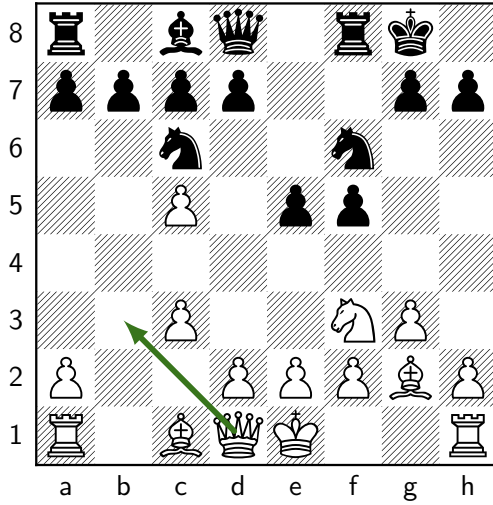
Figure 5: **Move 11, White: Rh1.** (a) Our position; Melanie plays Rh1 (g1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3679` (ply 20) [structural distance  $d=64$ , QAP obj=42]: the master played a4; under  $\mu$  our *Rg1* plays the role of that game’s *Pa3* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Bc1*, our *Pb2* as their *Qd1*; transferred move g1h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3680` (ply 20) [structural distance  $d=64$ , QAP obj=42]: the master played a4; under  $\mu$  our *Rg1* plays the role of that game’s *Pa3* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Bc1*, our *Pb2* as their *Qd1*; transferred move g1h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1928` (ply 20) [structural distance  $d=46$ , QAP obj=67]: the master played Ne5; under  $\mu$  our *Bc3* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *Ra1*, our *Nf3* as their *Rh1*; transferred move c3b4. Synthesis: 2 retrieved analogs transferred to Rh1 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



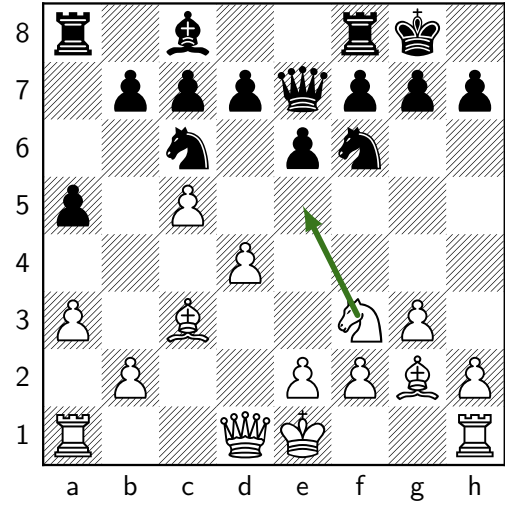
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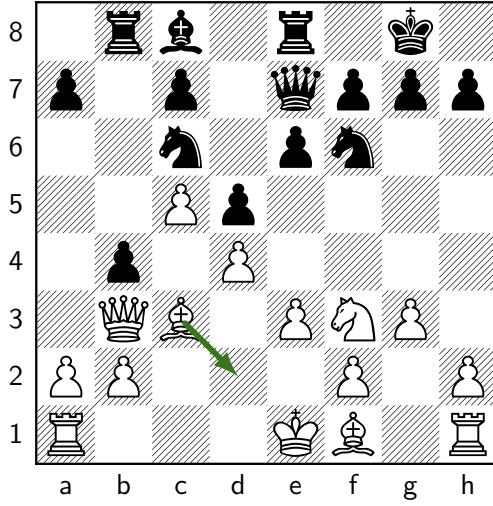


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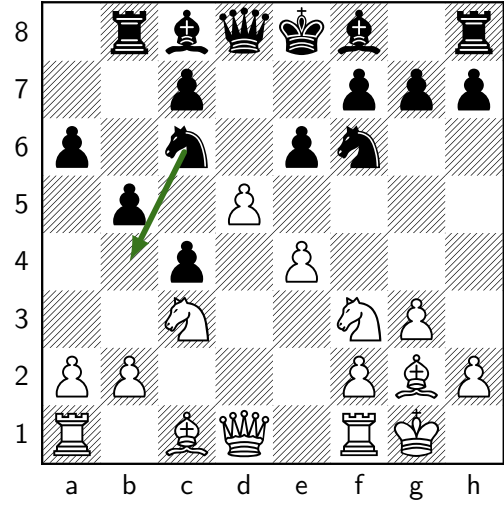


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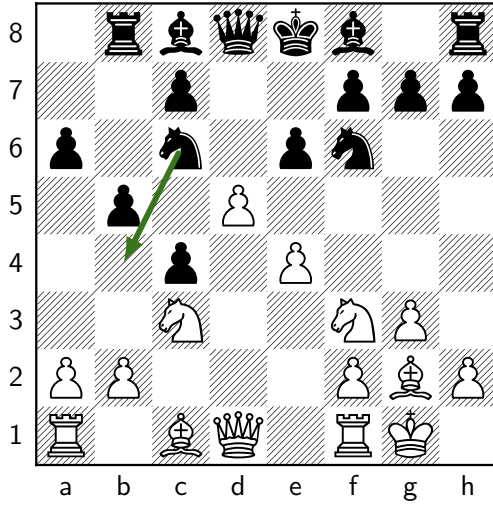
Figure 6: **Move 12, White: Qb3.** (a) Our position; Melanie plays Qb3 (d1b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#50` (ply 14) [structural distance  $d=62$ , QAP obj=28]: the master played Qb3+; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our Bf1 as their Ra1, our pe6 as their Bc1; transferred move d1b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#51` (ply 14) [structural distance  $d=62$ , QAP obj=28]: the master played Qb3+; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our Bf1 as their Ra1, our pe6 as their Bc1; transferred move d1b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1928` (ply 20) [structural distance  $d=63$ , QAP obj=63]: the master played Ne5; under  $\mu$  our Bc3 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Qd1, our Pa2 as their Rh1; transferred move c3b4. Synthesis: 2 retrieved analogs transferred to Qb3 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



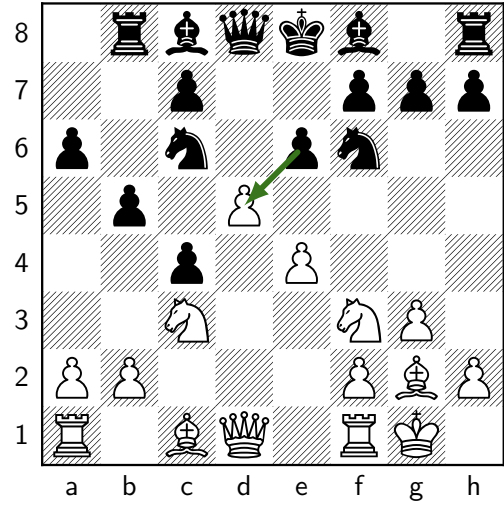
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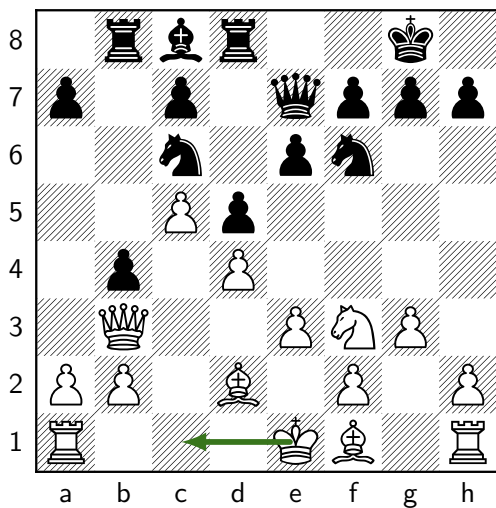
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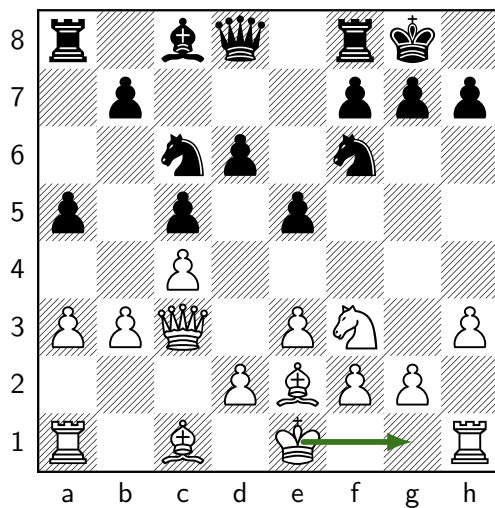
(d)

Figure 7: **Move 13, White: Bd2.** (a) Our position; Melanie plays Bd2 (c3d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3068` (ply 17) [structural distance  $d=105$ , QAP obj=79]: the master played Nb4; under  $\mu$  our Bc3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Bc8, our Pa2 as their Rh8; transferred move c3b4. (c) Retrieved from `ccr1_top_p0000#3069` (ply 17) [structural distance  $d=105$ , QAP obj=79]: the master played Nb4; under  $\mu$  our Bc3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Bc8, our Pa2 as their Rh8; transferred move c3b4. (d) Retrieved from `ccr1_top_p0001#2275` (ply 17) [structural distance  $d=105$ , QAP obj=79]: the master played exd5; under  $\mu$  our Bc3 plays the role of that game’s Pe3 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Bc8, our Pa2 as their Rh8; transferred move c3b4. Synthesis: no retrieved analog transferred to Bd2 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

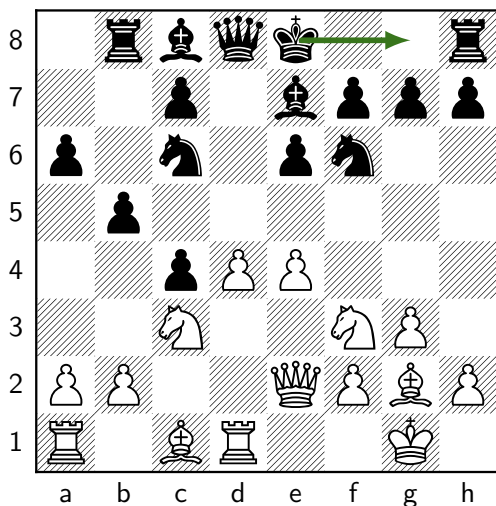




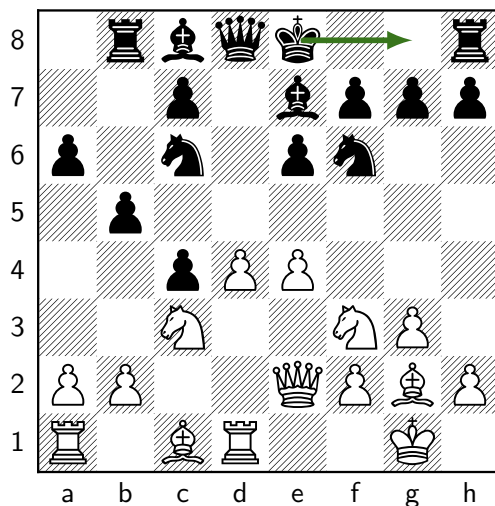
(a)



(b)

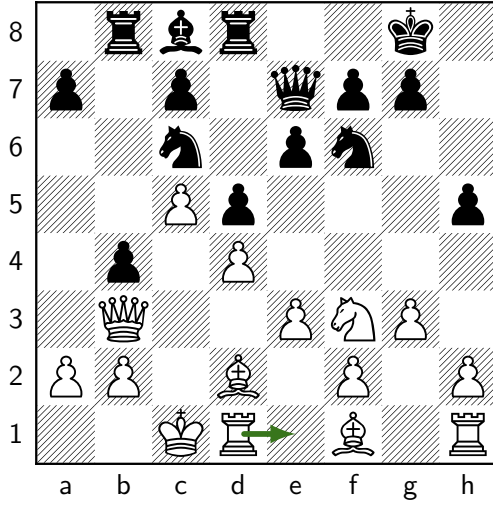


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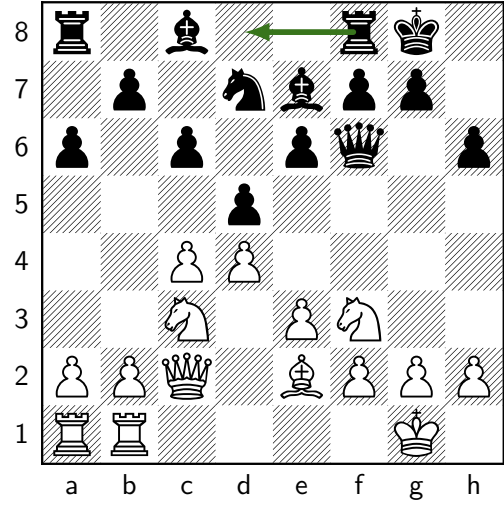


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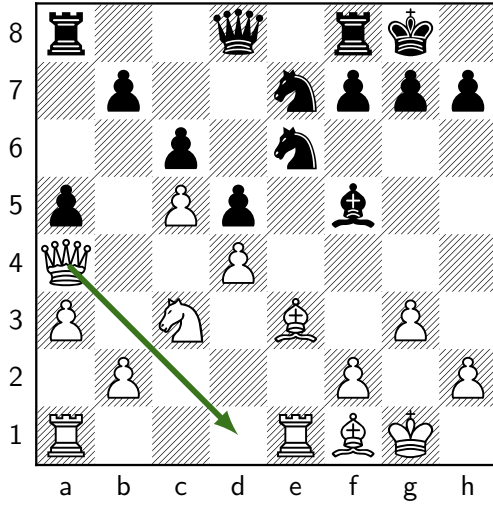
Figure 8: **Move 14, White: 0-0-0.** (a) Our position; Melanie plays 0-0-0 (e1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#658` (ply 20) [structural distance  $d=110$ , QAP obj=69]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bf1* as their *Ra1*, our *Pe3* as their *Bc1*; transferred move e1c1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1574` (ply 19) [structural distance  $d=106$ , QAP obj=62]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pa2* as their *Rb8*, our *Pe3* as their *Bc8*; transferred move e1c1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2406` (ply 19) [structural distance  $d=106$ , QAP obj=62]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pa2* as their *Rb8*, our *Pe3* as their *Bc8*; transferred move e1c1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to 0-0-0 (summed QAP objective 358, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



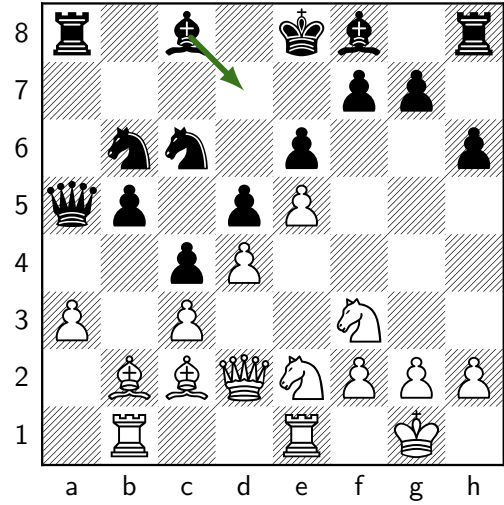
(a)



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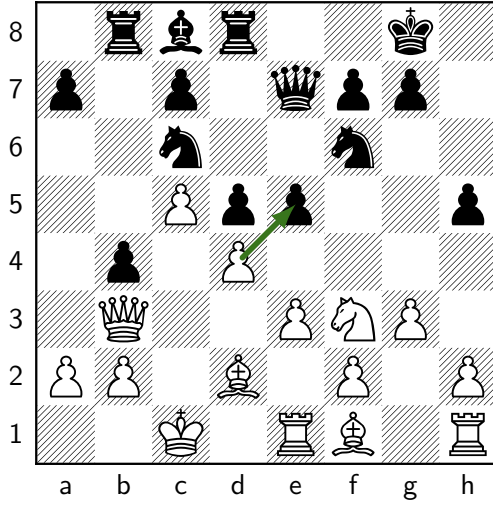


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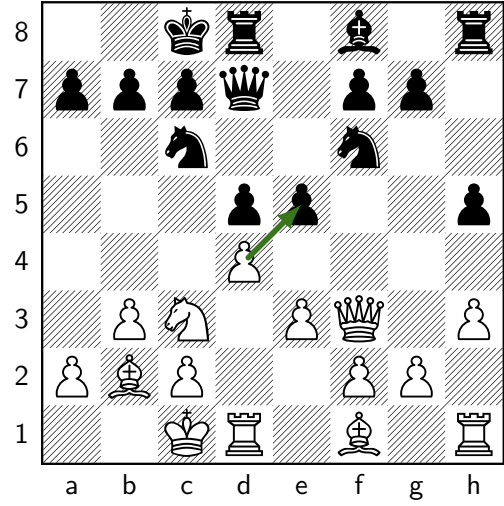


(d)

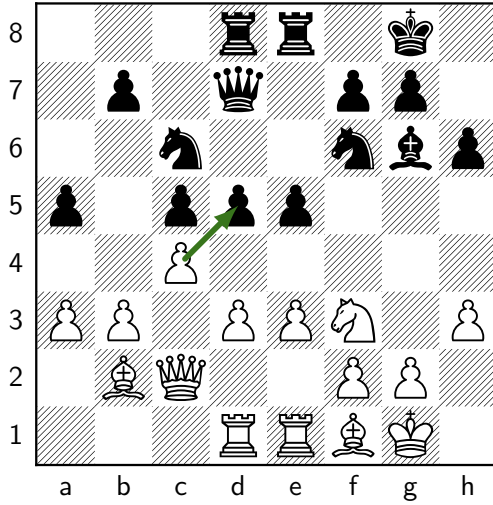
Figure 9: **Move 15, White: Re1.** (a) Our position; Melanie plays **Re1** (**d1e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2383` (ply 21) [structural distance  $d=86$ , QAP obj=63]: the master played **Rd8**; under  $\mu$  our **Rd1** plays the role of that game’s **Rf1**; cross-type roles: our **Nf3** as their **Ra8**, our **Pa2** as their **Nd7**; transferred move **d1e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1665` (ply 40) [structural distance  $d=100$ , QAP obj=102]: the master played **Qd1**; under  $\mu$  our **Qb3** plays the role of that game’s **Qa4**; cross-type roles: our **Pf2** as their **Ra1**, our **Nf3** as their **Re1**; transferred move **b3a3**. (d) Retrieved from `ccr1_top_p0001#2481` (ply 33) [structural distance  $d=105$ , QAP obj=79]: the master played **Bd7**; under  $\mu$  our **Pe3** plays the role of that game’s **Bc1** (a **cross-type** correspondence); cross-type roles: our **Rh1** as their **Bf8**, our **Nf3** as their **Rh8**; transferred move **e3e4**. Synthesis: 1 retrieved analog transferred to **Re1** (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



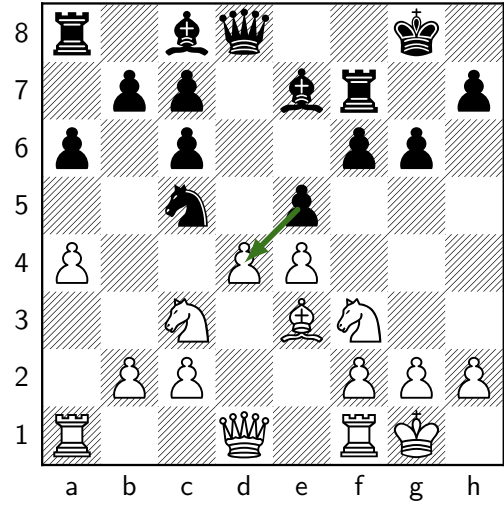
(a)



(b)



(c)



(d)

Figure 10: **Move 16, White: dxe5.** (a) Our position; Melanie plays dxe5 (d4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3051` (ply 18) [structural distance  $d=90$ , QAP obj=129]: the master played dxe5; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Pd4$ ; cross-type roles: our  $Nf3$  as their  $Rd1$ , our  $Rh1$  as their  $Bf1$ ; transferred move d4e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#658` (ply 34) [structural distance  $d=131$ , QAP obj=78]: the master played cxd5; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Pc4$ ; cross-type roles: our  $nc6$  as their  $Rd1$ , our  $Bf1$  as their  $Re1$ ; transferred move d4e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3048` (ply 25) [structural distance  $d=100$ , QAP obj=38]: the master played exd4; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Pe4$ ; cross-type roles: our  $ph5$  as their  $Ra8$ , our  $Pe3$  as their  $Bc8$ ; transferred move d4e5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to dxe5 (summed QAP objective 245, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

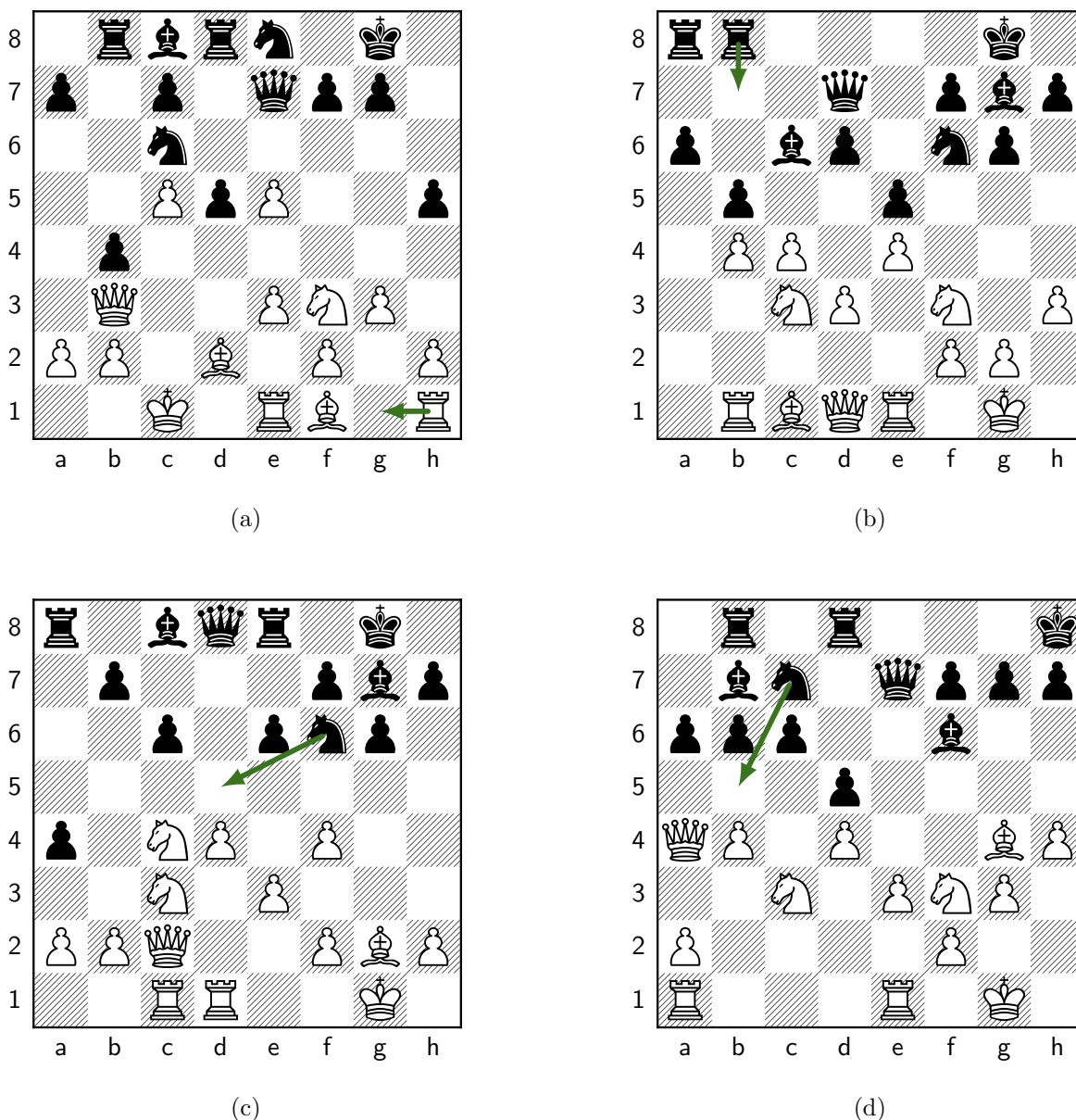
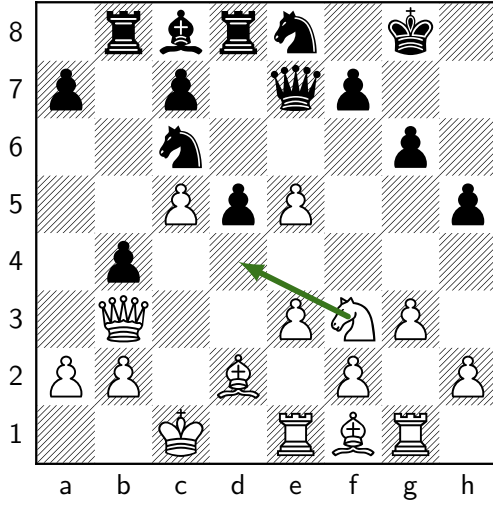
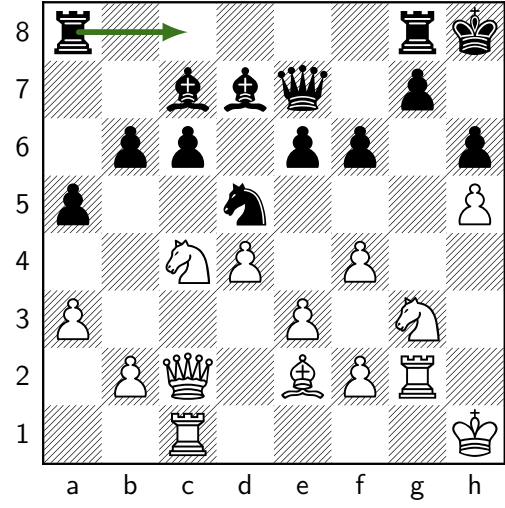


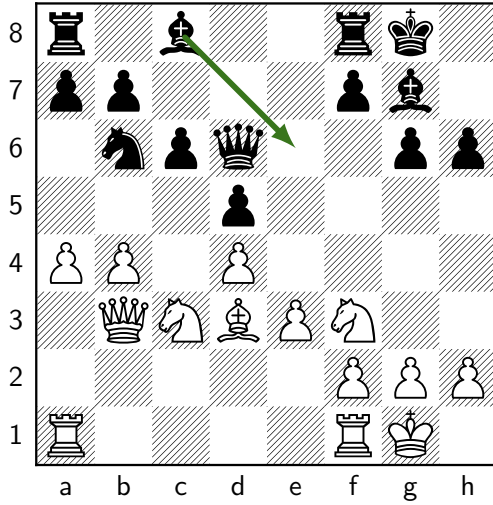
Figure 11: **Move 17, White: Rg1.** (a) Our position; Melanie plays Rg1 (h1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3419` (ply 31) [structural distance  $d=109$ , QAP obj=66]: the master played Rb7; under  $\mu$  our Rh1 plays the role of that game’s Rb1; cross-type roles: our Pa2 as their Ra8, our Pb2 as their Qd7; transferred move h1g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3659` (ply 31) [structural distance  $d=112$ , QAP obj=49]: the master played Nd5; under  $\mu$  our Rh1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Ra8, our Pa2 as their Bc8; transferred move h1g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#391` (ply 39) [structural distance  $d=117$ , QAP obj=92]: the master played Nb5; under  $\mu$  our Pe3 plays the role of that game’s Nc2 (a **cross-type** correspondence); cross-type roles: our Bd2 as their Rd8, our Pg3 as their Bb7; transferred move e3e4. Synthesis: 2 retrieved analogs transferred to Rg1 (summed QAP objective 115, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



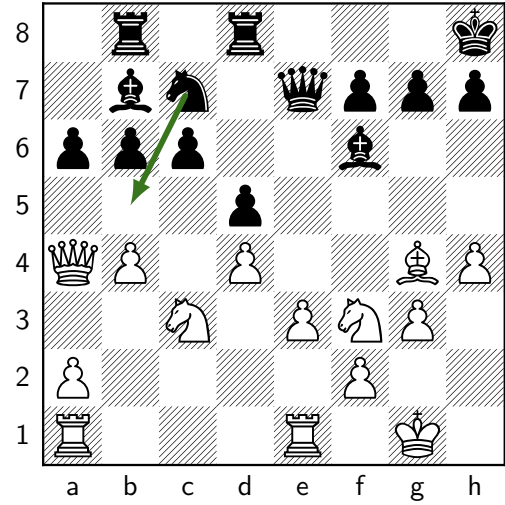
(a)



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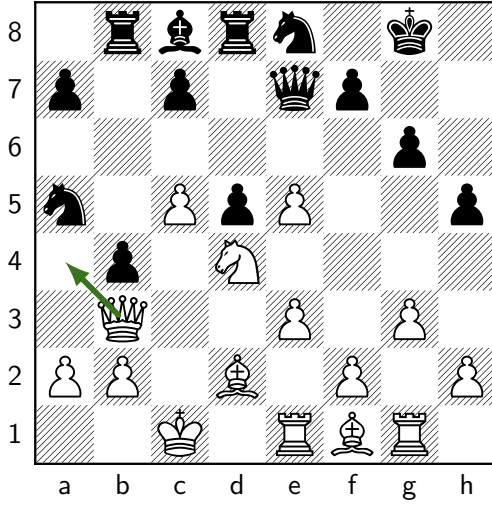


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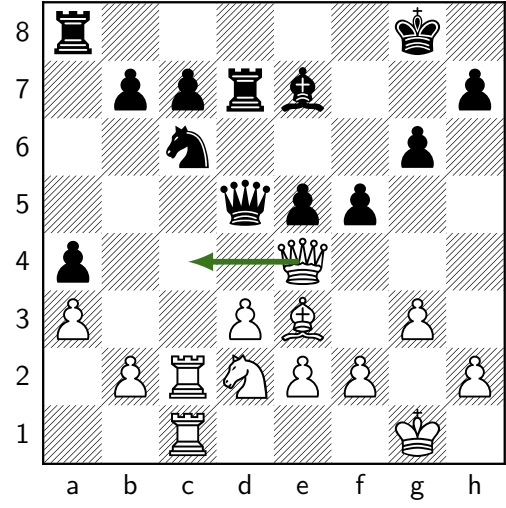


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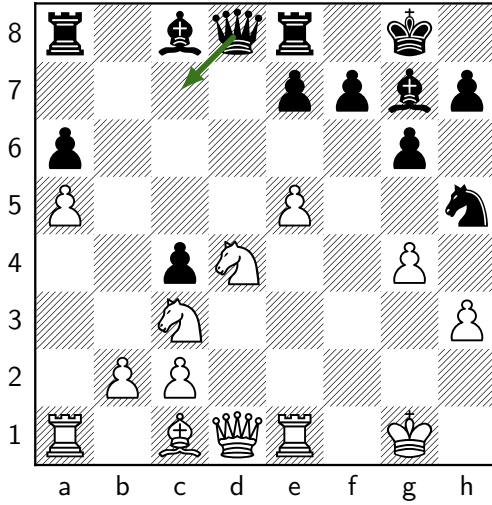
Figure 12: **Move 18, White: Nd4**. (a) Our position; Melanie plays Nd4 (f3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2163` (ply 49) [structural distance  $d=98$ , QAP obj=71]: the master played Rac8; under  $\mu$  our  $Nf3$  plays the role of that game’s  $Ra1$  (a **cross-type** correspondence); cross-type roles: our  $Bd2$  as their  $Rg8$ , our  $Qb3$  as their  $Bc7$ ; transferred move f3d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1106` (ply 27) [structural distance  $d=101$ , QAP obj=58]: the master played Be6; under  $\mu$  our  $Nf3$  plays the role of that game’s  $Bc1$  (a **cross-type** correspondence); cross-type roles: our  $Pf2$  as their  $Ra8$ , our  $Pb2$  as their  $Rf8$ ; transferred move f3d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#391` (ply 39) [structural distance  $d=100$ , QAP obj=93]: the master played Nb5; under  $\mu$  our  $Pe3$  plays the role of that game’s  $Nc2$  (a **cross-type** correspondence); cross-type roles: our  $Nf3$  as their  $Rb8$ , our  $Bd2$  as their  $Rd8$ ; transferred move e3e4. Synthesis: 2 retrieved analogs transferred to Nd4 (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



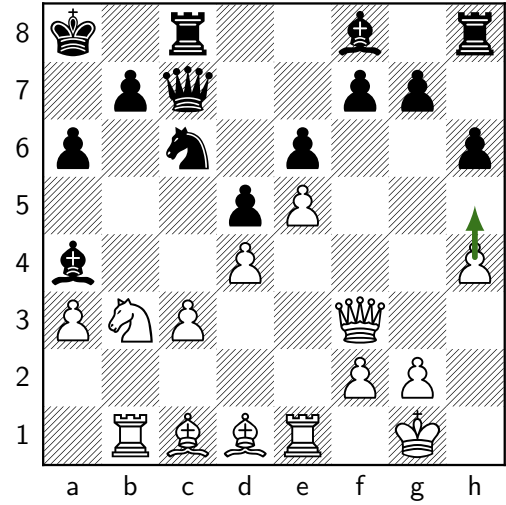
(a)



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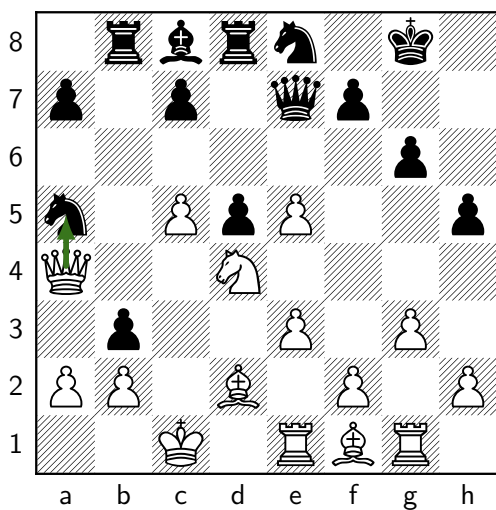


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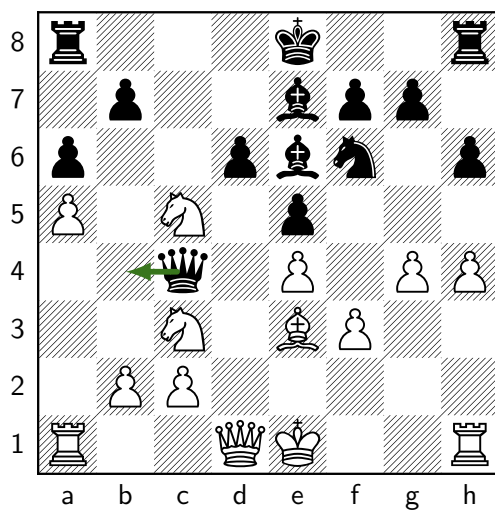


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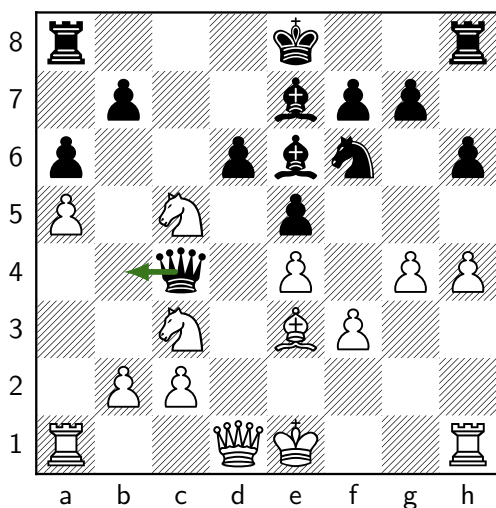
Figure 13: **Move 19, White: Qa4.** (a) Our position; Melanie plays Qa4 (b3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0002#936` (ply 38) [structural distance  $d=104$ , QAP obj=79]: the master played Qc4; under  $\mu$  our Qb3 plays the role of that game’s Qe4; cross-type roles: our Pg3 as their Rc2, our Pb2 as their Nd2; transferred move b3c3. (c) Retrieved from `ccl1_top_p0002#3819` (ply 37) [structural distance  $d=105$ , QAP obj=40]: the master played Qc7; under  $\mu$  our Qb3 plays the role of that game’s Qd1; cross-type roles: our Pe3 as their Ra8, our Pa2 as their Bc8; transferred move b3a3. (d) Retrieved from `ccl1_top_p0001#632` (ply 40) [structural distance  $d=99$ , QAP obj=40]: the master played h5; under  $\mu$  our Bf1 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our Nd4 as their Rb1, our Pa2 as their Bc1; transferred move f1h3. Synthesis: no retrieved analog transferred to Qa4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



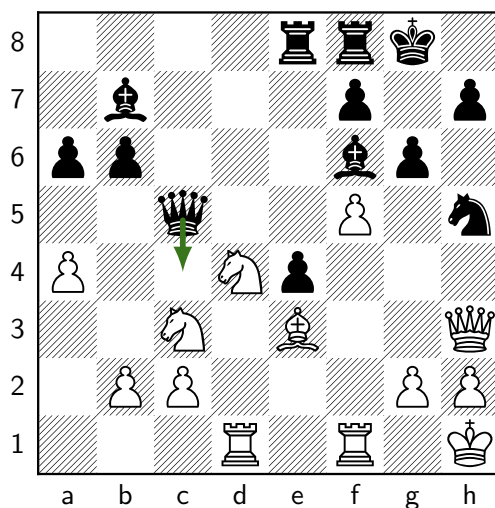
(a)



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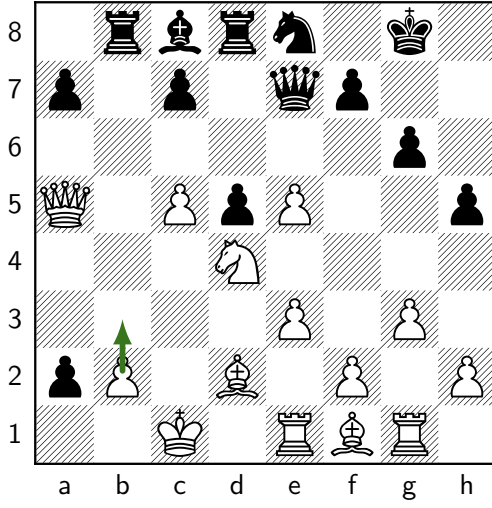
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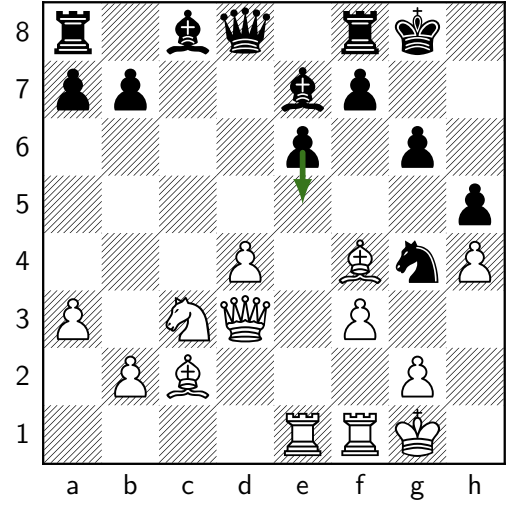
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Figure 14: **Move 20, White: Qxa5.** (a) Our position; Melanie plays Qxa5 (a4a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1511` (ply 33) [structural distance  $d=112$ , QAP obj=68]: the master played Qb4; under  $\mu$  our Qa4 plays the role of that game's Qc5; cross-type roles: our Pe5 as their Ra8, our Nd4 as their Be7; transferred move a4b4. (c) Retrieved from `ccr1_top_p0001#412` (ply 29) [structural distance  $d=112$ , QAP obj=68]: the master played Qb4; under  $\mu$  our Qa4 plays the role of that game's Qc5; cross-type roles: our Pe5 as their Ra8, our Nd4 as their Be7; transferred move a4b4. (d) Retrieved from `ccr1_top_p0002#243` (ply 47) [structural distance  $d=104$ , QAP obj=68]: the master played Qc4; under  $\mu$  our Qa4 plays the role of that game's Qc4; cross-type roles: our Pe3 as their Rf8, our Pb2 as their Bb7; transferred move a4a3. Synthesis: no retrieved analog transferred to Qxa5 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

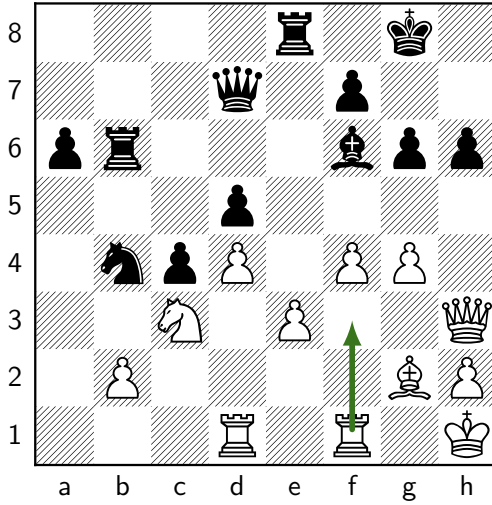




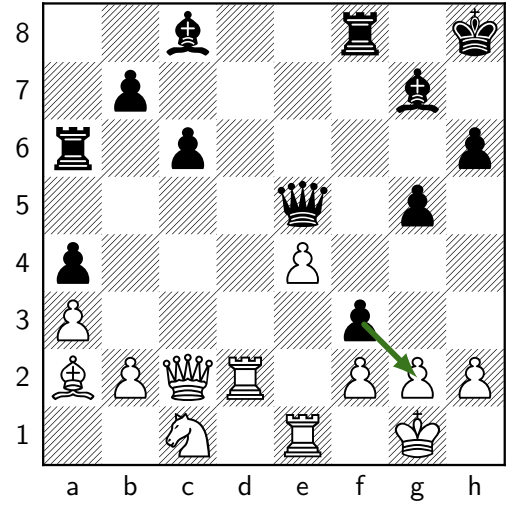
(a)



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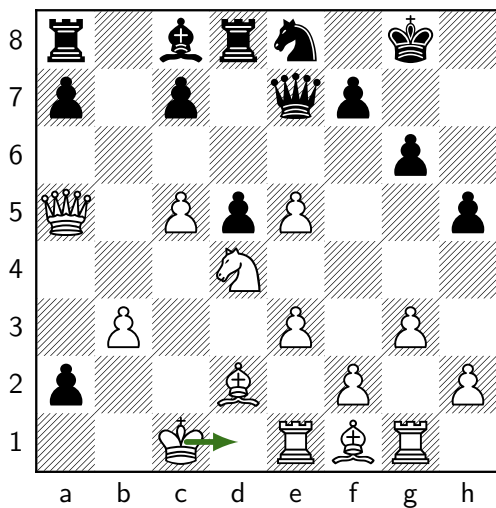
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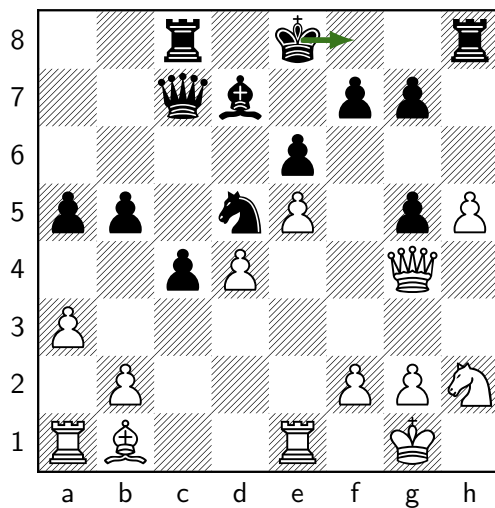
(d)

Figure 15: **Move 21, White: b3.** (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3316` (ply 35) [structural distance  $d=90$ , QAP obj=21]: the master played e5; under  $\mu$  our *Pb2* plays the role of that game’s *Pe3*; cross-type roles: our *Pc5* as their *Bc8*, our *Bd2* as their *Rf8*; transferred move b2b3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2663` (ply 50) [structural distance  $d=92$ , QAP obj=105]: the master played Rf3; under  $\mu$  our *Re1* plays the role of that game’s *Rf1*; cross-type roles: our *bc8* as their *Pb2*, our *qe7* as their *Bg2*; transferred move e1e2. (d) Retrieved from `ccrl_top_p0000#1344` (ply 47) [structural distance  $d=65$ , QAP obj=97]: the master played fxg2; under  $\mu$  our *Pe5* plays the role of that game’s *Pf6*; cross-type roles: our *rb8* as their *Pb7*, our *bc8* as their *Ra6*; transferred move e5e6. Synthesis: 1 retrieved analog transferred to b3 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

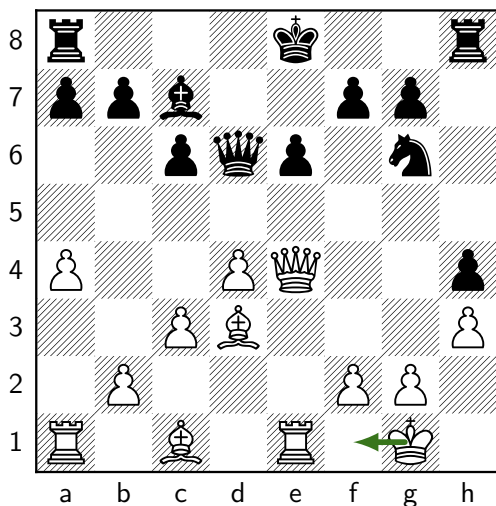




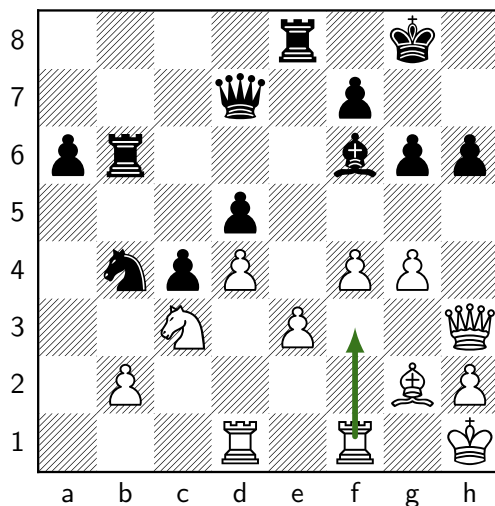
(a)



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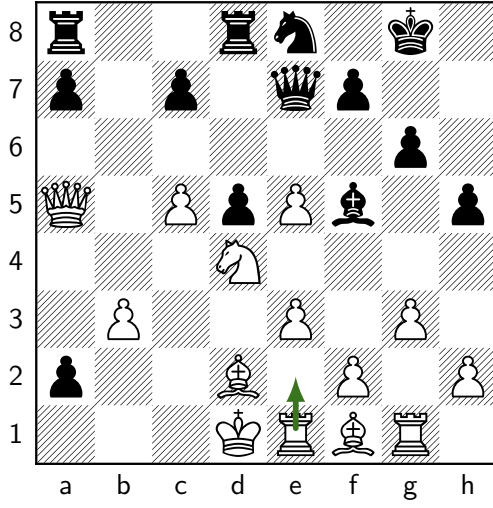


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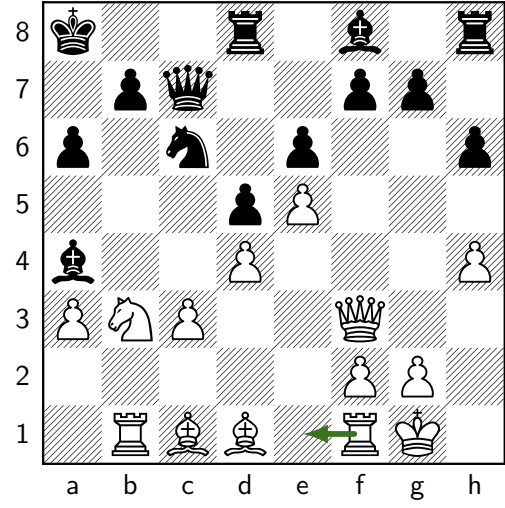


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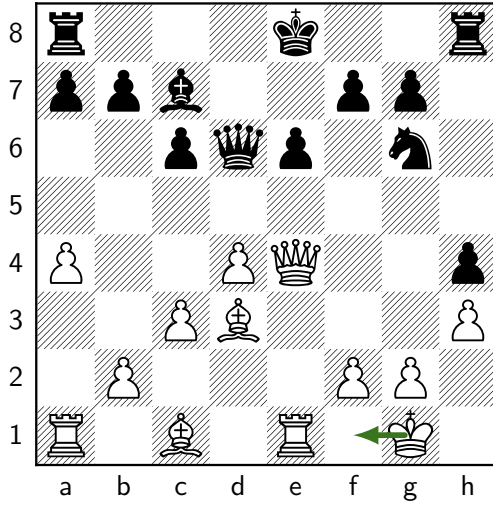
Figure 16: **Move 22, White: Kd1.** (a) Our position; Melanie plays Kd1 (c1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3736` (ply 45) [structural distance  $d=80$ , QAP obj=95]: the master played Kf8; under  $\mu$  our Kc1 plays the role of that game's Ke1; cross-type roles: our Bd2 as their Rc8, our Qa5 as their Rh8; transferred move c1d1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2148` (ply 36) [structural distance  $d=79$ , QAP obj=79]: the master played Kf1; under  $\mu$  our Kc1 plays the role of that game's Kg1; cross-type roles: our pd5 as their Ra1, our Re1 as their Bc1; transferred move c1d1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2663` (ply 50) [structural distance  $d=99$ , QAP obj=109]: the master played Rf3; under  $\mu$  our Re1 plays the role of that game's Rf1; cross-type roles: our ne8 as their Rd1, our bc8 as their Pb2; transferred move e1e2. Synthesis: 2 retrieved analogs transferred to Kd1 (summed QAP objective 174, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



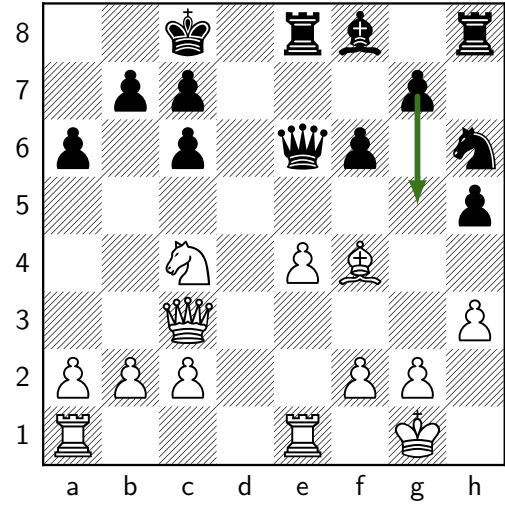
(a)



(b)

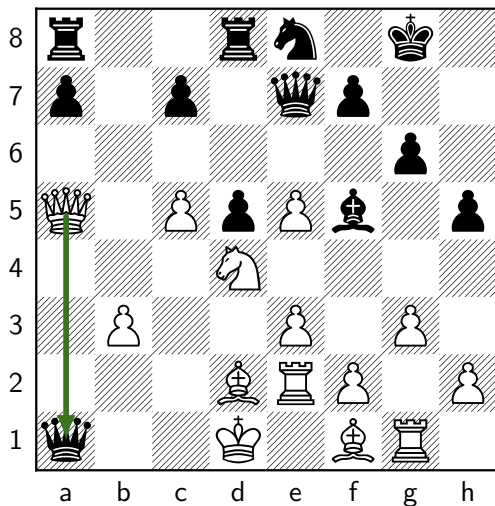


(c)

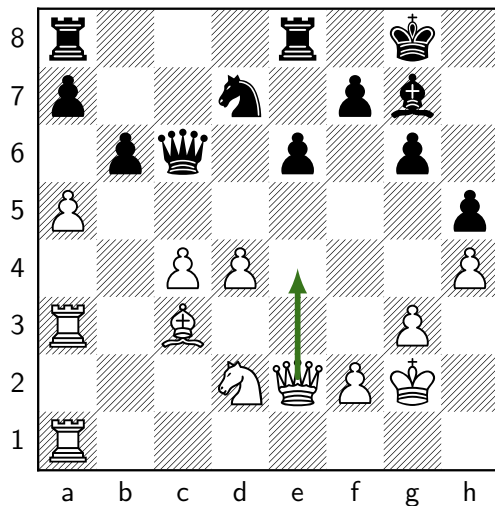


(d)

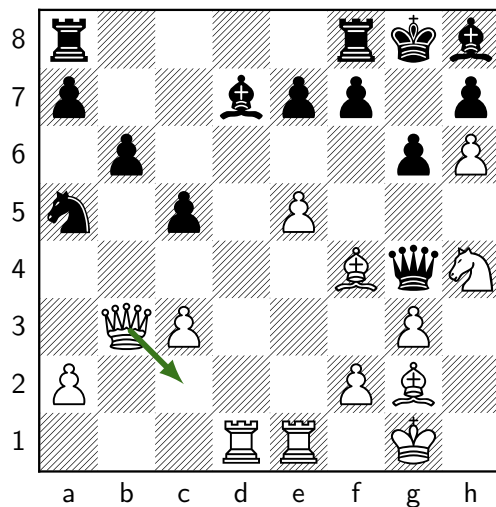
Figure 17: **Move 23, White: Re2.** (a) Our position; Melanie plays Re2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#632` (ply 38) [structural distance  $d=86$ , QAP obj=82]: the master played Re1; under  $\mu$  our Re1 plays the role of that game’s Rf1; cross-type roles: our Pe5 as their Bc1, our bf5 as their Nb3; transferred move e1e2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2148` (ply 36) [structural distance  $d=84$ , QAP obj=77]: the master played Kf1; under  $\mu$  our Kd1 plays the role of that game’s Kg1; cross-type roles: our Nd4 as their Ra1, our pd5 as their Bc1; transferred move d1c1. (d) Retrieved from `ccrl_top_p0002#87` (ply 33) [structural distance  $d=81$ , QAP obj=77]: the master played g5; under  $\mu$  our Pg3 plays the role of that game’s Pg2; cross-type roles: our Bd2 as their Re8, our Pe3 as their Bf8; transferred move g3g4. Synthesis: 1 retrieved analog transferred to Re2 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



(a)

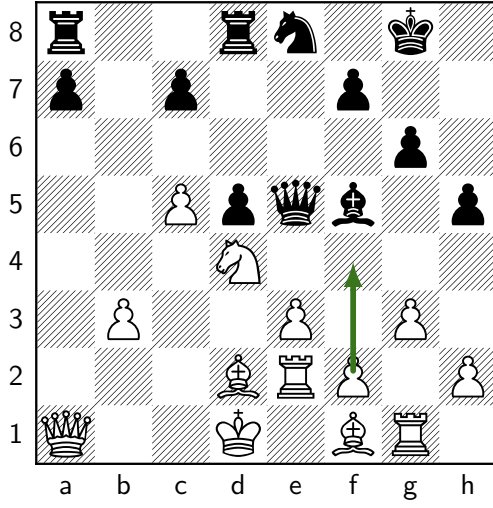


(b)

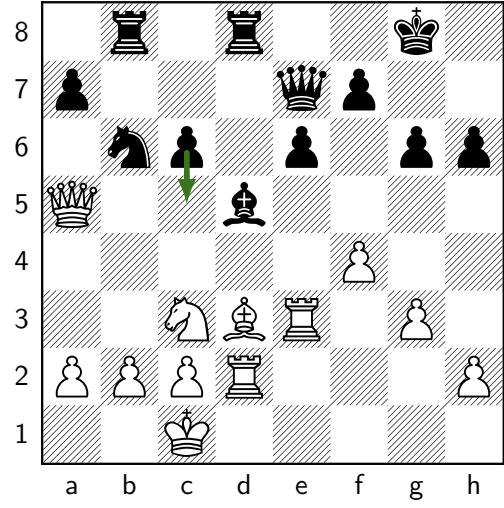


(c)

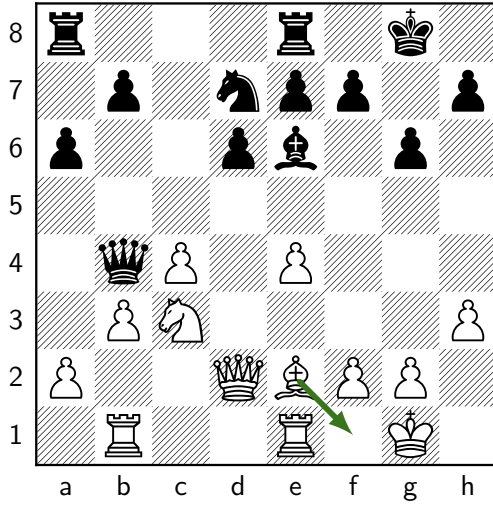
Figure 18: **Move 24, White: Qxa1**. (a) Our position; Melanie plays Qxa1 (a5a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#36` (ply 46) [structural distance  $d=114$ , QAP obj=77]: the master played Qe4; under  $\mu$  our Qa5 plays the role of that game's Qe2; cross-type roles: our Bf1 as their Ra1, our Bd2 as their Nd2; transferred move a5a1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2458` (ply 44) [structural distance  $d=108$ , QAP obj=34]: the master played Qc2; under  $\mu$  our Qa5 plays the role of that game's Qb3; cross-type roles: our Rg1 as their Pa2, our Pb3 as their Bg2; transferred move a5a1 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Qxa1 (summed QAP objective 110, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



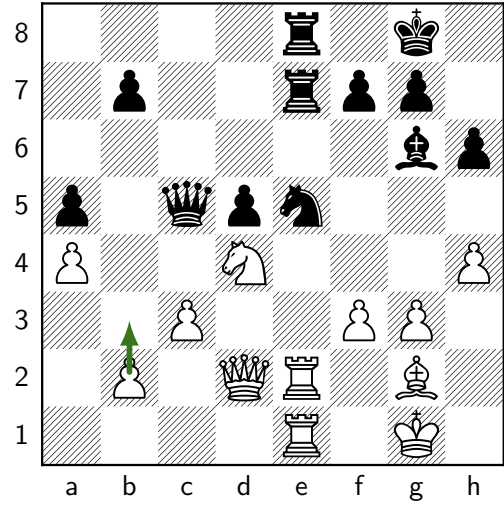
(a)



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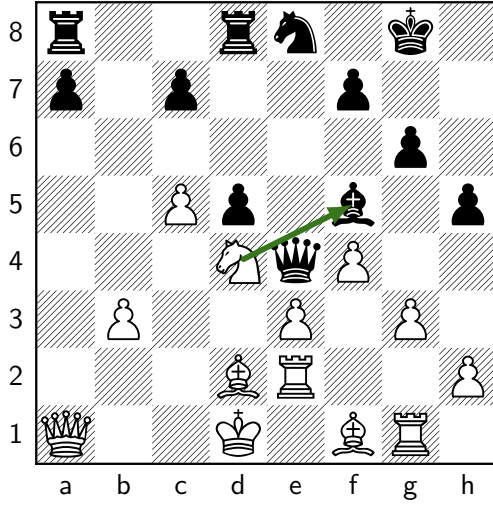


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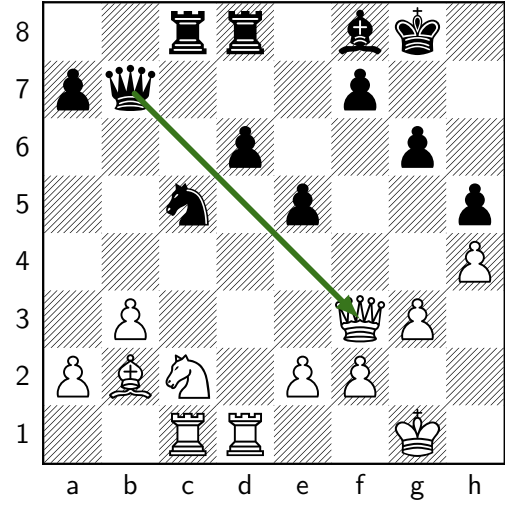


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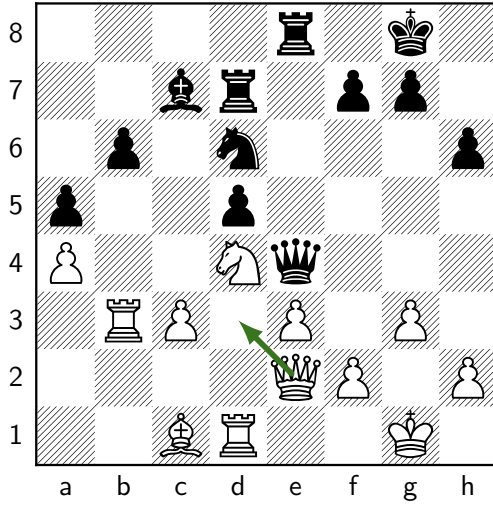
Figure 19: **Move 25, White: f4.** (a) Our position; Melanie plays f4 (f2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2718` (ply 47) [structural distance  $d=76$ , QAP obj=82]: the master played c5; under  $\mu$  our *Pf2* plays the role of that game’s *Pc3*; cross-type roles: our *Qa1* as their *Rd8*, our *Bd2* as their *Qe7*; transferred move f2f4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1167` (ply 40) [structural distance  $d=78$ , QAP obj=115]: the master played Bf1; under  $\mu$  our *Bd2* plays the role of that game’s *Be2*; cross-type roles: our *Qa1* as their *Re1*, our *Rg1* as their *Qd2*; transferred move d2e1. (d) Retrieved from `ccrl_top_p0002#2480` (ply 54) [structural distance  $d=78$ , QAP obj=88]: the master played b3; under  $\mu$  our *Ph2* plays the role of that game’s *Pb2*; cross-type roles: our *Qa1* as their *Re1*, our *Rg1* as their *Qd2*; transferred move h2h4. Synthesis: 1 retrieved analog transferred to f4 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



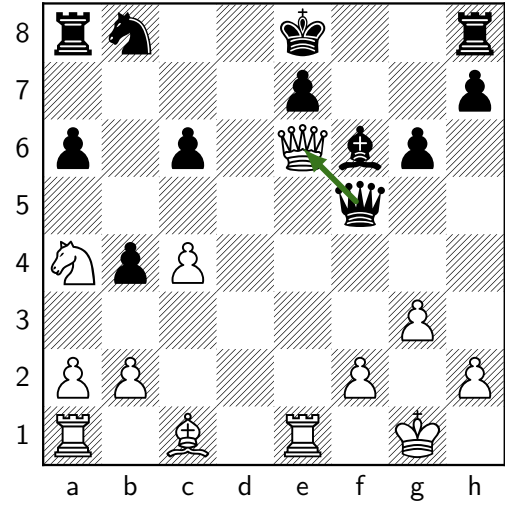
(a)



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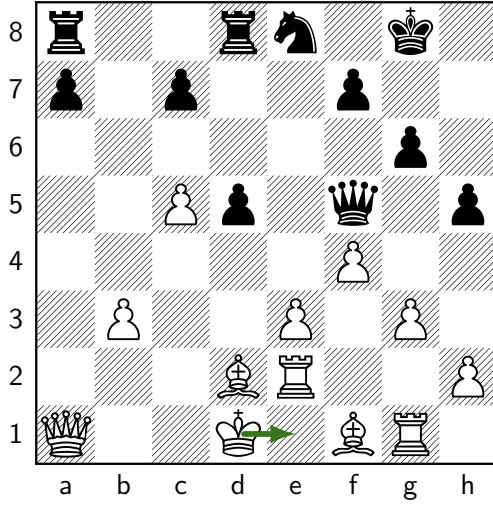


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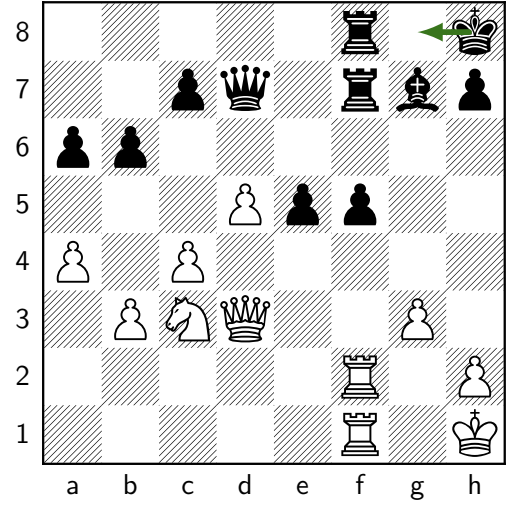


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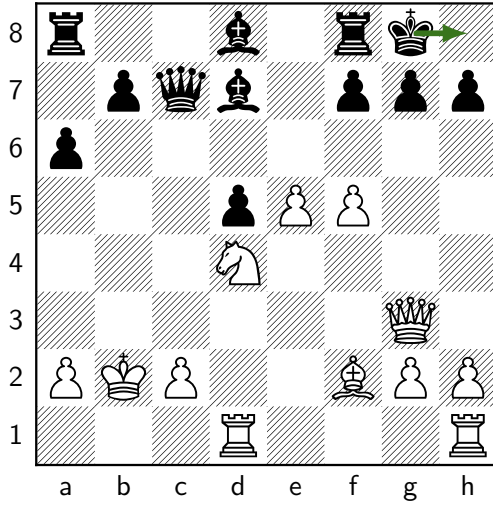
Figure 20: **Move 26, White: Nxf5**. (a) Our position; Melanie plays Nxf5 (d4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2314` (ply 55) [structural distance  $d=70$ , QAP obj=43]: the master played Qxf3; under  $\mu$  our Nd4 plays the role of that game’s Qb2 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Rc8, our Pc5 as their Bf8; transferred move d4f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2533` (ply 60) [structural distance  $d=68$ , QAP obj=92]: the master played Qd3; under  $\mu$  our Bd2 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Re2 as their Bc1, our Qa1 as their Rd1; transferred move d2a5. (d) Retrieved from `ccr1_top_p0002#3488` (ply 37) [structural distance  $d=79$ , QAP obj=72]: the master played Qxe6; under  $\mu$  our Nd4 plays the role of that game’s Qf4 (a **cross-type** correspondence); cross-type roles: our ph5 as their Ra8, our Re2 as their Nb8; transferred move d4f3. Synthesis: 1 retrieved analog transferred to Nxf5 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



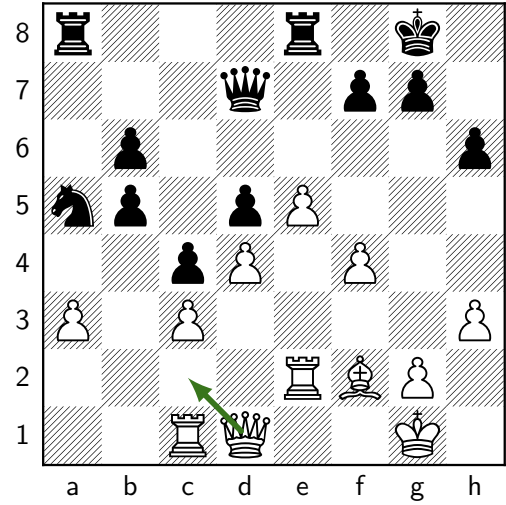
(a)



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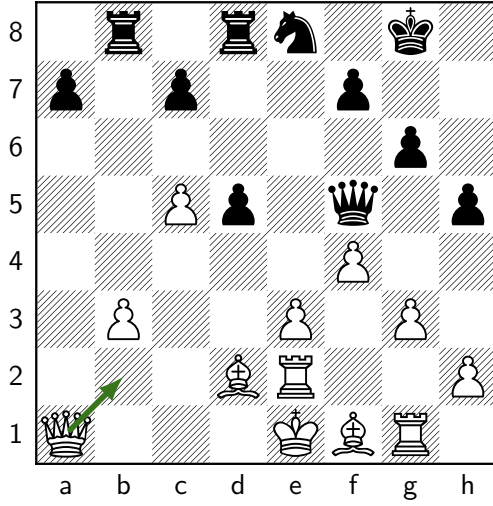


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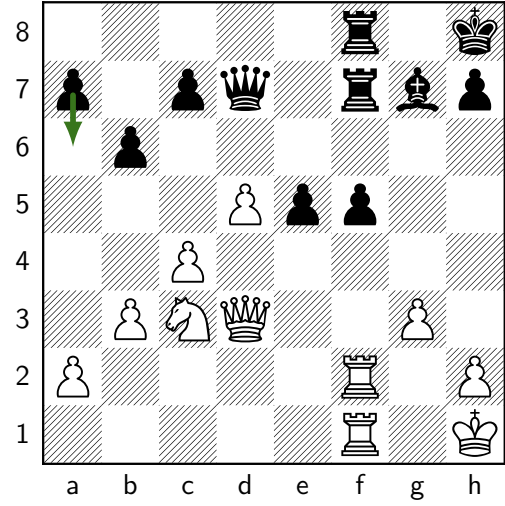


(d)

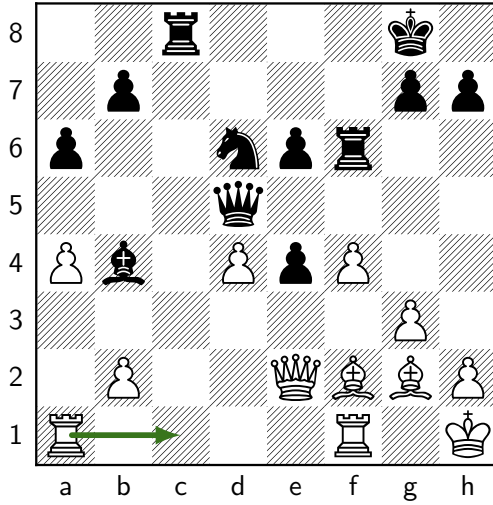
Figure 21: **Move 27, White: Ke1.** (a) Our position; Melanie plays Ke1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#202` (ply 63) [structural distance  $d=78$ , QAP obj=41]: the master played Kg8; under  $\mu$  our Kd1 plays the role of that game’s Kh1; cross-type roles: our Qa1 as their Rf8, our Bf1 as their Pc7; transferred move d1e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3551` (ply 41) [structural distance  $d=80$ , QAP obj=27]: the master played Kh8; under  $\mu$  our Kd1 plays the role of that game’s Kg1; cross-type roles: our pf7 as their Bd8, our Pe3 as their Rf8; transferred move d1c1. (d) Retrieved from `ccr1_top_p0000#3535` (ply 50) [structural distance  $d=75$ , QAP obj=18]: the master played Qc2; under  $\mu$  our Qa1 plays the role of that game’s Qd1; cross-type roles: our Bf1 as their Re2, our Pc5 as their Bf2; transferred move a1a4. Synthesis: 1 retrieved analog transferred to Ke1 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



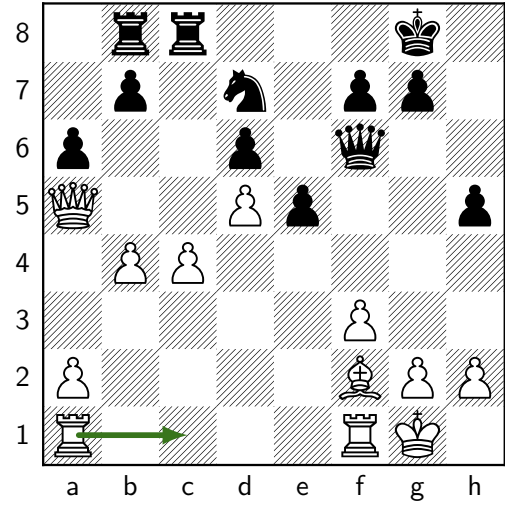
(a)



(b)



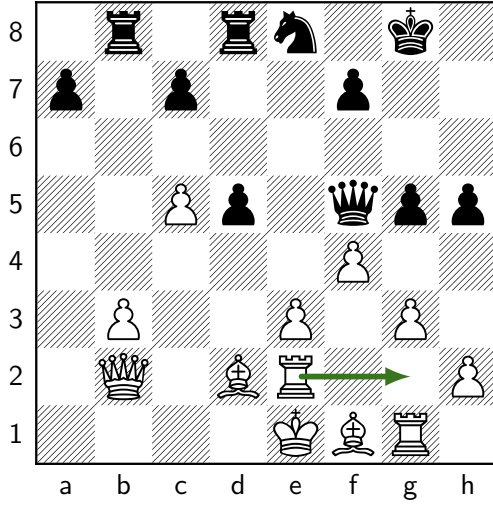
(c)



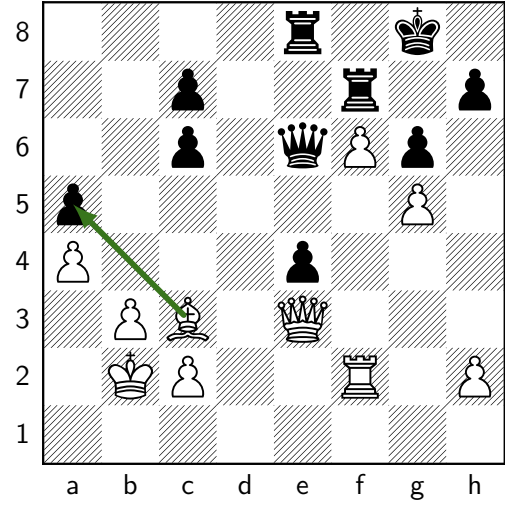
(d)

Figure 22: **Move 28, White: Qb2.** (a) Our position; Melanie plays Qb2 (a1b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#202` (ply 61) [structural distance  $d=80$ , QAP obj=36]: the master played a6; under  $\mu$  our *Qa1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Bd2* as their *Rf8*, our *pg6* as their *Rf7*; transferred move a1b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1356` (ply 48) [structural distance  $d=88$ , QAP obj=38]: the master played Rac1; under  $\mu$  our *Qa1* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Bd2* as their *Rf1*, our *Pg3* as their *Bf2*; transferred move a1b1. (d) Retrieved from `ccr1_top_p0002#1052` (ply 40) [structural distance  $d=86$ , QAP obj=36]: the master played Rac1; under  $\mu$  our *Qa1* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Bf1* as their *Ph2*, our *ne8* as their *ph5*; transferred move a1c1. Synthesis: 1 retrieved analog transferred to Qb2 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

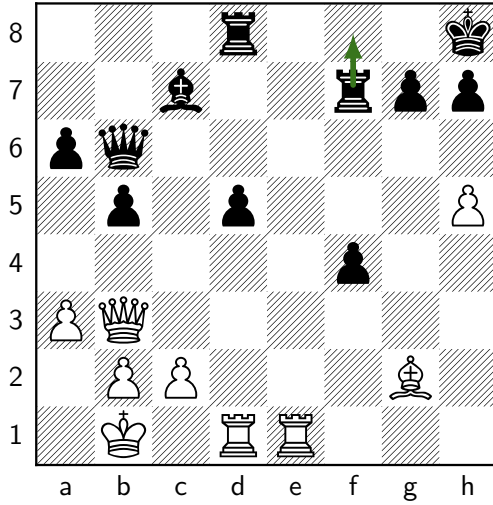




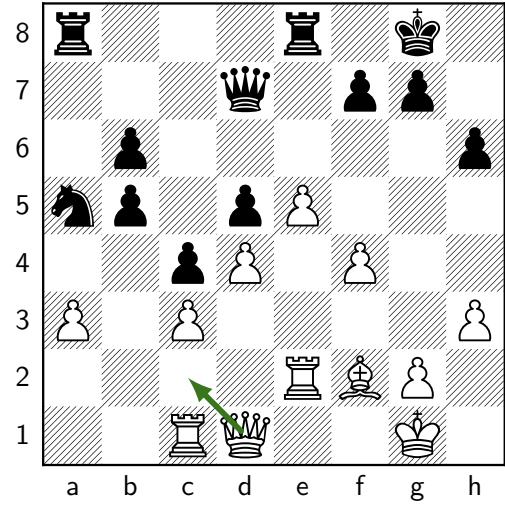
(a)



(b)

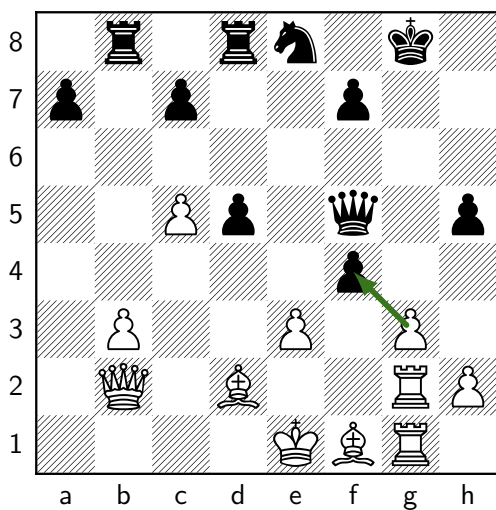


(c)

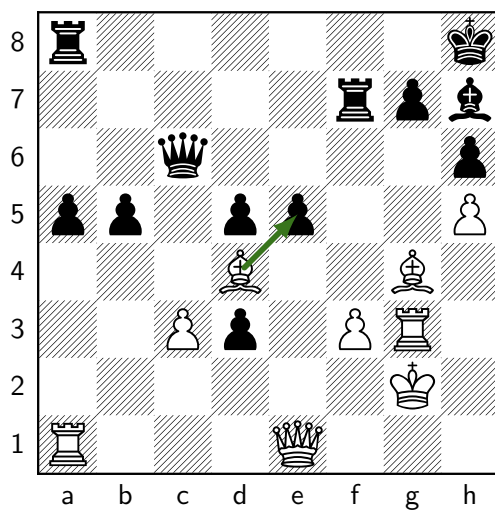


(d)

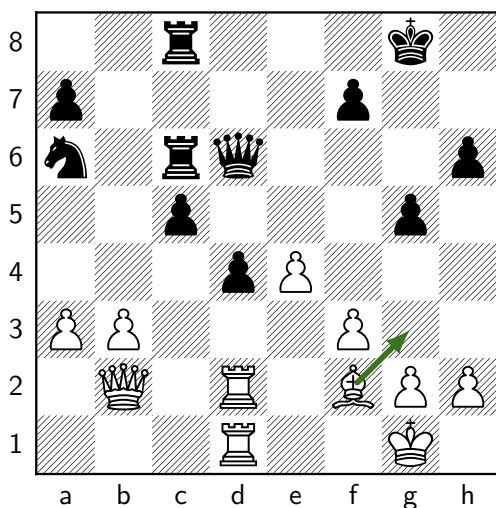
Figure 23: **Move 29, White: Reg2.** (a) Our position; Melanie plays Reg2 (e2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1442` (ply 70) [structural distance  $d=96$ , QAP obj=102]: the master played Bxa5; under  $\mu$  our *Re2* plays the role of that game’s *Bc3* (a **cross-type** correspondence); cross-type roles: our *Bd2* as their *Rf2*, our *pg5* as their *qe6*; transferred move e2g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2313` (ply 55) [structural distance  $d=107$ , QAP obj=58]: the master played Rff8; under  $\mu$  our *Re2* plays the role of that game’s *Rf2*; cross-type roles: our *Bd2* as their *Rd8*, our *Rg1* as their *Bc7*; transferred move e2g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3535` (ply 50) [structural distance  $d=106$ , QAP obj=19]: the master played Qc2; under  $\mu$  our *Re2* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qb2* as their *Re2*, our *Rg1* as their *Pa3*; transferred move e2g2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Reg2 (summed QAP objective 179, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



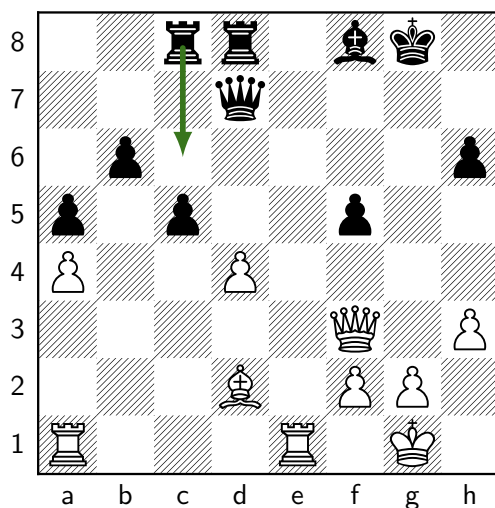
(a)



(b)

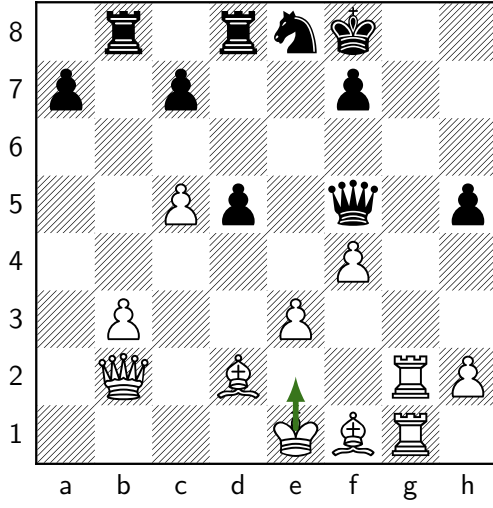


(c)

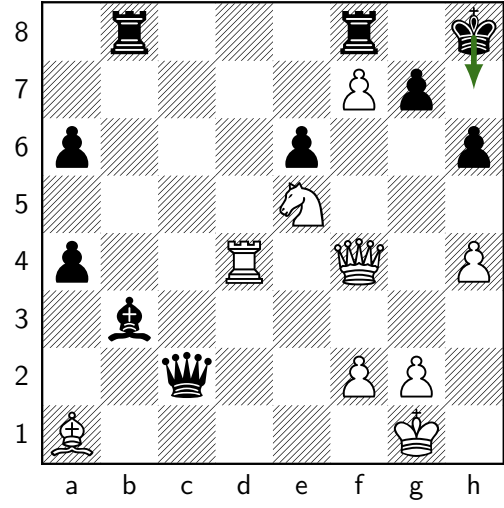


(d)

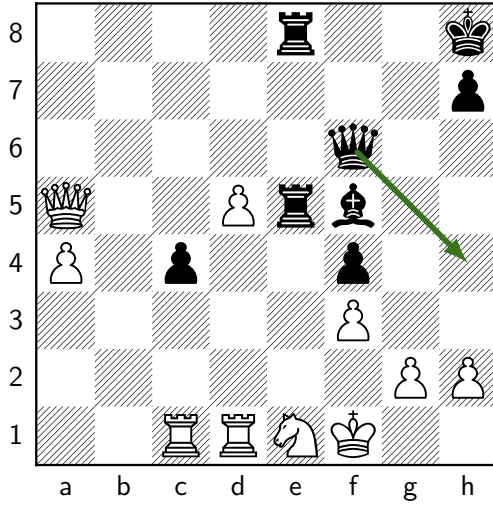
Figure 24: **Move 30, White: gxf4+.** (a) Our position; Melanie plays gxf4+ (g3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3717` (ply 68) [structural distance  $d=92$ , QAP obj=38]: the master played Bxe5; under  $\mu$  our  $Pg3$  plays the role of that game’s  $Bd4$  (a **cross-type** correspondence); cross-type roles: our  $rd8$  as their  $Qe1$ , our  $Rg2$  as their  $Pc3$ ; transferred move g3f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1201` (ply 60) [structural distance  $d=98$ , QAP obj=31]: the master played Bg3; under  $\mu$  our  $Pg3$  plays the role of that game’s  $Bf2$  (a **cross-type** correspondence); cross-type roles: our  $Bd2$  as their  $Rd1$ , our  $Pe3$  as their  $Qb2$ ; transferred move g3f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1970` (ply 53) [structural distance  $d=99$ , QAP obj=67]: the master played Rc6; under  $\mu$  our  $Bd2$  plays the role of that game’s  $Rc1$  (a **cross-type** correspondence); cross-type roles: our  $Ph2$  as their  $Bf8$ , our  $Pe3$  as their  $Qd7$ ; transferred move d2c3. Synthesis: 2 retrieved analogs transferred to gxf4+ (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



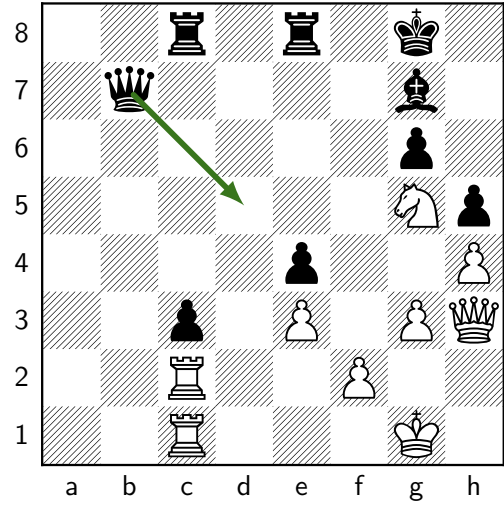
(a)



(b)

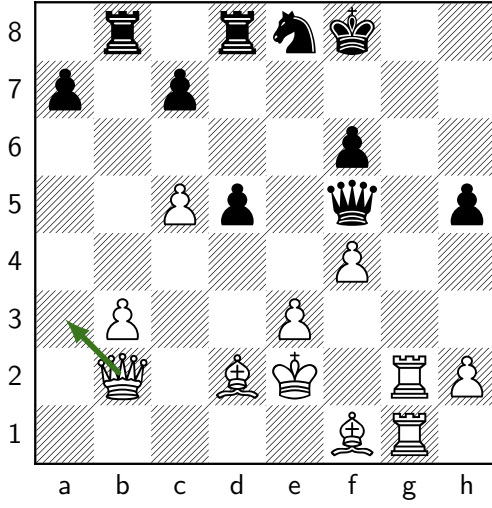


(c)

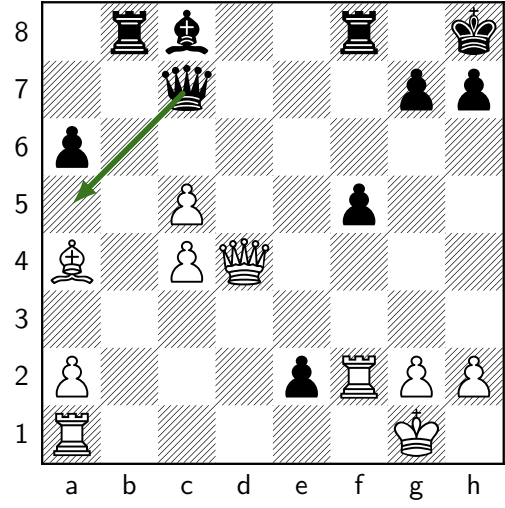


(d)

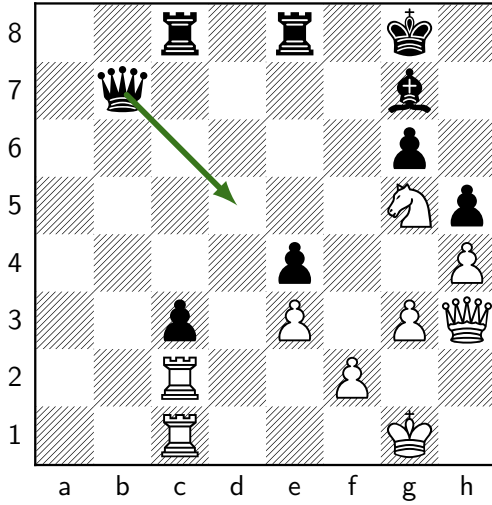
Figure 25: **Move 31, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1576` (ply 67) [structural distance  $d=95$ , QAP obj=44]: the master played Kh7; under  $\mu$  our *Ke1* plays the role of that game’s *Kh1*; cross-type roles: our *ph5* as their *Rb8*, our *qf5* as their *Rf8*; transferred move e1e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2987` (ply 67) [structural distance  $d=95$ , QAP obj=56]: the master played Qh4; under  $\mu$  our *Bd2* plays the role of that game’s *Qf3* (a **cross-type** correspondence); cross-type roles: our *pf7* as their *Re8*, our *Bf1* as their *Ph7*; transferred move d2b4. (d) Retrieved from `ccr1_top_p0001#2736` (ply 77) [structural distance  $d=91$ , QAP obj=53]: the master played Qd5; under  $\mu$  our *Pe3* plays the role of that game’s *Qb2* (a **cross-type** correspondence); cross-type roles: our *Pf4* as their *Rc8*, our *Bd2* as their *Re8*; transferred move e3e4. Synthesis: 1 retrieved analog transferred to Ke2 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



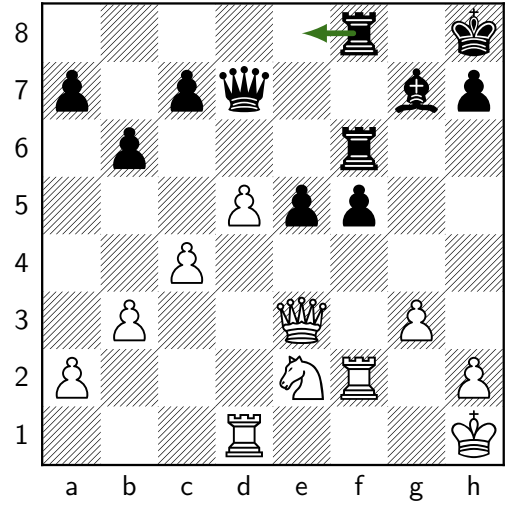
(a)



(b)

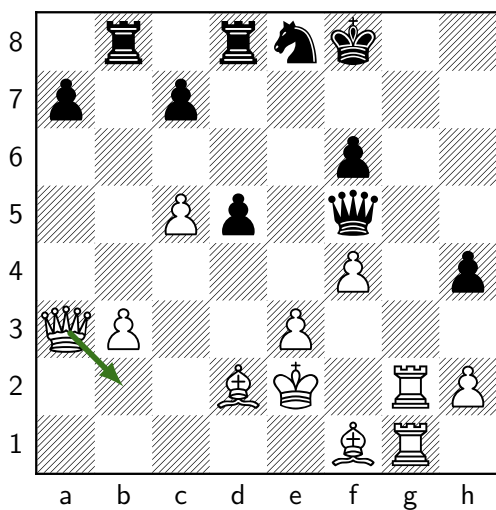


(c)

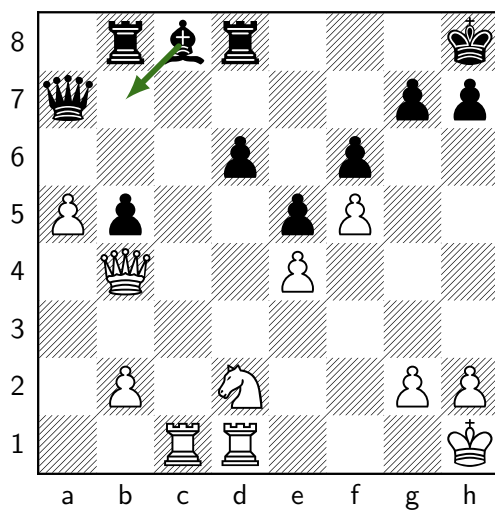


(d)

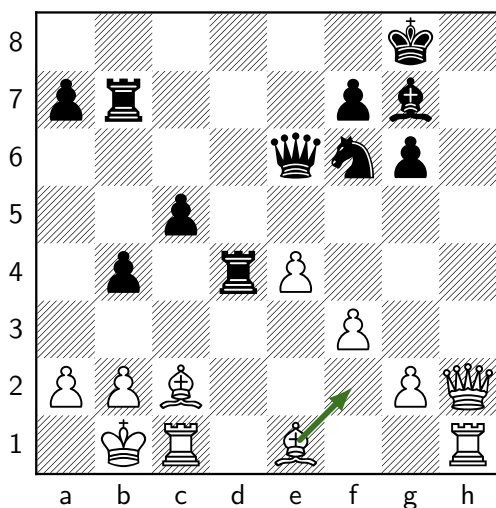
Figure 26: **Move 32, White: Qa3.** (a) Our position; Melanie plays Qa3 (b2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3793` (ply 47) [structural distance  $d=88$ , QAP obj=44]: the master played Qa5; under  $\mu$  our Qb2 plays the role of that game’s Qc2; cross-type roles: our Bd2 as their Rb8, our Pb3 as their Bc8; transferred move b2a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2736` (ply 77) [structural distance  $d=88$ , QAP obj=52]: the master played Qd5; under  $\mu$  our Qb2 plays the role of that game’s Qb2; cross-type roles: our Pf4 as their Rc8, our pa7 as their Bg7; transferred move b2d4. (d) Retrieved from `ccr1_top_p0001#202` (ply 55) [structural distance  $d=82$ , QAP obj=38]: the master played Re8; under  $\mu$  our Rg2 plays the role of that game’s Rf1; cross-type roles: our Bd2 as their Ph7, our ne8 as their Rf6; transferred move g2g3. Synthesis: 1 retrieved analog transferred to Qa3 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



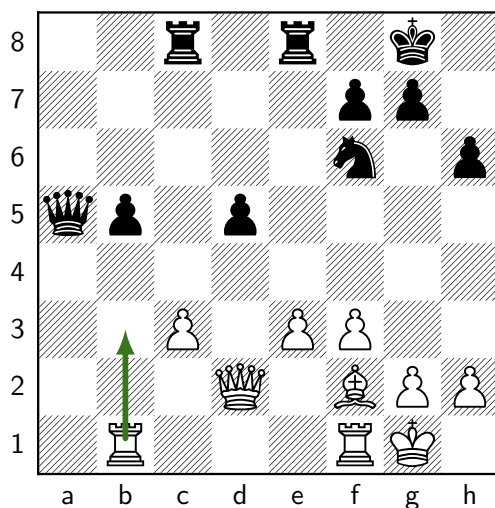
(a)



(b)

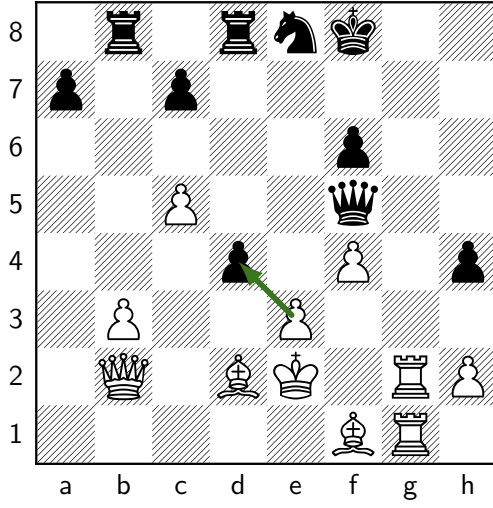


(c)

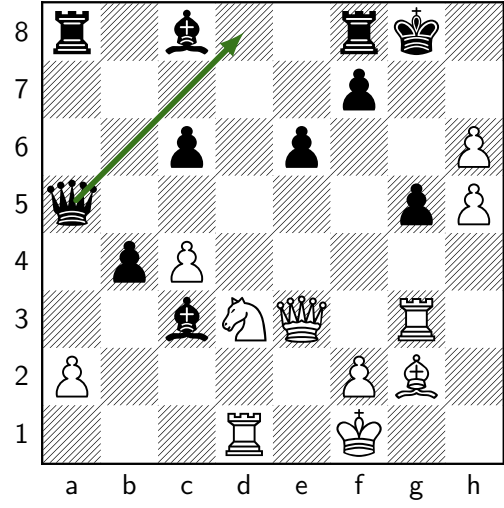


(d)

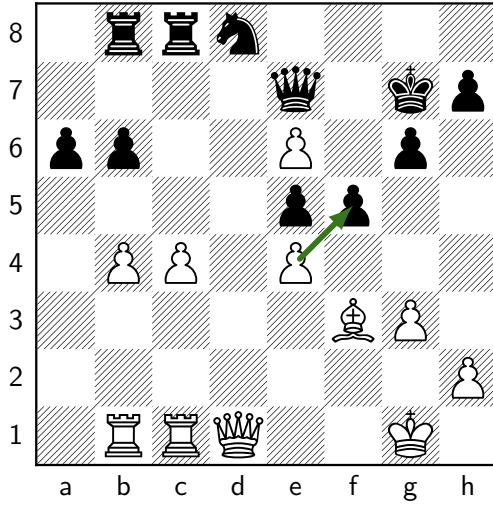
Figure 27: **Move 33, White: Qb2.** (a) Our position; Melanie plays Qb2 (a3b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#379` (ply 49) [structural distance  $d=78$ , QAP obj=52]: the master played Bb7; under  $\mu$  our  $Qa3$  plays the role of that game's Bc1 (a **cross-type** correspondence); cross-type roles: our Pe3 as their Rb8, our pf6 as their Qa7; transferred move a3b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1831` (ply 50) [structural distance  $d=85$ , QAP obj=49]: the master played Bf2; under  $\mu$  our  $Qa3$  plays the role of that game's Be1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Rc1, our Rg2 as their Bc2; transferred move a3b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2355` (ply 58) [structural distance  $d=69$ , QAP obj=33]: the master played Rb3; under  $\mu$  our Ph2 plays the role of that game's Rb1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Rf1, our Rg2 as their Bf2; transferred move h2h3. Synthesis: 2 retrieved analogs transferred to Qb2 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



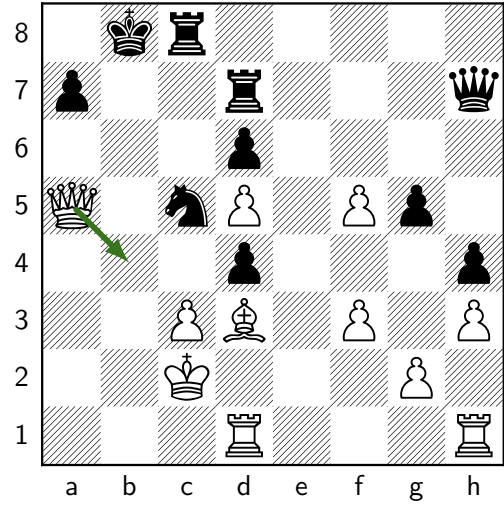
(a)



(b)

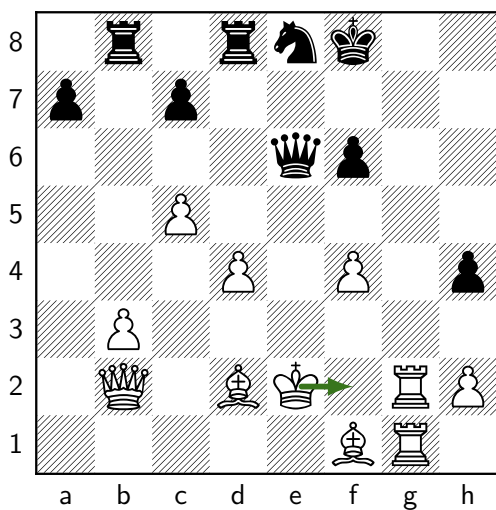


(c)

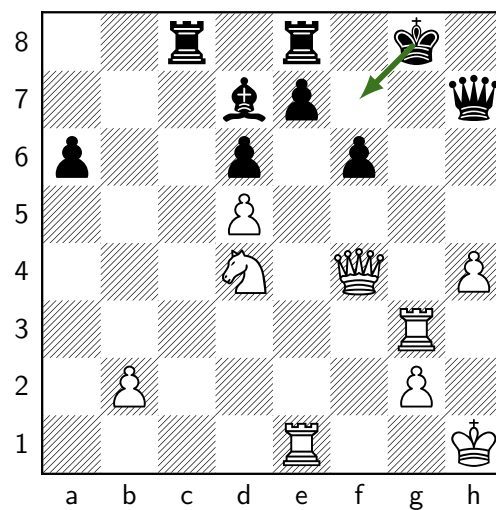


(d)

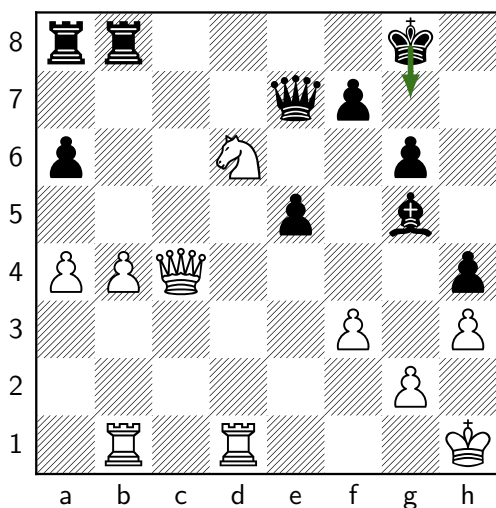
Figure 28: **Move 34, White: exd4.** (a) Our position; Melanie plays exd4 (e3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#2025 (ply 53) [structural distance  $d=95$ , QAP obj=51]: the master played Qd8; under  $\mu$  our Pe3 plays the role of that game’s Qa4 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Ra8, our ph4 as their Bc8; transferred move e3d4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#1845 (ply 56) [structural distance  $d=81$ , QAP obj=73]: the master played exf5; under  $\mu$  our Pe3 plays the role of that game’s Pe4; cross-type roles: our Bf1 as their Rc1, our pa7 as their Qd1; transferred move e3e4. (d) Retrieved from ccr1\_top\_p0002#3499 (ply 64) [structural distance  $d=74$ , QAP obj=66]: the master played Qb4+; under  $\mu$  our Qb2 plays the role of that game’s Qa5; cross-type roles: our Bd2 as their Rd1, our Pe3 as their Bd3; transferred move b2a3. Synthesis: 1 retrieved analog transferred to exd4 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



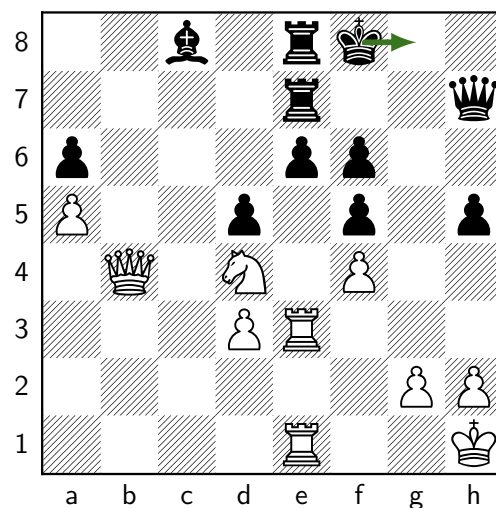
(a)



(b)



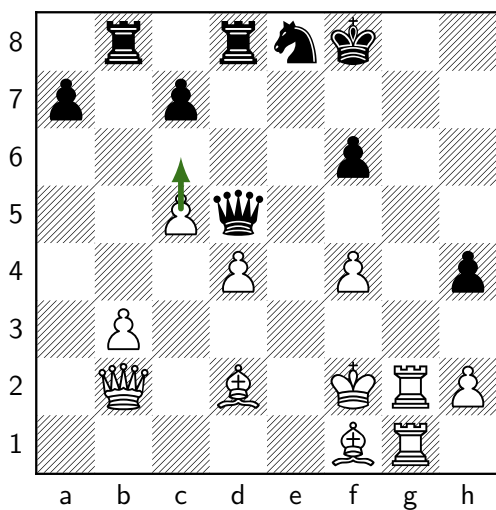
(c)



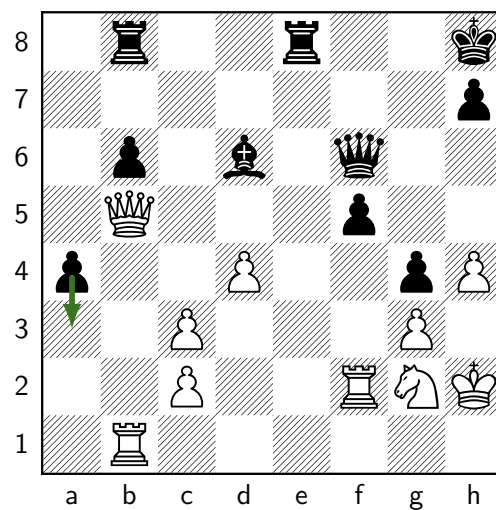
(d)

Figure 29: **Move 35, White: Kf2.** (a) Our position; Melanie plays Kf2 (e2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2544` (ply 77) [structural distance  $d=104$ , QAP obj=83]: the master played Kf7; under  $\mu$  our *Ke2* plays the role of that game’s *Kg1*; cross-type roles: our *ph4* as their *Rc8*, our *Pc5* as their *Bd7*; transferred move e2f2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1670` (ply 65) [structural distance  $d=96$ , QAP obj=31]: the master played Kg7; under  $\mu$  our *Ke2* plays the role of that game’s *Kg1*; cross-type roles: our *ne8* as their *Ra8*, our *qe6* as their *nd6*; transferred move e2d1. (d) Retrieved from `ccrl_top_p0003#1357` (ply 65) [structural distance  $d=99$ , QAP obj=12]: the master played Kg8; under  $\mu$  our *Ke2* plays the role of that game’s *Kf1*; cross-type roles: our *Rg2* as their *Bc8*, our *Bf1* as their *Re8*; transferred move e2d1. Synthesis: 1 retrieved analog transferred to Kf2 (summed QAP objective 83, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

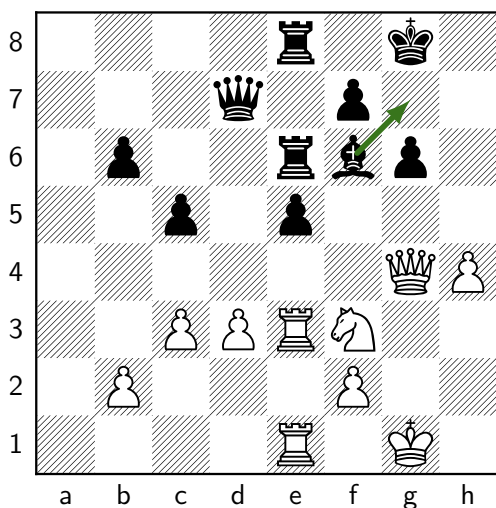




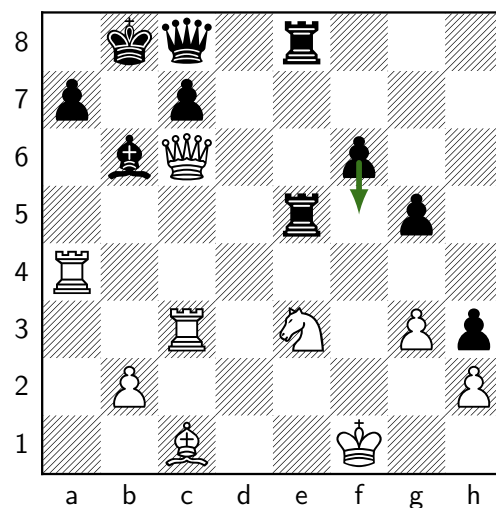
(a)



(b)

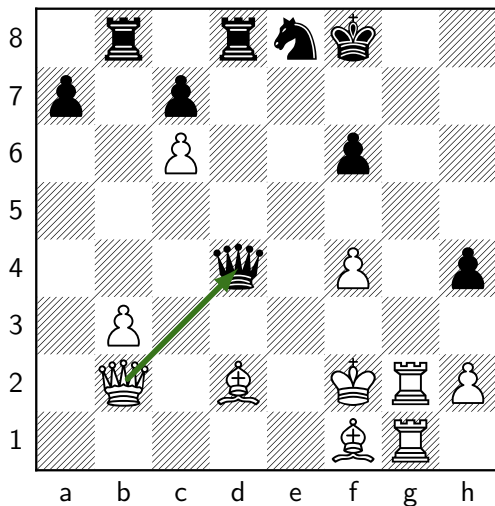


(c)

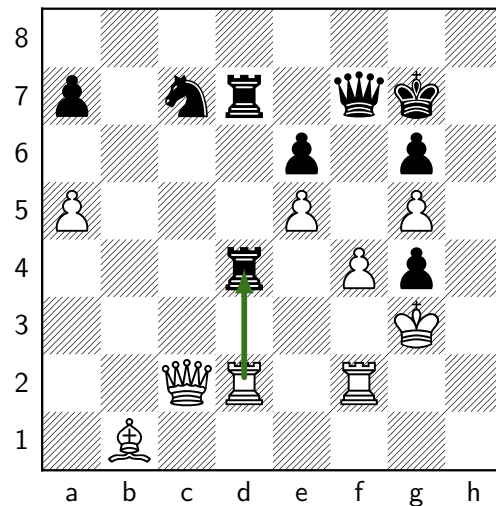


(d)

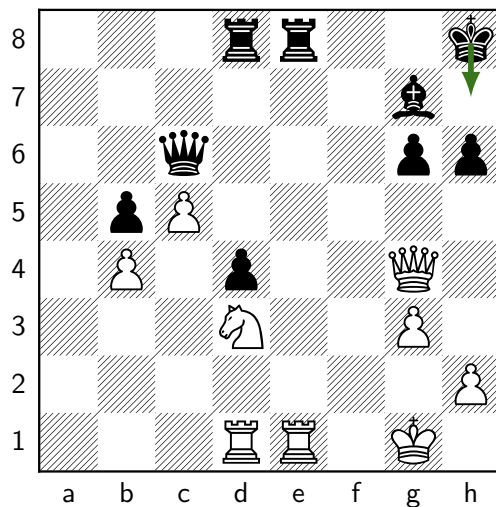
Figure 30: **Move 36, White: c6.** (a) Our position; Melanie plays c6 (c5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2549` (ply 61) [structural distance  $d=94$ , QAP obj=104]: the master played a3; under  $\mu$  our  $Pc5$  plays the role of that game’s  $Pa5$ ; cross-type roles: our  $Qb2$  as their  $Rb8$ , our  $Bd2$  as their  $Ph7$ ; transferred move c5c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2585` (ply 63) [structural distance  $d=100$ , QAP obj=26]: the master played Bg7; under  $\mu$  our  $Pc5$  plays the role of that game’s  $Bf3$  (a **cross-type** correspondence); cross-type roles: our  $Qb2$  as their  $Re8$ , our  $Rg1$  as their  $Qd7$ ; transferred move c5c6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1899` (ply 73) [structural distance  $d=105$ , QAP obj=84]: the master played f5; under  $\mu$  our  $Pb3$  plays the role of that game’s  $Pf3$ ; cross-type roles: our  $ne8$  as their  $Pa7$ , our  $Pc5$  as their  $Bb6$ ; transferred move b3b4. Synthesis: 2 retrieved analogs transferred to c6 (summed QAP objective 130, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

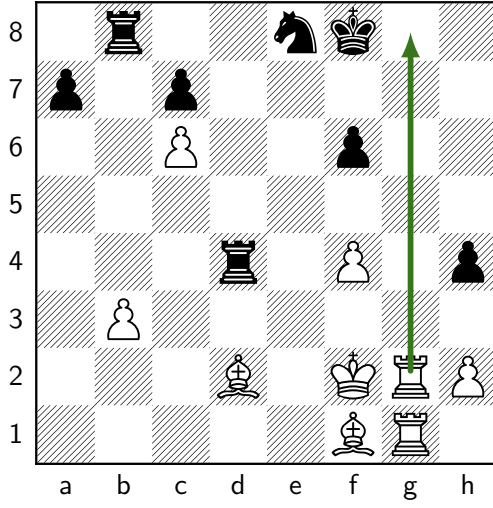


(b)

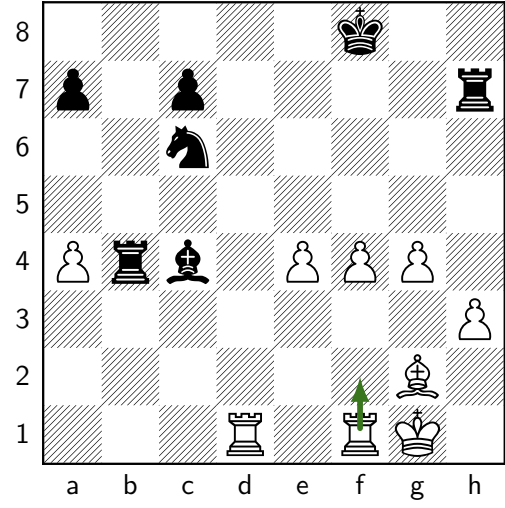


(c)

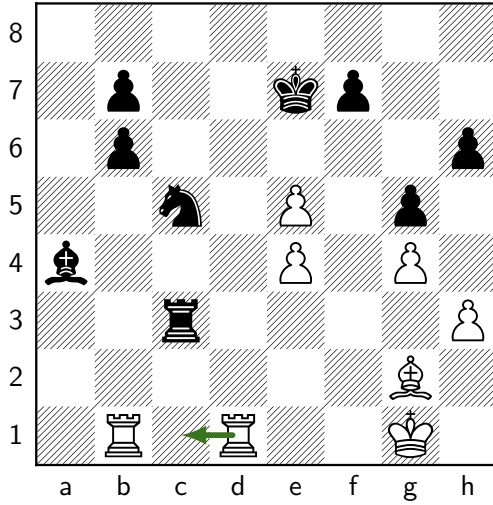
Figure 31: **Move 37, White: Qxd4.** (a) Our position; Melanie plays Qxd4 (b2d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2683` (ply 66) [structural distance  $d=98$ , QAP obj=19]: the master played Rxd4; under  $\mu$  our *Qb2* plays the role of that game's *Rd2* (a **cross-type** correspondence); cross-type roles: our *Pf4* as their *Bb1*, our *Rg2* as their *Qc2*; transferred move b2d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1015` (ply 79) [structural distance  $d=88$ , QAP obj=13]: the master played Kh7; under  $\mu$  our *Kf2* plays the role of that game's *Kh1*; cross-type roles: our *Pc6* as their *Rd8*, our *Rg1* as their *Bg7*; transferred move f2e2. Synthesis: 1 retrieved analog transferred to Qxd4 (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



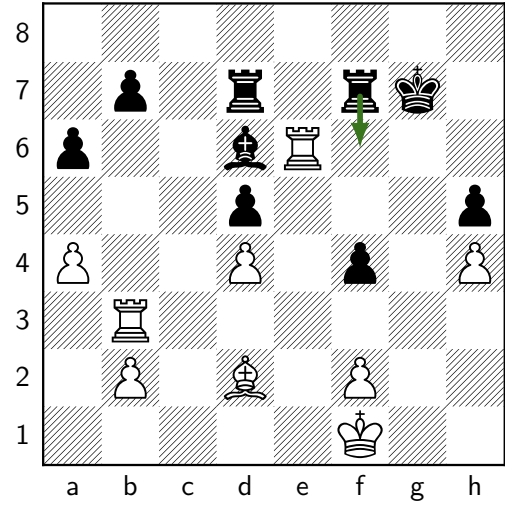
(a)



(b)

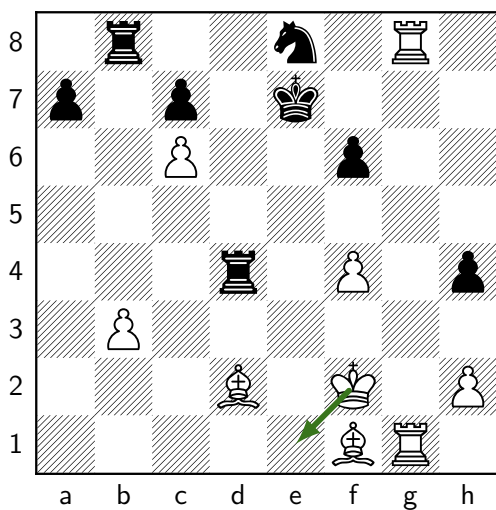


(c)

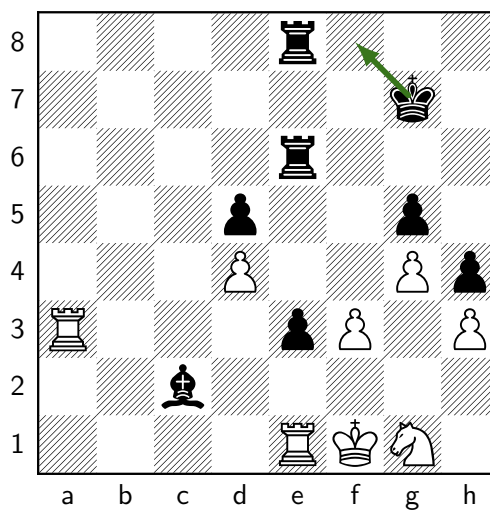


(d)

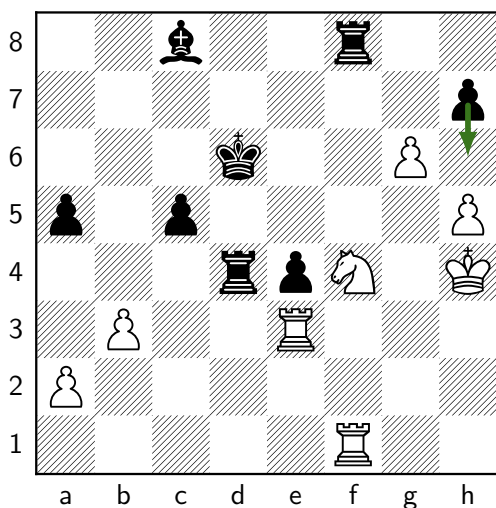
Figure 32: **Move 38, White: Rg8+.** (a) Our position; Melanie plays Rg8+ (g2g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#50` (ply 66) [structural distance  $d=67$ , QAP obj=20]: the master played Rf2; under  $\mu$  our Bd2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Rd1, our Rg1 as their Bg2; transferred move d2e1. (c) Retrieved from `ccr1_top_p0003#1337` (ply 72) [structural distance  $d=75$ , QAP obj=19]: the master played Rdc1; under  $\mu$  our Bd2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Rb1, our Pb3 as their Bg2; transferred move d2b4. (d) Retrieved from `ccr1_top_p0003#1093` (ply 75) [structural distance  $d=67$ , QAP obj=14]: the master played Rf6; under  $\mu$  our Rg2 plays the role of that game’s Rf2; cross-type roles: our Bf1 as their Pb7, our pf6 as their Rd7; transferred move g2g3. Synthesis: no retrieved analog transferred to Rg8+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



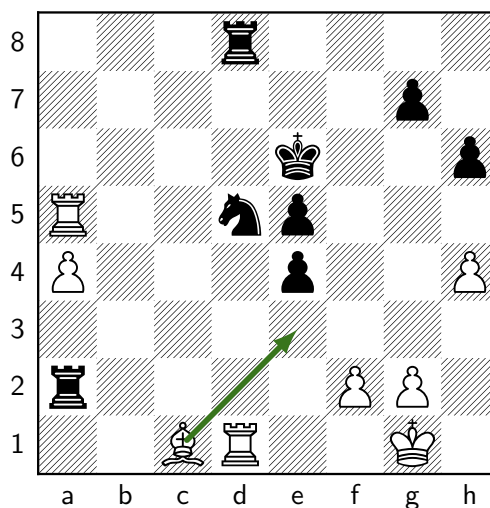
(a)



(b)

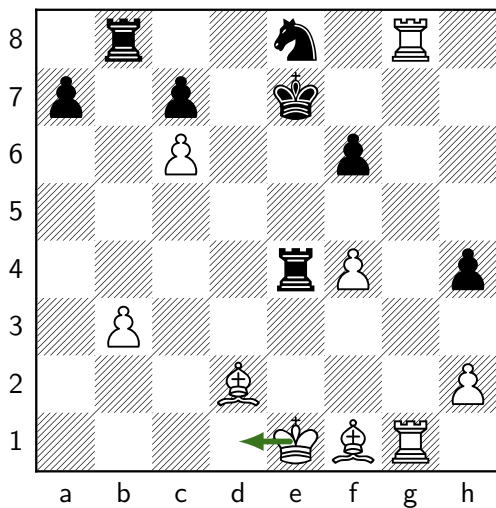


(c)

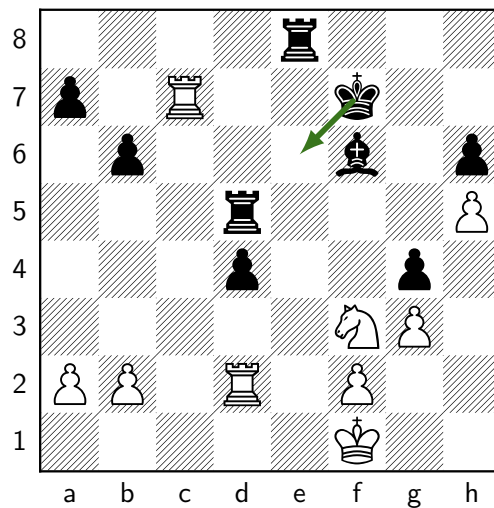


(d)

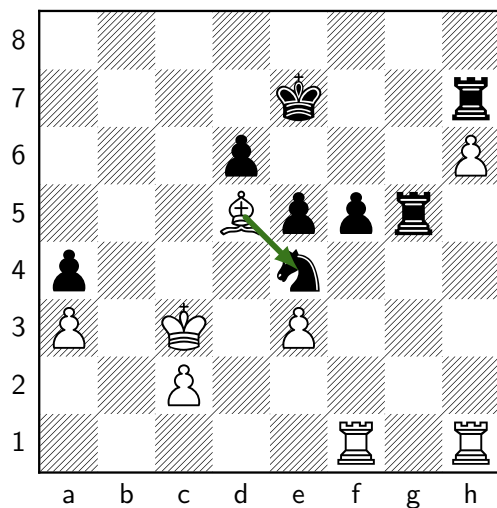
Figure 33: **Move 39, White: Ke1.** (a) Our position; Melanie plays Ke1 (f2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2683` (ply 93) [structural distance  $d=58$ , QAP obj=17]: the master played Kf8; under  $\mu$  our Kf2 plays the role of that game’s Kg2; cross-type roles: our pf6 as their Re8, our rd4 as their Pg5; transferred move f2e1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#336` (ply 91) [structural distance  $d=57$ , QAP obj=42]: the master played h6; under  $\mu$  our Pf4 plays the role of that game’s Ph2; cross-type roles: our Rg1 as their Bc8, our Bf1 as their Pa5; transferred move f4f5. (d) Retrieved from `ccrl_top_p0000#3205` (ply 66) [structural distance  $d=59$ , QAP obj=40]: the master played Be3; under  $\mu$  our Rg1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Pg2, our Bd2 as their Pa4; transferred move g1h1. Synthesis: 1 retrieved analog transferred to Ke1 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

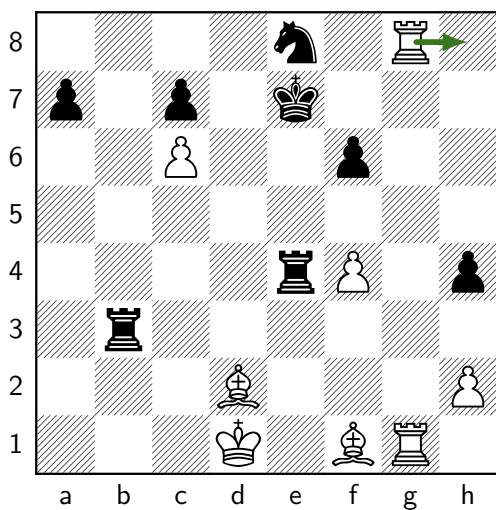


(b)

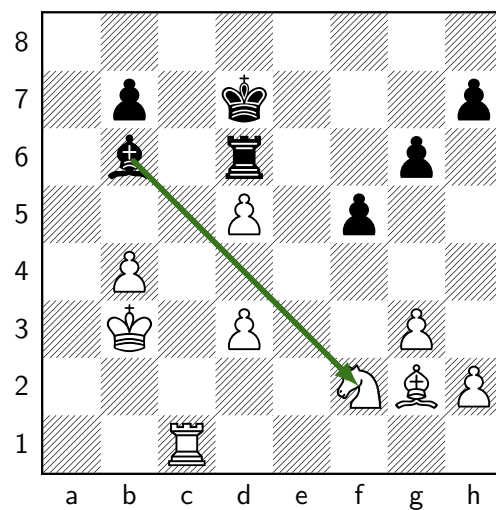


(c)

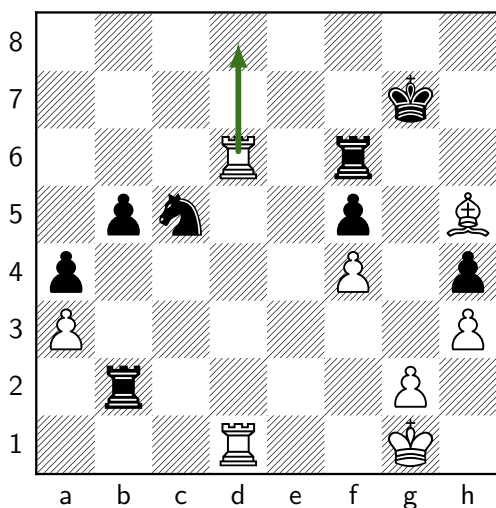
Figure 34: **Move 40, White: Kd1.** (a) Our position; Melanie plays Kd1 (e1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2623` (ply 77) [structural distance  $d=65$ , QAP obj=78]: the master played Ke6; under  $\mu$  our *Ke1* plays the role of that game’s *Kf2*; cross-type roles: our *Bf1* as their *Re8*, our *Bd2* as their *Pb6*; transferred move e1d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2876` (ply 70) [structural distance  $d=66$ , QAP obj=66]: the master played Bxe4; under  $\mu$  our *Bf1* plays the role of that game’s *Bd5*; cross-type roles: our *Pc6* as their *Rf1*, our *Bd2* as their *Pc2*; transferred move f1e2. Synthesis: 2 retrieved analogs transferred to Kd1 (summed QAP objective 132, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



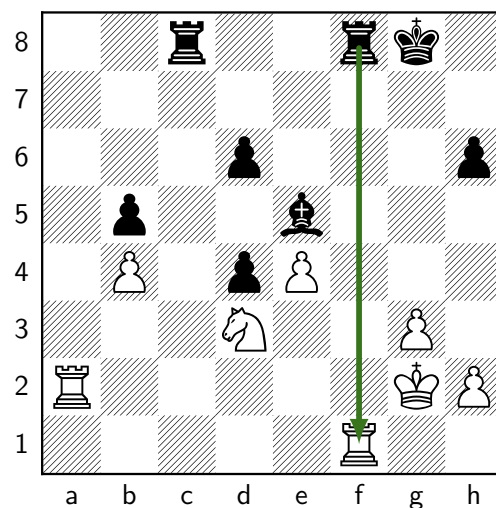
(a)



(b)

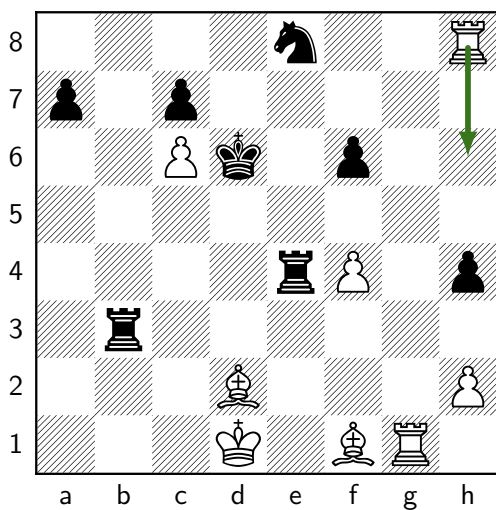


(c)

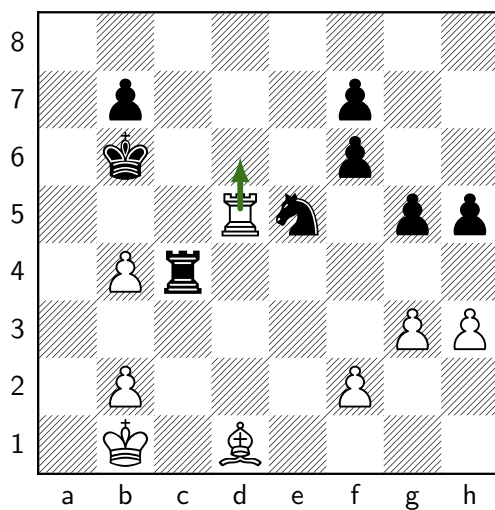


(d)

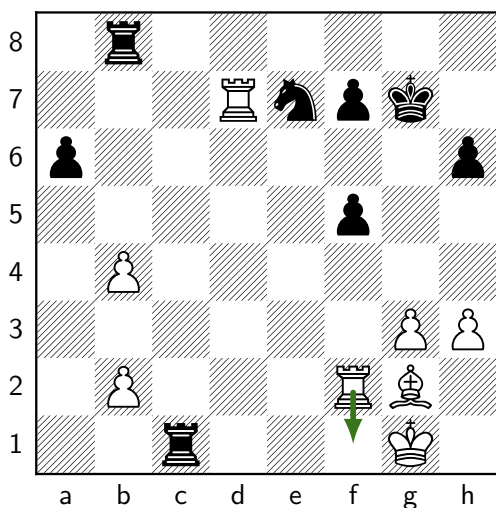
Figure 35: **Move 41, White: Rh8.** (a) Our position; Melanie plays Rh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3774` (ply 69) [structural distance  $d=55$ , QAP obj=24]: the master played Bxf2; under  $\mu$  our Rg8 plays the role of that game’s Bb3 (a **cross-type** correspondence); cross-type roles: our Rg1 as their Pb7, our Bd2 as their Ph7; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2974` (ply 92) [structural distance  $d=45$ , QAP obj=15]: the master played Rd8; under  $\mu$  our Rg8 plays the role of that game’s Rd6; cross-type roles: our rb3 as their Pg2, our Bf1 as their ph4; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#343` (ply 69) [structural distance  $d=42$ , QAP obj=28]: the master played Rxf1; under  $\mu$  our Pf4 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Rc8, our Bd2 as their Ph6; transferred move f4f5. Synthesis: 2 retrieved analogs transferred to Rh8 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



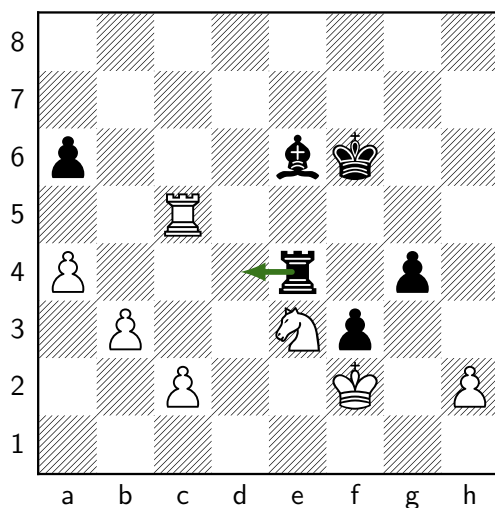
(a)



(b)



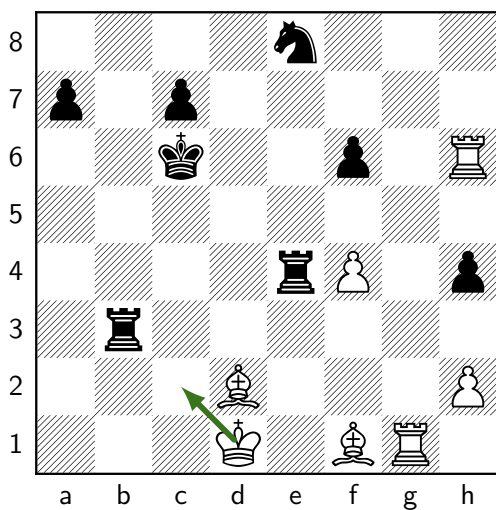
(c)



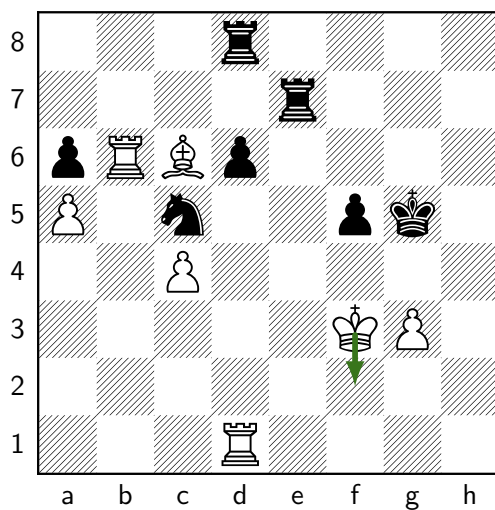
(d)

Figure 36: **Move 42, White: Rh6.** (a) Our position; Melanie plays Rh6 (h8h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2760` (ply 88) [structural distance  $d=51$ , QAP obj=15]: the master played Rd6+; under  $\mu$  our Rh8 plays the role of that game’s Rd5; cross-type roles: our Rg1 as their Bd1, our Bd2 as their Pb2; transferred move h8h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3017` (ply 84) [structural distance  $d=56$ , QAP obj=2]: the master played Rf1; under  $\mu$  our Rh8 plays the role of that game’s Rf2; cross-type roles: our Bd2 as their Pb2, our Pf4 as their Bg2; transferred move h8h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3129` (ply 87) [structural distance  $d=50$ , QAP obj=17]: the master played Rd4; under  $\mu$  our Rh8 plays the role of that game’s Re5; cross-type roles: our re4 as their pb3; transferred move h8h7. Synthesis: 2 retrieved analogs transferred to Rh6 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

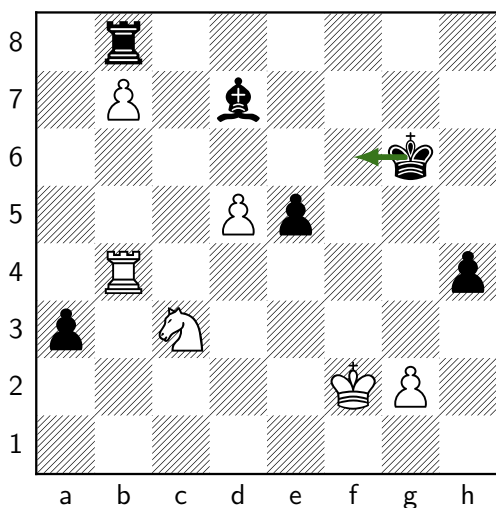




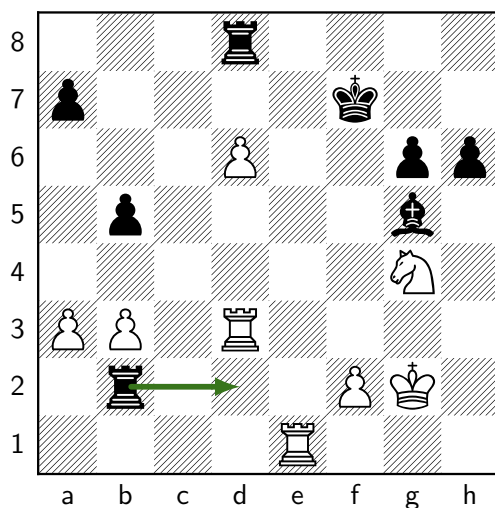
(a)



(b)

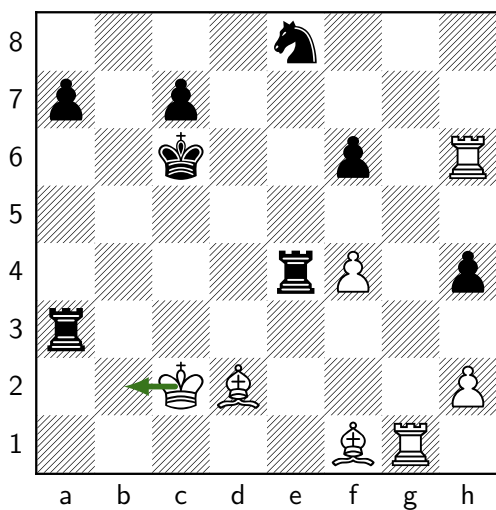


(c)

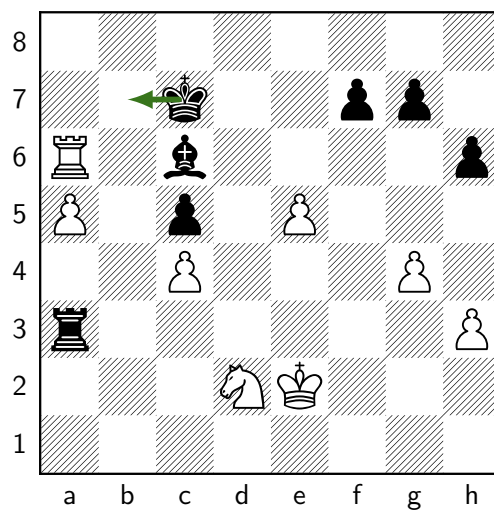


(d)

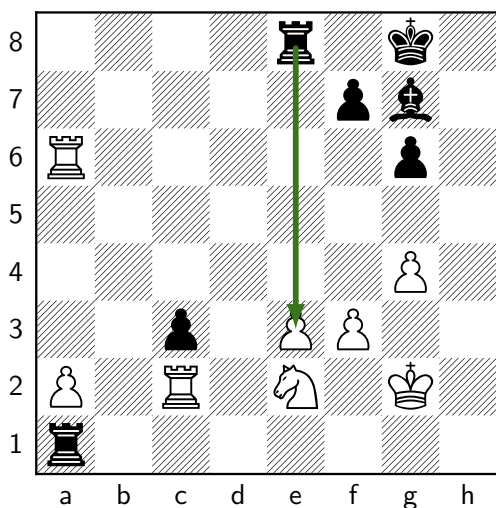
Figure 37: **Move 43, White: Kc2.** (a) Our position; Melanie plays Kc2 (d1c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#978` (ply 72) [structural distance  $d=35$ , QAP obj=29]: the master played Kf2; under  $\mu$  our Kd1 plays the role of that game’s Kf3; cross-type roles: our Rg1 as their Pc4, our Bd2 as their Pa5; transferred move d1c2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3471` (ply 97) [structural distance  $d=48$ , QAP obj=23]: the master played Kf6; under  $\mu$  our Kd1 plays the role of that game’s Kg3; cross-type roles: our rb3 as their Bd7, our ne8 as their rb4; transferred move d1c2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2122` (ply 71) [structural distance  $d=41$ , QAP obj=65]: the master played Rd2; under  $\mu$  our Rh6 plays the role of that game’s Rb7; cross-type roles: our Bf1 as their Pa7, our Rg1 as their Pg6; transferred move h6h7. Synthesis: 2 retrieved analogs transferred to Kc2 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



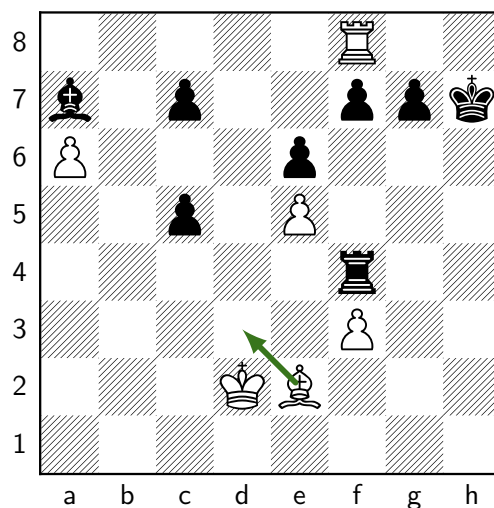
(a)



(b)

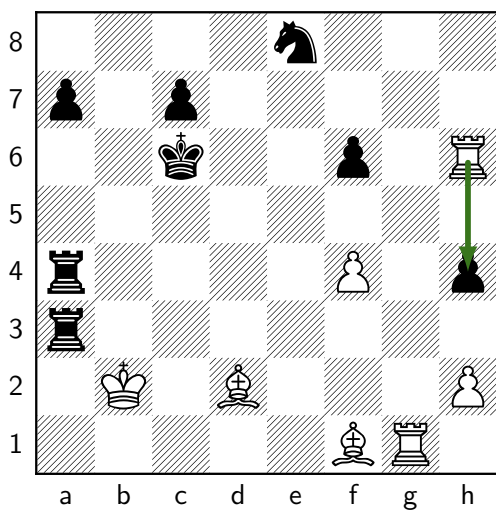


(c)

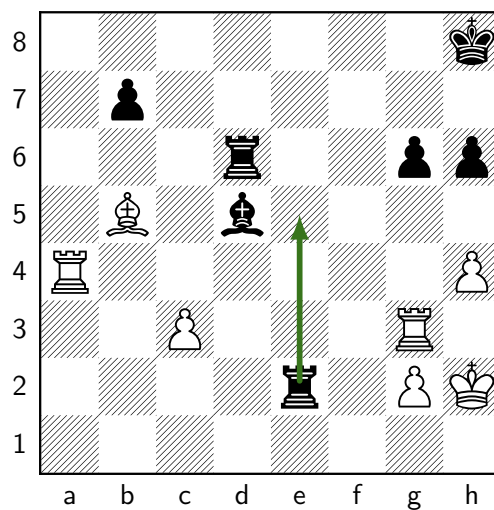


(d)

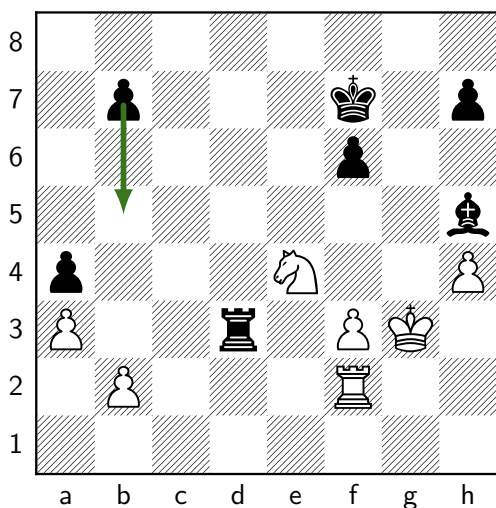
Figure 38: **Move 44, White: Kb2.** (a) Our position; Melanie plays Kb2 (c2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3687` (ply 97) [structural distance  $d=46$ , QAP obj=39]: the master played Kb7; under  $\mu$  our *Kc2* plays the role of that game's *Kc2*; cross-type roles: our *Bd2* as their *Pf7*, our *Pf4* as their *Bc6*; transferred move c2b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1587` (ply 83) [structural distance  $d=46$ , QAP obj=45]: the master played Rxe3; under  $\mu$  our *Ph2* plays the role of that game's *Re1* (a **cross-type** correspondence); cross-type roles: our *ne8* as their *ra6*, our *Rg1* as their *Pg6*; transferred move h2h3. (d) Retrieved from `ccr1_top_p0000#312` (ply 94) [structural distance  $d=49$ , QAP obj=42]: the master played Bd3+; under  $\mu$  our *Bd2* plays the role of that game's *Be2*; cross-type roles: our *ra3* as their *pc5*, our *ne8* as their *Pe5*; transferred move d2c1. Synthesis: 1 retrieved analog transferred to Kb2 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



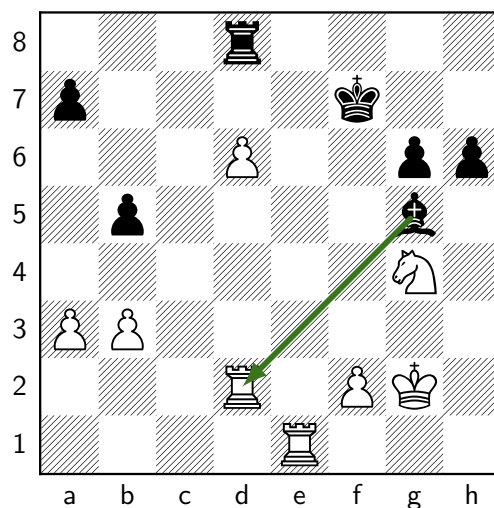
(a)



(b)

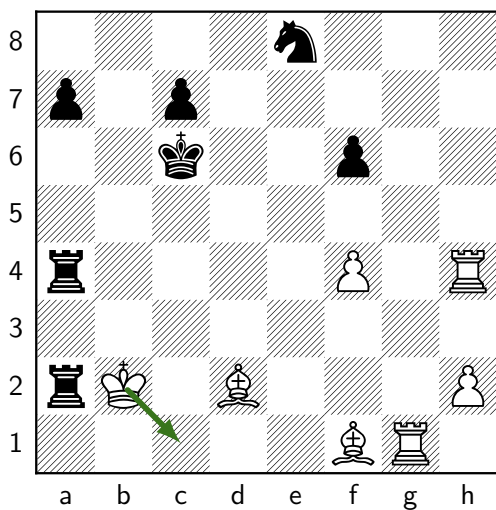


(c)

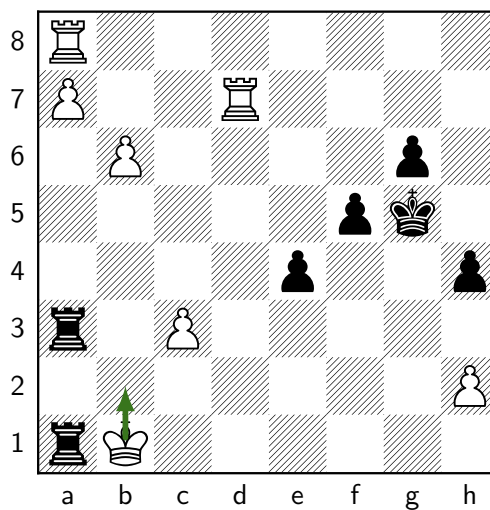


(d)

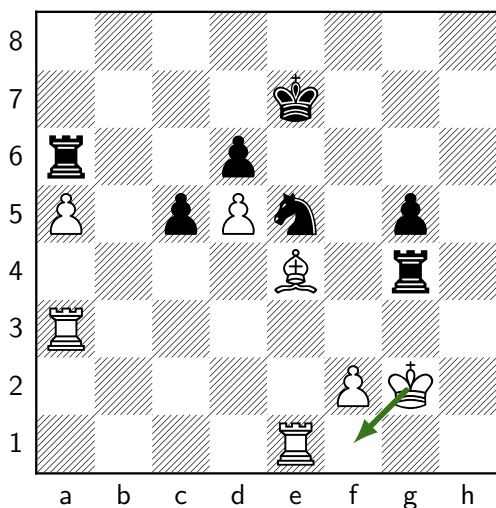
Figure 39: **Move 45, White: Rxh4**. (a) Our position; Melanie plays Rxh4 (h6h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2516` (ply 95) [structural distance  $d=39$ , QAP obj=24]: the master played Re5; under  $\mu$  our Rh6 plays the role of that game’s Re7; cross-type roles: our Bd2 as their Pb7, our Bf1 as their Rd6; transferred move h6f6. (c) Retrieved from `ccr1_top_p0002#1548` (ply 93) [structural distance  $d=50$ , QAP obj=23]: the master played b5; under  $\mu$  our Bf1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Rg1 as their Ph7, our Pf4 as their Bh5; transferred move f1e2. (d) Retrieved from `ccr1_top_p0003#2122` (ply 73) [structural distance  $d=47$ , QAP obj=23]: the master played Bxd2; under  $\mu$  our Bf1 plays the role of that game’s Bg4; cross-type roles: our Bd2 as their Pa7, our Ra4 as their Pg6; transferred move f1a6. Synthesis: no retrieved analog transferred to Rxh4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



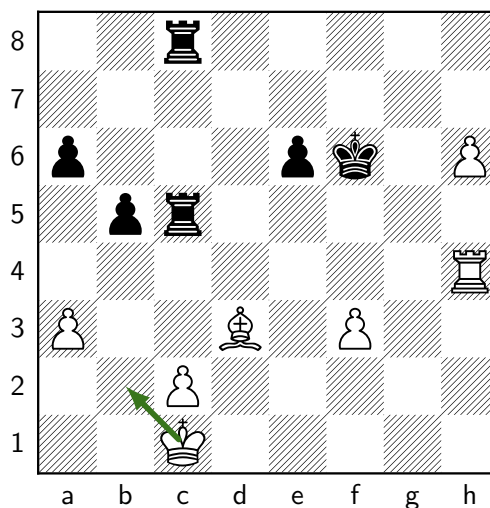
(a)



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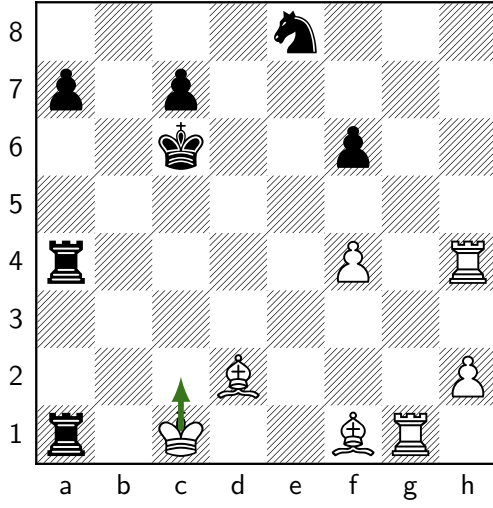


(c)

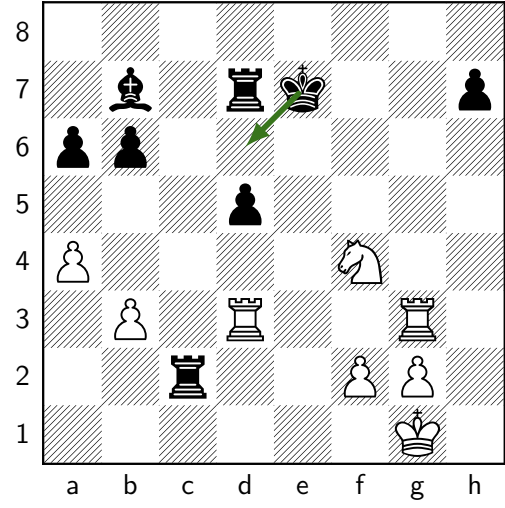


(d)

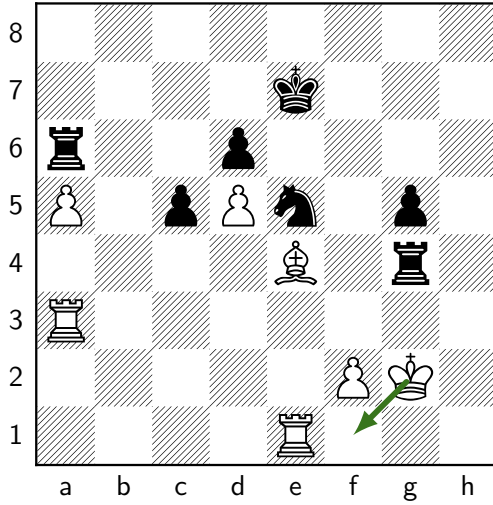
Figure 40: **Move 46, White: Kc1.** (a) Our position; Melanie plays Kc1 (b2c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1776` (ply 104) [structural distance  $d=50$ , QAP obj=95]: the master played Kb2; under  $\mu$  our Kb2 plays the role of that game’s Kb1; cross-type roles: our ne8 as their Ph2, our Bf1 as their Pc3; transferred move b2b3. (c) Retrieved from `ccr1_top_p0002#2661` (ply 98) [structural distance  $d=42$ , QAP obj=36]: the master played Kf1; under  $\mu$  our Kb2 plays the role of that game’s Kg2; cross-type roles: our pc7 as their Re1, our Bf1 as their Pf2; transferred move b2b1. (d) Retrieved from `ccr1_top_p0001#2223` (ply 84) [structural distance  $d=50$ , QAP obj=18]: the master played Kb2; under  $\mu$  our Kb2 plays the role of that game’s Kc1; cross-type roles: our Bf1 as their Pa3, our Rg1 as their Pf3; transferred move b2b1. Synthesis: no retrieved analog transferred to Kc1 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



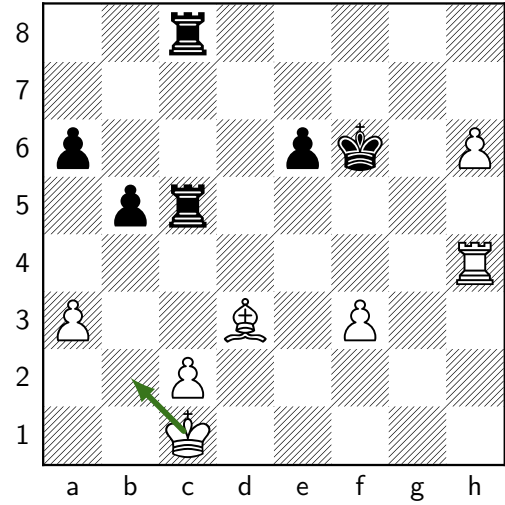
(a)



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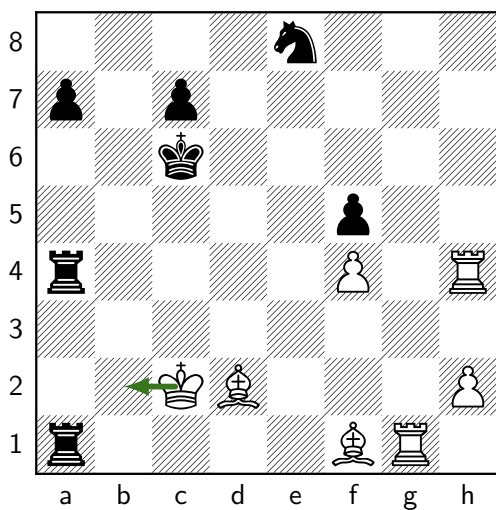


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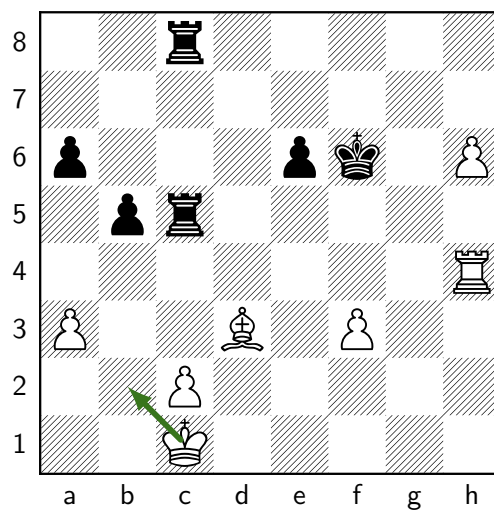


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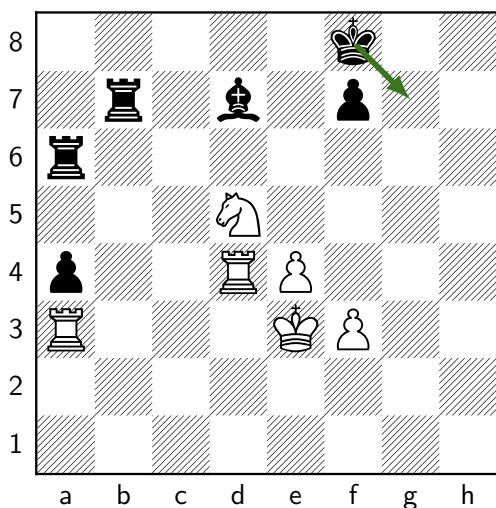
Figure 41: **Move 47, White: Kc2.** (a) Our position; Melanie plays Kc2 (c1c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2511` (ply 77) [structural distance  $d=54$ , QAP obj=15]: the master played Kd6; under  $\mu$  our *Kc1* plays the role of that game's *Ke2*; cross-type roles: our *Rh4* as their *Bb7*, our *Bd2* as their *Rd7*; transferred move c1c2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2661` (ply 98) [structural distance  $d=38$ , QAP obj=70]: the master played Kf1; under  $\mu$  our *Kc1* plays the role of that game's *Kg2*; cross-type roles: our *pc7* as their *Re1*, our *ne8* as their *Pf2*; transferred move c1b2. (d) Retrieved from `ccrl_top_p0001#2223` (ply 84) [structural distance  $d=55$ , QAP obj=18]: the master played Kb2; under  $\mu$  our *Kc1* plays the role of that game's *Kc1*; cross-type roles: our *Bf1* as their *Pa3*, our *Rg1* as their *Pf3*; transferred move c1b2. Synthesis: 1 retrieved analog transferred to Kc2 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



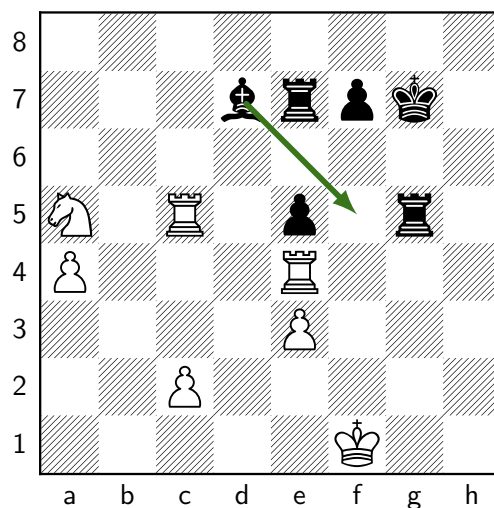
(a)



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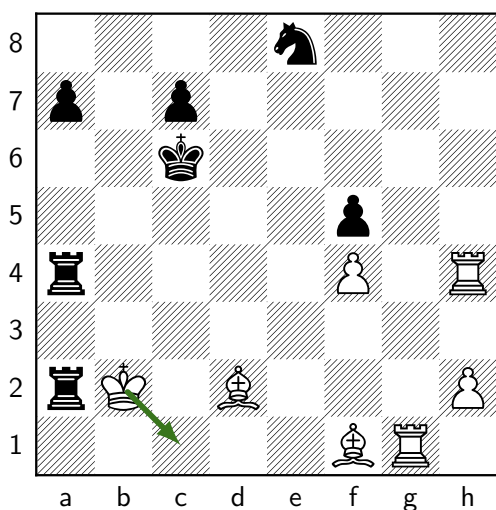


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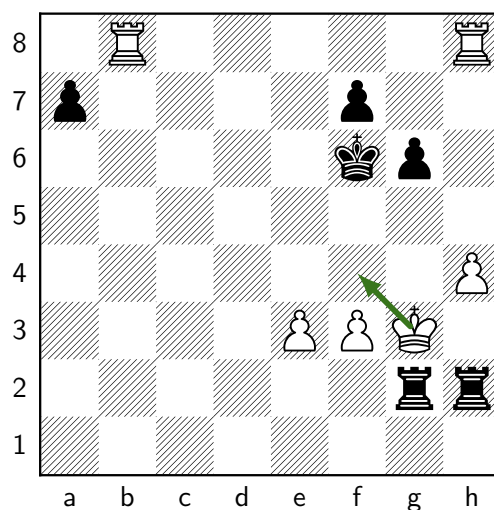


(d)

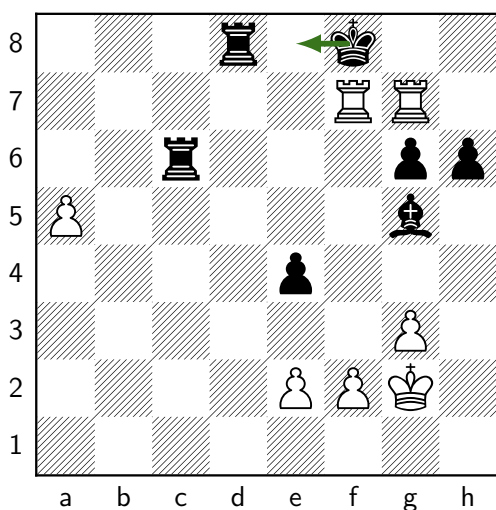
Figure 42: **Move 48, White: Kb2.** (a) Our position; Melanie plays Kb2 (c2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#2223 (ply 84) [structural distance  $d=44$ , QAP obj=29]: the master played Kb2; under  $\mu$  our Kc2 plays the role of that game's Kc1; cross-type roles: our Rh4 as their Pa3, our ra1 as their Bd3; transferred move c2b2 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#2991 (ply 79) [structural distance  $d=48$ , QAP obj=28]: the master played Kg7; under  $\mu$  our Kc2 plays the role of that game's Kf1; cross-type roles: our Ph2 as their Rb7, our ne8 as their Pf7; transferred move c2b2 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#2491 (ply 97) [structural distance  $d=49$ , QAP obj=44]: the master played Bf5; under  $\mu$  our Bd2 plays the role of that game's Bd2; cross-type roles: our pf5 as their na5, our Bf1 as their Pe5; transferred move d2e3. Synthesis: 2 retrieved analogs transferred to Kb2 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



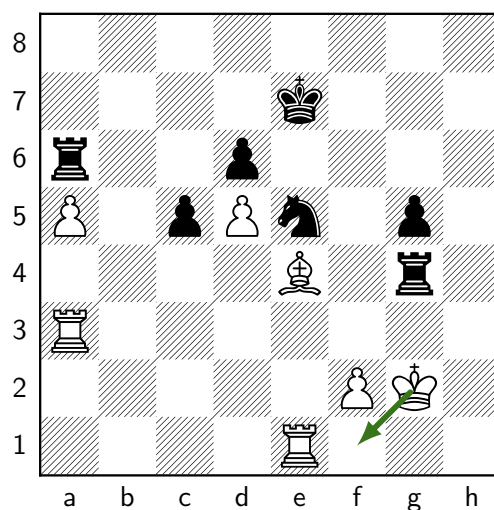
(a)



(b)

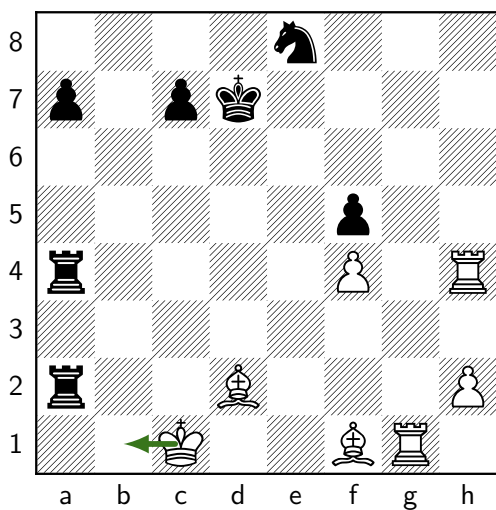


(c)

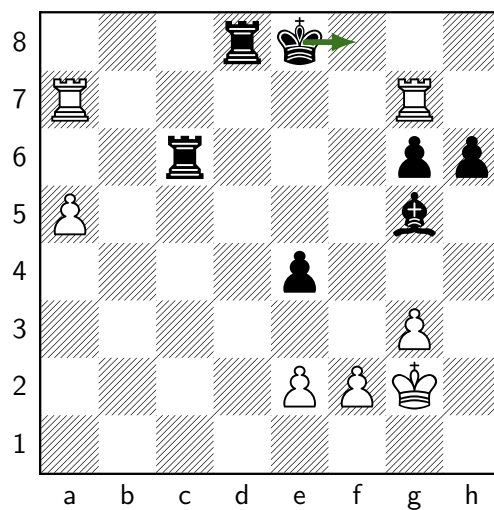


(d)

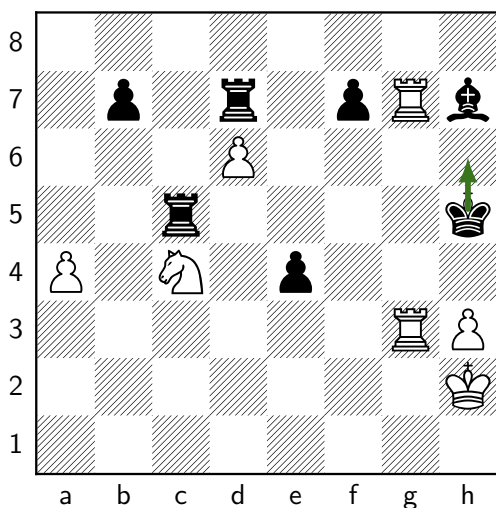
Figure 43: **Move 49, White: Kc1.** (a) Our position; Melanie plays Kc1 (b2c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top.p0000#2043` (ply 100) [structural distance  $d=43$ , QAP obj=81]: the master played Kf4; under  $\mu$  our Kb2 plays the role of that game's Kg3; cross-type roles: our Rg1 as their Pf3, our Bf1 as their Rb8; transferred move b2c3. (c) Retrieved from `ccl1_top.p0000#1650` (ply 111) [structural distance  $d=38$ , QAP obj=69]: the master played Ke8; under  $\mu$  our Kb2 plays the role of that game's Kf1; cross-type roles: our Ph2 as their Rd8, our Bd2 as their Rc6; transferred move b2b1. (d) Retrieved from `ccl1_top.p0002#2661` (ply 98) [structural distance  $d=44$ , QAP obj=36]: the master played Kf1; under  $\mu$  our Kb2 plays the role of that game's Kg2; cross-type roles: our Bf1 as their Re1, our ne8 as their Pf2; transferred move b2b1. Synthesis: no retrieved analog transferred to Kc1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



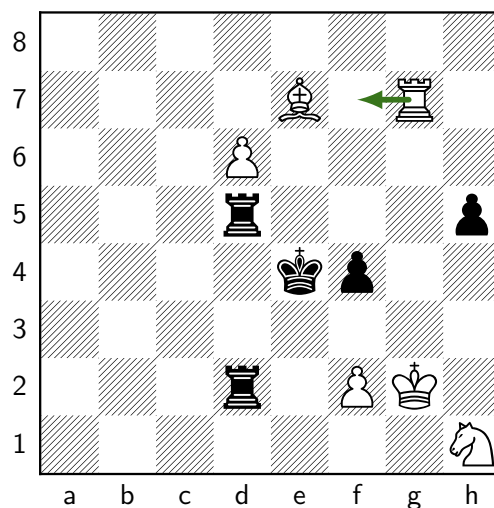
(a)



(b)



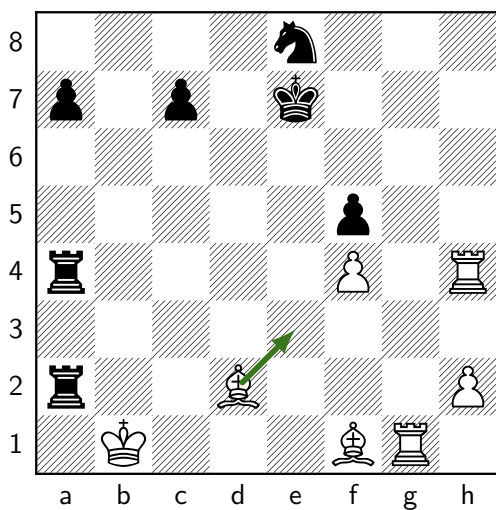
(c)



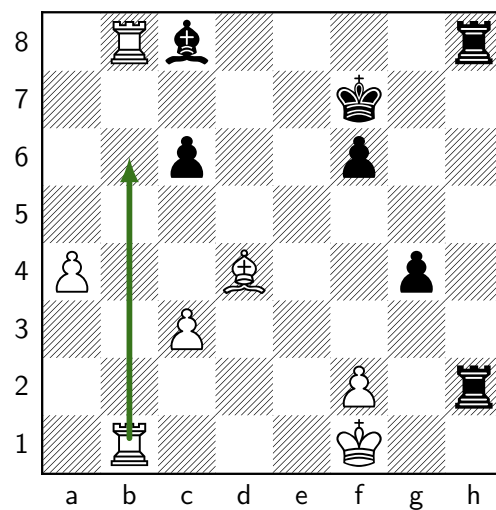
(d)

Figure 44: **Move 50, White: Kb1.** (a) Our position; Melanie plays Kb1 (**c1b1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1650` (ply 113) [structural distance  $d=35$ , QAP obj=20]: the master played Kf8; under  $\mu$  our Kc1 plays the role of that game's Ke1; cross-type roles: our Bf1 as their Rc6, our Bd2 as their Pg6; transferred move c1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1799` (ply 87) [structural distance  $d=44$ , QAP obj=12]: the master played Kh6; under  $\mu$  our Kc1 plays the role of that game's Kh4; cross-type roles: our Bf1 as their Rd7, our Bd2 as their Pf7; transferred move c1b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#342` (ply 114) [structural distance  $d=48$ , QAP obj=33]: the master played Rf7; under  $\mu$  our Rg1 plays the role of that game's Rg7; cross-type roles: our Rh4 as their Nh1, our Bd2 as their Pd6; transferred move g1g8. Synthesis: 2 retrieved analogs transferred to Kb1 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

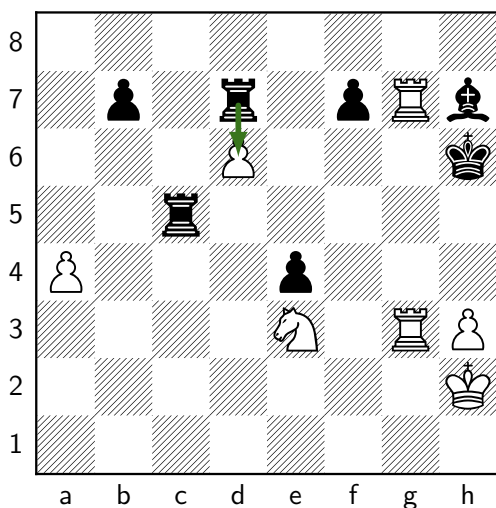




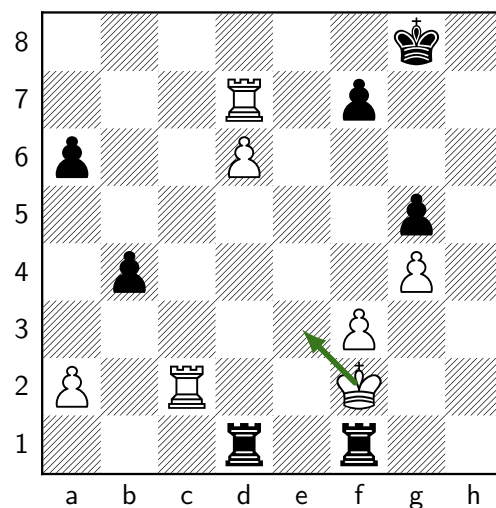
(a)



(b)

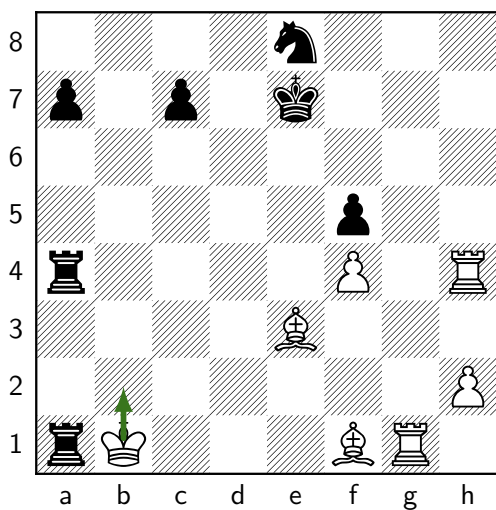


(c)

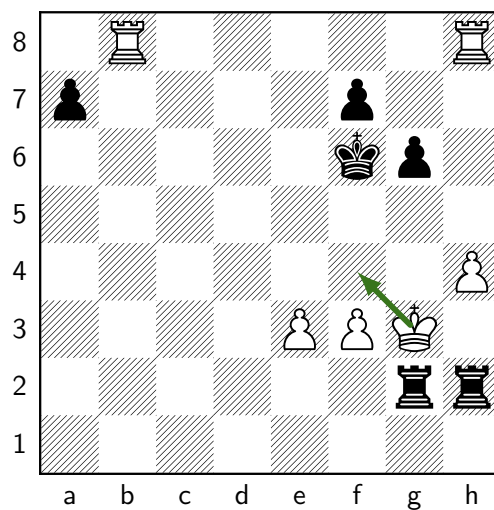


(d)

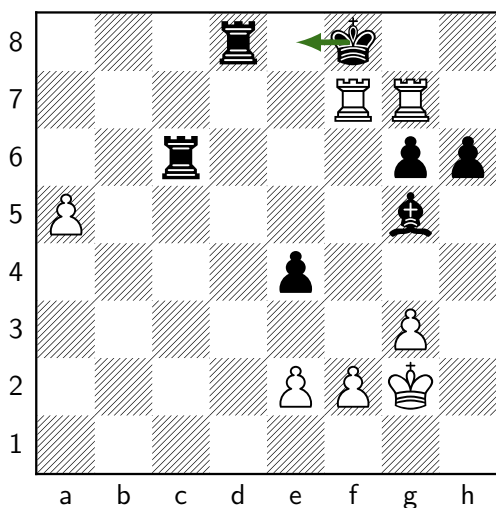
Figure 45: **Move 51, White: Be3**. (a) Our position; Melanie plays Be3 (d2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2221` (ply 84) [structural distance  $d=46$ , QAP obj=8]: the master played R1b6; under  $\mu$  our Bd2 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Rg1 as their Pf2, our Ph2 as their rh2; transferred move d2e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1799` (ply 89) [structural distance  $d=29$ , QAP obj=34]: the master played Rxd6; under  $\mu$  our Rh4 plays the role of that game’s Rd2; cross-type roles: our Rg1 as their Pb7, our Bf1 as their Rc5; transferred move h4h5. (d) Retrieved from `ccr1_top_p0002#1891` (ply 86) [structural distance  $d=46$ , QAP obj=28]: the master played Ke3; under  $\mu$  our Kb1 plays the role of that game’s Kf2; cross-type roles: our Bf1 as their Pa2, our Bd2 as their Rc2; transferred move b1c1. Synthesis: 1 retrieved analog transferred to Be3 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



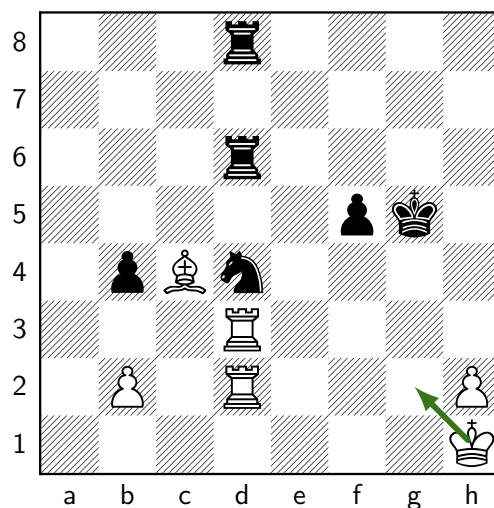
(a)



(b)



(c)



(d)

Figure 46: **Move 52, White: Kb2.** (a) Our position; Melanie plays Kb2 (b1b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2043` (ply 100) [structural distance  $d=55$ , QAP obj=80]: the master played Kf4; under  $\mu$  our Kb1 plays the role of that game’s Kg3; cross-type roles: our Rg1 as their Pe3, our Rh4 as their pf7; transferred move b1b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1650` (ply 111) [structural distance  $d=39$ , QAP obj=69]: the master played Ke8; under  $\mu$  our Kb1 plays the role of that game’s Kf1; cross-type roles: our pc7 as their Rd8, our Be3 as their Rc6; transferred move b1b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3242` (ply 88) [structural distance  $d=52$ , QAP obj=4]: the master played Kg2; under  $\mu$  our Kb1 plays the role of that game’s Kh1; cross-type roles: our Bf1 as their Pb2, our Be3 as their Ph2; transferred move b1b2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kb2 (summed QAP objective 303, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

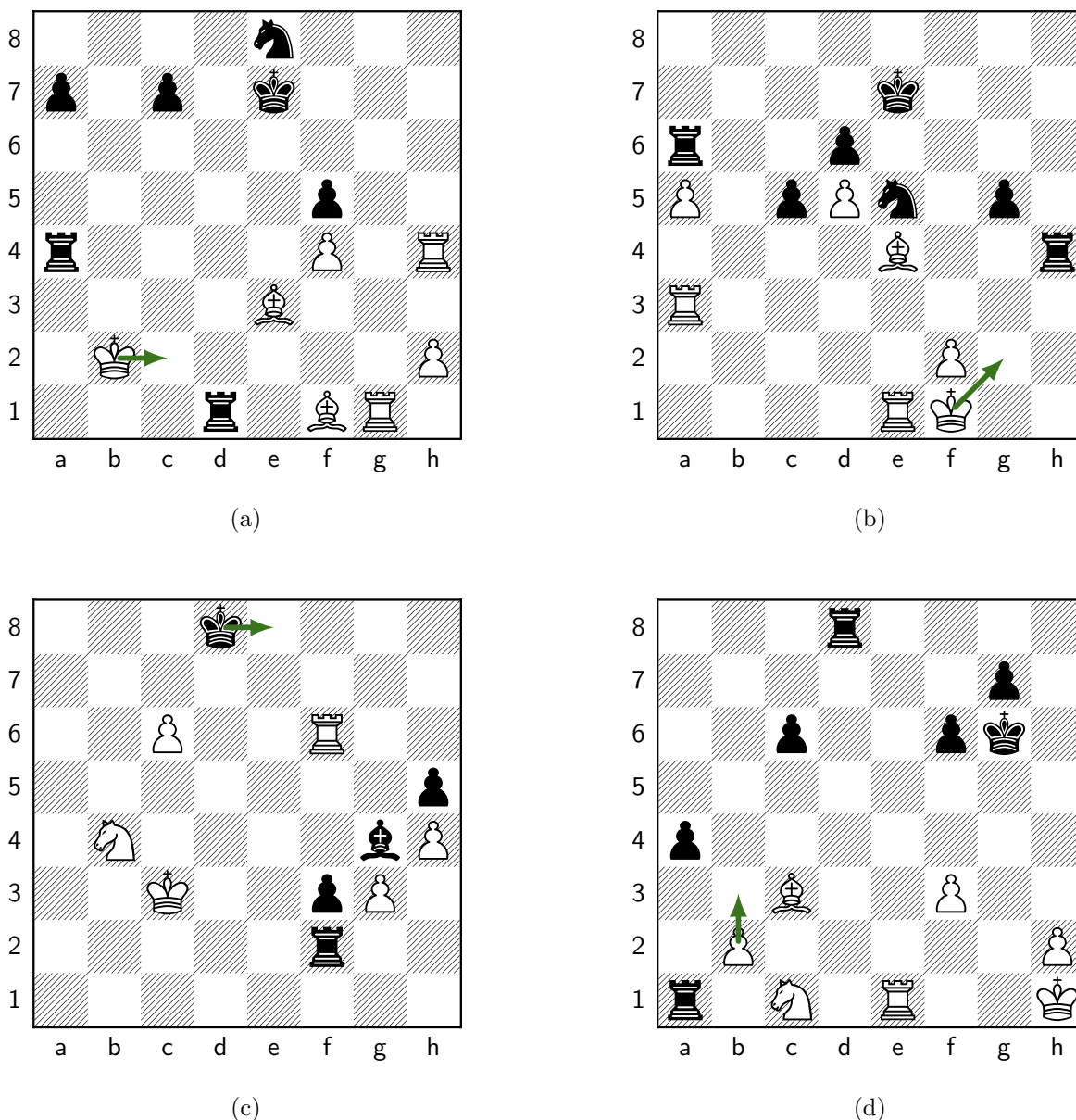
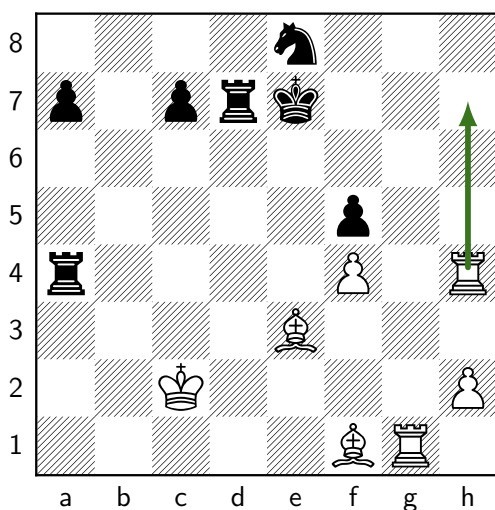
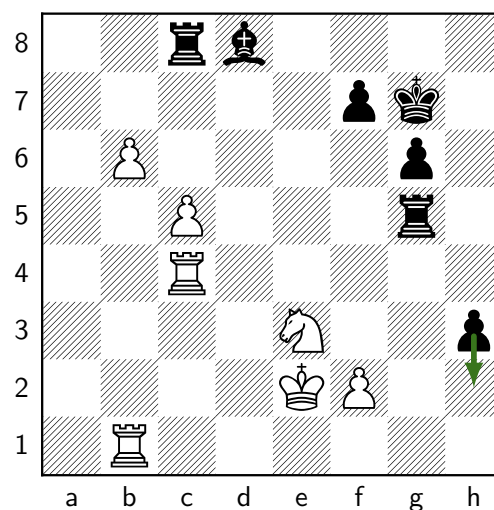


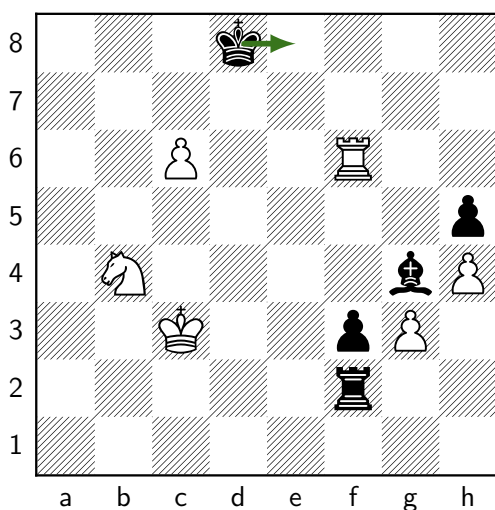
Figure 47: **Move 53, White: Kc2.** (a) Our position; Melanie plays Kc2 (b2c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#2661 (ply 100) [structural distance  $d=53$ , QAP obj=30]: the master played Kg2; under  $\mu$  our Kb2 plays the role of that game's Kf1; cross-type roles: our Bf1 as their Pa5, our Rh4 as their pc5; transferred move b2c2 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#1255 (ply 111) [structural distance  $d=47$ , QAP obj=16]: the master played Ke8; under  $\mu$  our Kb2 plays the role of that game's Kd1; cross-type roles: our Be3 as their Ph5, our Rg1 as their Bg4; transferred move b2c2 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#1684 (ply 90) [structural distance  $d=49$ , QAP obj=44]: the master played b3; under  $\mu$  our Be3 plays the role of that game's Pb2 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Nc1, our Rh4 as their Bc3; transferred move e3d2. Synthesis: 4 retrieved analogs transferred to Kc2 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



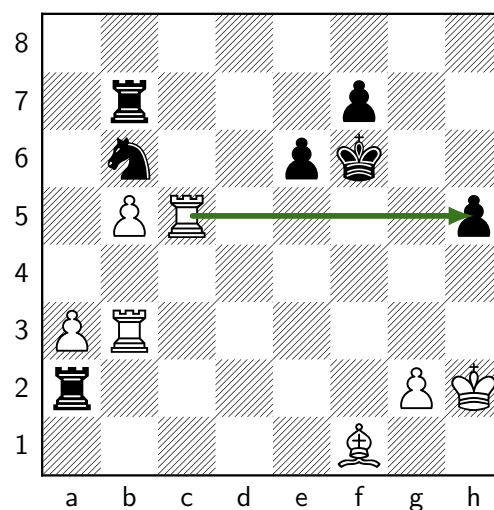
(a)



(b)

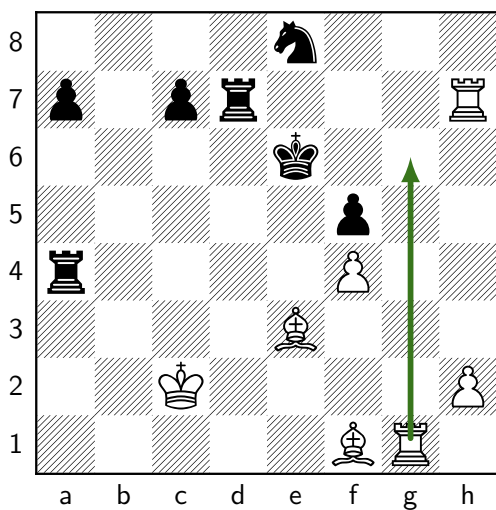


(c)

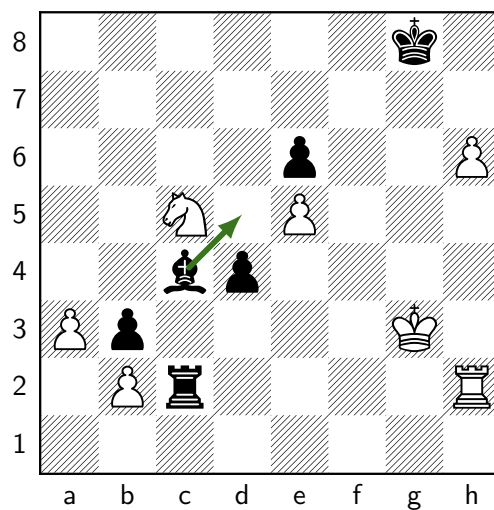


(d)

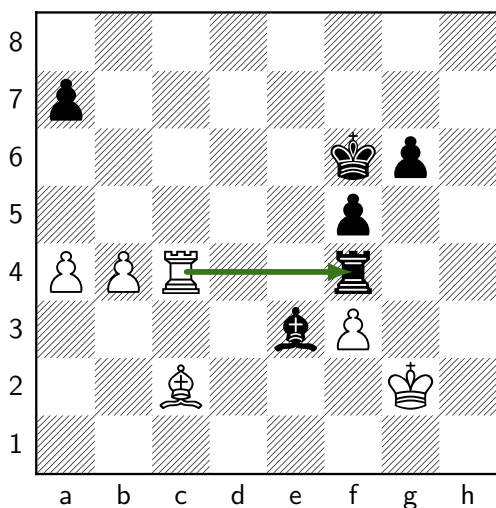
Figure 48: **Move 54, White: Rh7+.** (a) Our position; Melanie plays Rh7+ (h4h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0002#2111` (ply 91) [structural distance  $d=50$ , QAP obj=36]: the master played h2; under  $\mu$  our *Ph2* plays the role of that game’s *Ph6*; cross-type roles: our *Rg1* as their *Pg6*, our *Bf1* as their *ne3*; transferred move h2h3. (c) Retrieved from `ccr1_top.p0002#1255` (ply 111) [structural distance  $d=50$ , QAP obj=31]: the master played Ke8; under  $\mu$  our *Kc2* plays the role of that game’s *Kd1*; cross-type roles: our *ne8* as their *Ph5*, our *pf5* as their *nb4*; transferred move c2c1. (d) Retrieved from `ccr1_top.p0002#3915` (ply 94) [structural distance  $d=49$ , QAP obj=21]: the master played Rxh5; under  $\mu$  our *Rh4* plays the role of that game’s *Rc5*; cross-type roles: our *Bf1* as their *Pa3*, our *ne8* as their *Rb3*; transferred move h4h5. Synthesis: no retrieved analog transferred to Rh7+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



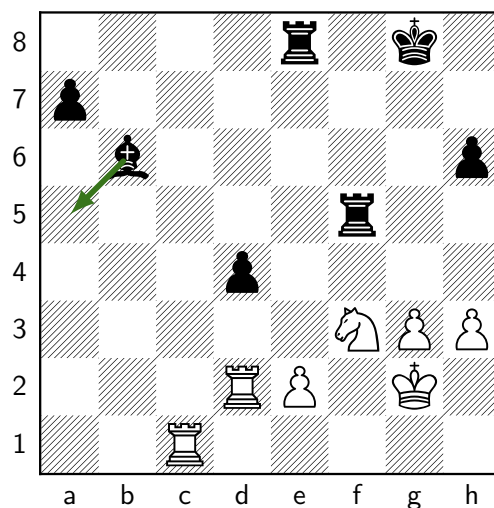
(a)



(b)

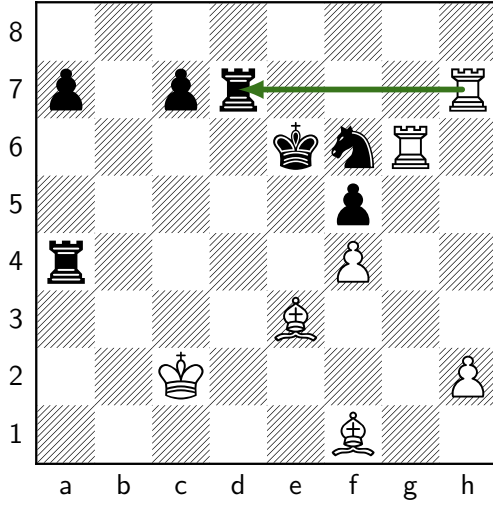


(c)

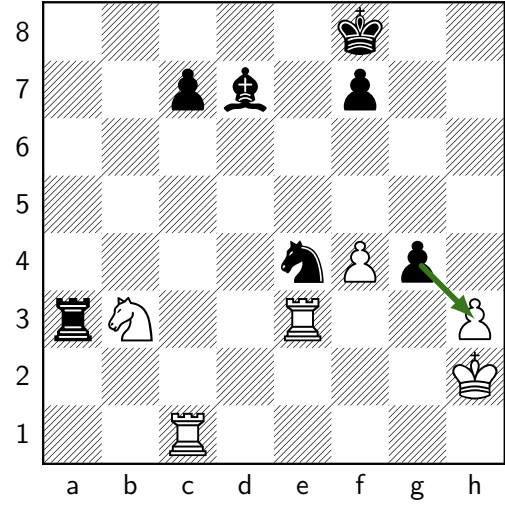


(d)

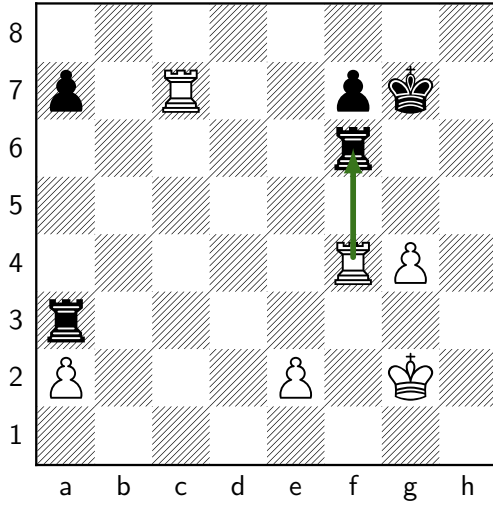
Figure 49: **Move 55, White: Rg6+.** (a) Our position; Melanie plays Rg6+ (g1g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#347` (ply 121) [structural distance  $d=48$ , QAP obj=47]: the master played Bd5; under  $\mu$  our Ph2 plays the role of that game’s Bc5 (a **cross-type** correspondence); cross-type roles: our Rh7 as their Pe6, our rd7 as their nc5; transferred move h2h3. (c) Retrieved from `ccr1_top_p0000#2809` (ply 100) [structural distance  $d=45$ , QAP obj=36]: the master played Rxf4; under  $\mu$  our Rh7 plays the role of that game’s Rc4; cross-type roles: our ne8 as their be3, our Rh7 as their Pf3; transferred move h7h5. (d) Retrieved from `ccr1_top_p0002#3261` (ply 109) [structural distance  $d=37$ , QAP obj=22]: the master played Ba5; under  $\mu$  our Bf1 plays the role of that game’s Bb3; cross-type roles: our Be3 as their Re8, our Rg1 as their Ph6; transferred move f1h3. Synthesis: no retrieved analog transferred to Rg6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



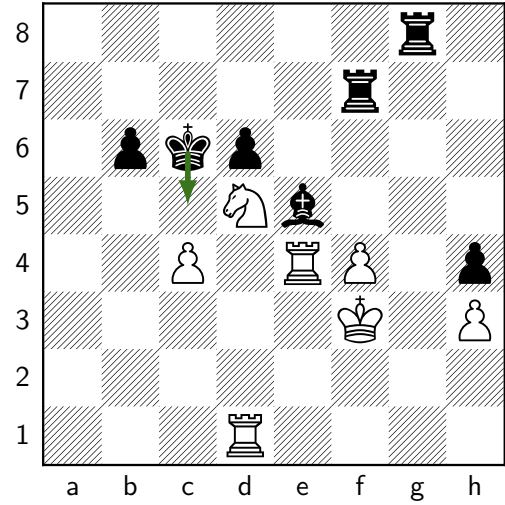
(a)



(b)

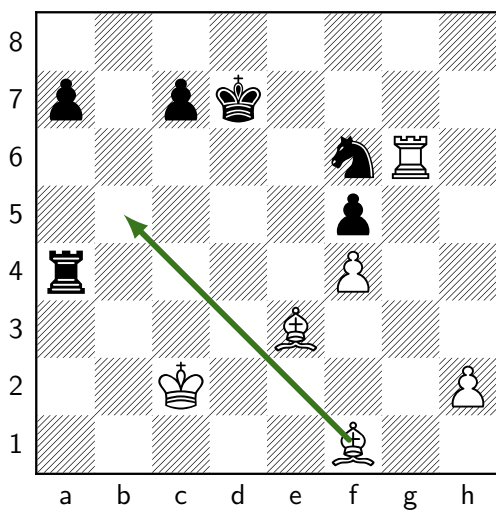


(c)

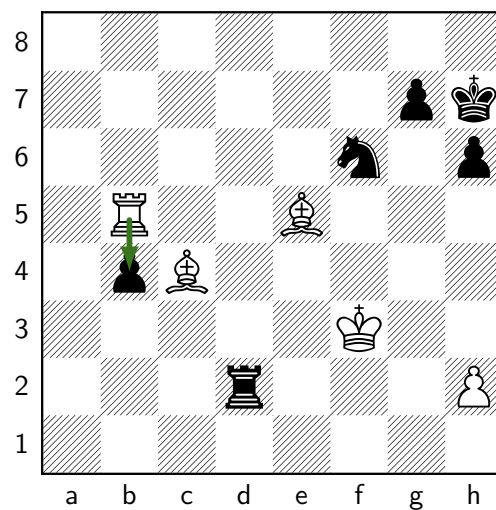


(d)

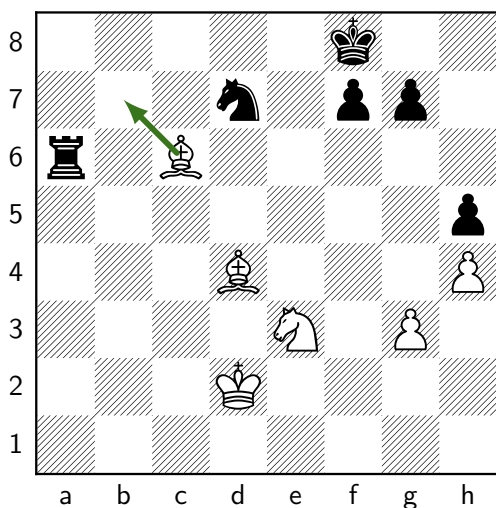
Figure 50: **Move 56, White: Rxd7.** (a) Our position; Melanie plays Rxd7 (h7d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#43` (ply 97) [structural distance  $d=45$ , QAP obj=60]: the master played `gxh3`; under  $\mu$  our *Rh7* plays the role of that game’s *Pg5* (a **cross-type** correspondence); cross-type roles: our *Bf1* as their *Pc7*, our *rd7* as their *ph3*; transferred move `h7d7` (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3990` (ply 104) [structural distance  $d=54$ , QAP obj=25]: the master played `Rxf6`; under  $\mu$  our *Rh7* plays the role of that game’s *Rf4*; cross-type roles: our *nf6* as their *Pa2*, our *Bf1* as their *pa7*; transferred move `h7d7` (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#793` (ply 99) [structural distance  $d=55$ , QAP obj=45]: the master played `Kc5`; under  $\mu$  our *Kc2* plays the role of that game’s *Kc3*; cross-type roles: our *Be3* as their *nd5*, our *Rh7* as their *Be5*; transferred move `c2c3`. Synthesis: 2 retrieved analogs transferred to Rxd7 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



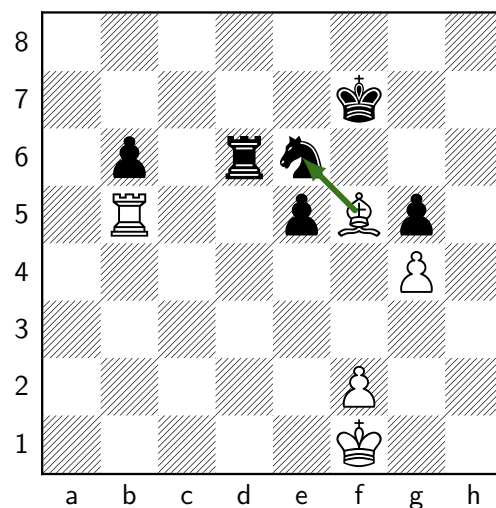
(a)



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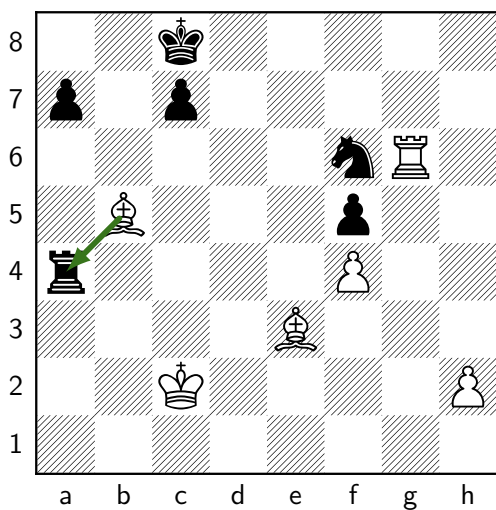


(c)

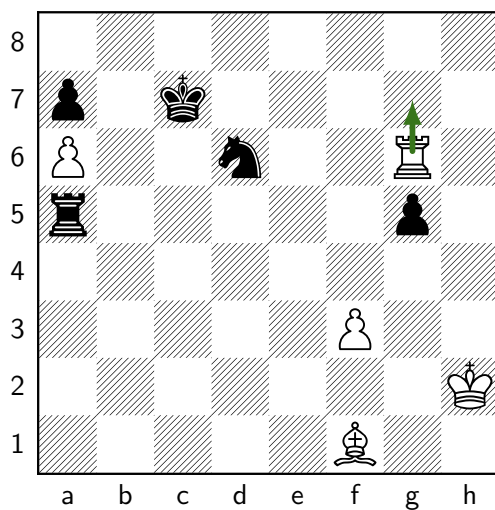


(d)

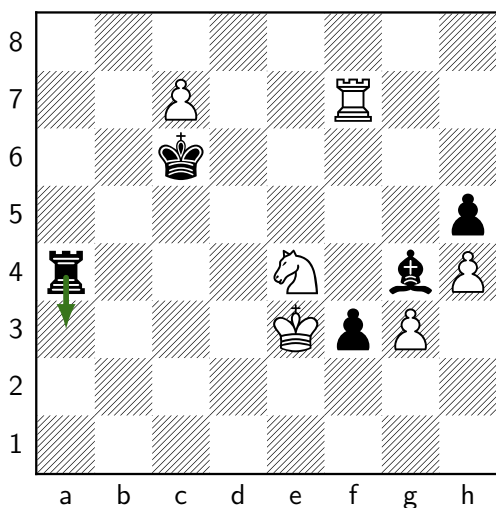
Figure 51: **Move 57, White: Bb5+.** (a) Our position; Melanie plays Bb5+ (f1b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0003#2853` (ply 102) [structural distance  $d=23$ , QAP obj=24]: the master played  $Rxb4$ ; under  $\mu$  our  $Bf1$  plays the role of that game’s  $Rb5$  (a **cross-type** correspondence); cross-type roles: our  $Rg6$  as their  $Be5$ ; transferred move  $f1h3$ . (c) Retrieved from `ccr1_top.p0000#920` (ply 106) [structural distance  $d=26$ , QAP obj=24]: the master played  $Bb7$ ; under  $\mu$  our  $Rg6$  plays the role of that game’s  $Bc6$  (a **cross-type** correspondence); cross-type roles: our  $Pf4$  as their  $Ne3$ , our  $Be3$  as their  $Pg3$ ; transferred move  $g6g7$ . (d) Retrieved from `ccr1_top.p0002#2924` (ply 108) [structural distance  $d=24$ , QAP obj=20]: the master played  $Bxe6+$ ; under  $\mu$  our  $Rg6$  plays the role of that game’s  $Bf5$  (a **cross-type** correspondence); cross-type roles: our  $Bf1$  as their  $Pf2$ , our  $Be3$  as their  $Rb5$ ; transferred move  $g6f6$ . Synthesis: no retrieved analog transferred to Bb5+ – the **search selector** chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



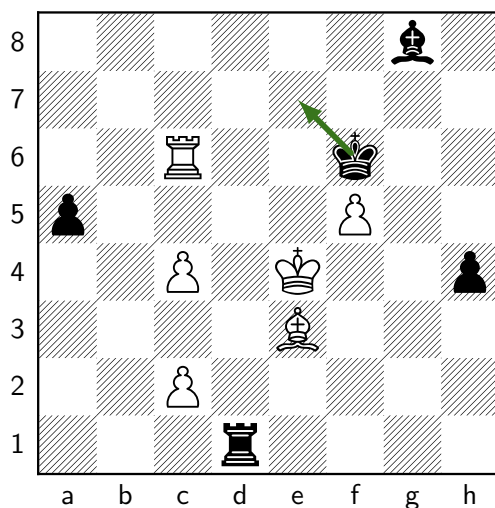
(a)



(b)



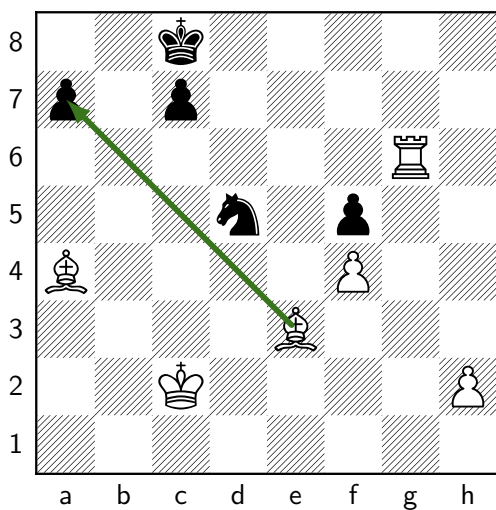
(c)



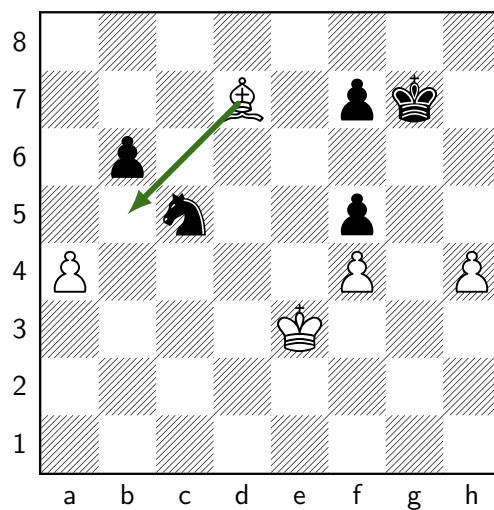
(d)

Figure 52: **Move 58, White: Bxa4**. (a) Our position; Melanie plays Bxa4 (b5a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#356` (ply 116) [structural distance  $d=29$ , QAP obj=5]: the master played Rg7+; under  $\mu$  our Bb5 plays the role of that game’s Rg6 (a **cross-type** correspondence); cross-type roles: our Rg6 as their Pf3, our pc7 as their nd6; transferred move b5a4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1255` (ply 131) [structural distance  $d=31$ , QAP obj=5]: the master played Ra3+; under  $\mu$  our Bb5 plays the role of that game’s Ra5 (a **cross-type** correspondence); cross-type roles: our Rg6 as their Ph5, our pc7 as their ne4; transferred move b5a4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#977` (ply 99) [structural distance  $d=31$ , QAP obj=13]: the master played Ke7; under  $\mu$  our Kc2 plays the role of that game’s Kf3; cross-type roles: our nf6 as their Bg8, our Bb5 as their rc6; transferred move c2b2. Synthesis: 2 retrieved analogs transferred to Bxa4 (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

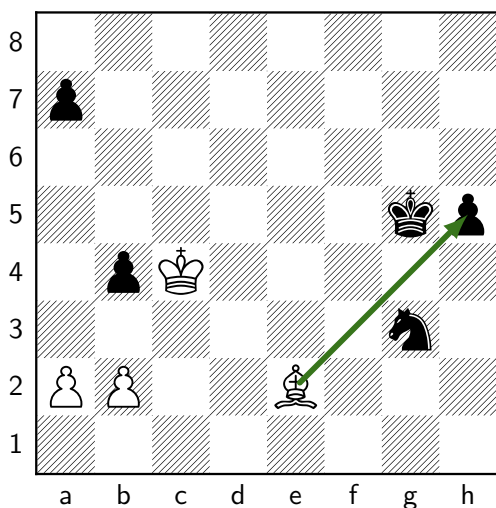




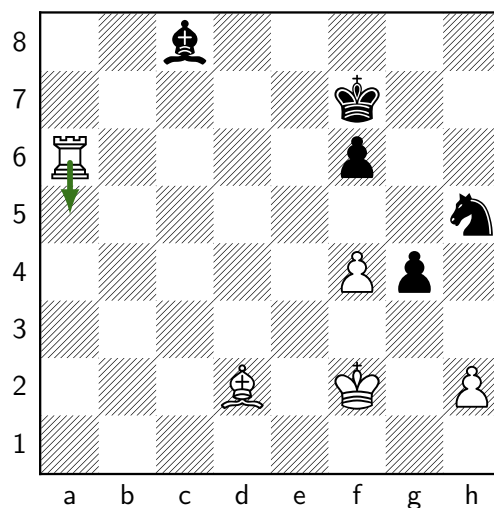
(a)



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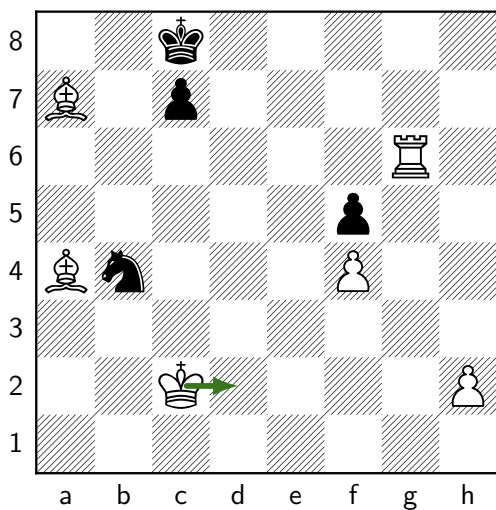


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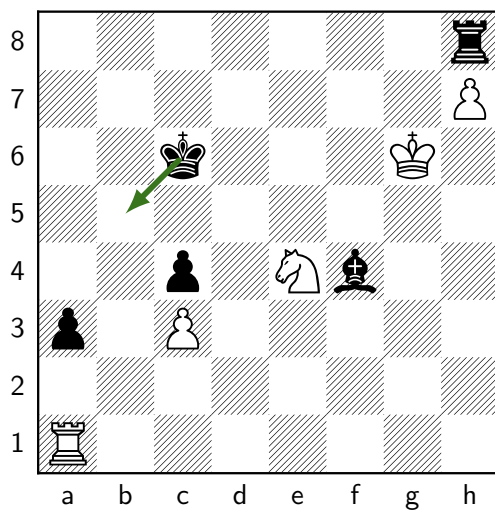


(d)

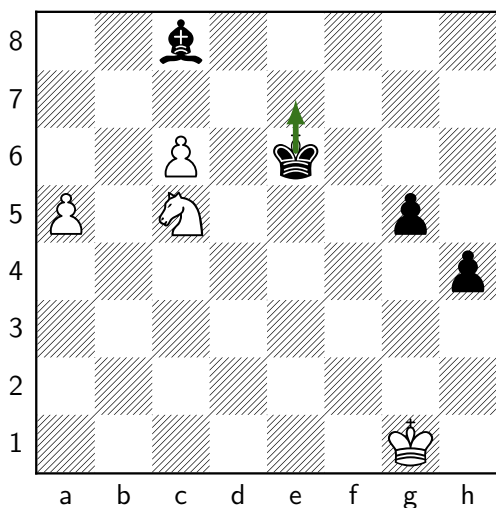
Figure 53: **Move 59, White: Bxa7**. (a) Our position; Melanie plays Bxa7 (e3a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0000#1162` (ply 110) [structural distance  $d=31$ , QAP obj=48]: the master played Bb5; under  $\mu$  our Be3 plays the role of that game’s Bd7; cross-type roles: our Rg6 as their Pf4; transferred move e3a7 (**endorses this move**). (c) Retrieved from `ccr1_top.p0000#629` (ply 102) [structural distance  $d=29$ , QAP obj=30]: the master played Bxh5; under  $\mu$  our Be3 plays the role of that game’s Be2; cross-type roles: our Ba4 as their Pa2, our Rg6 as their Pb2; transferred move e3a7 (**endorses this move**). (d) Retrieved from `ccr1_top.p0001#1032` (ply 108) [structural distance  $d=37$ , QAP obj=29]: the master played Ra5; under  $\mu$  our Be3 plays the role of that game’s Ra6 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Bd2, our Rg6 as their Pf4; transferred move e3a7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bxa7 (summed QAP objective 107, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



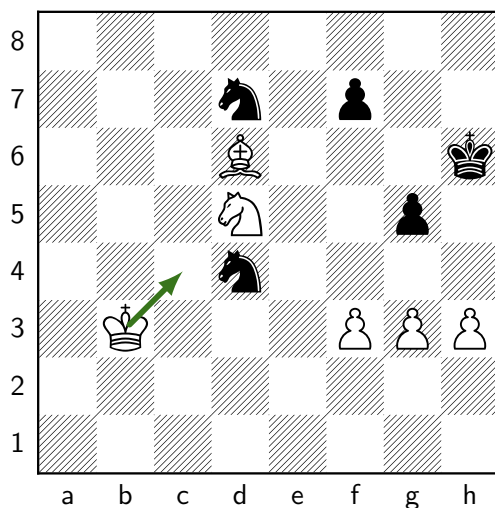
(a)



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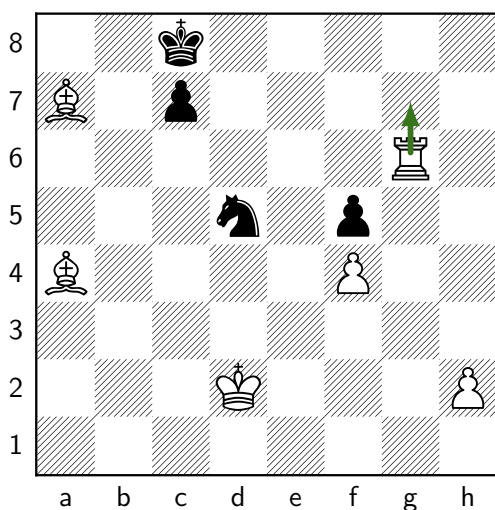


(c)

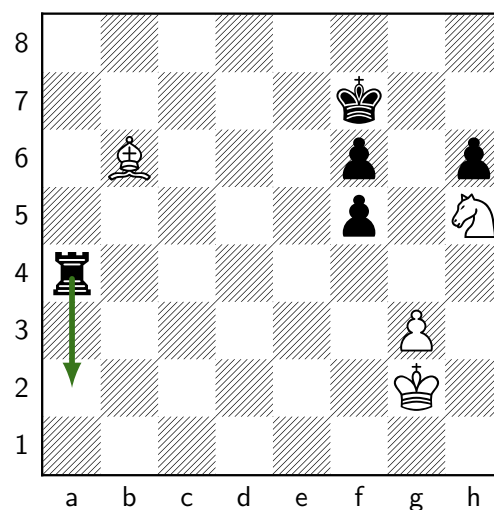


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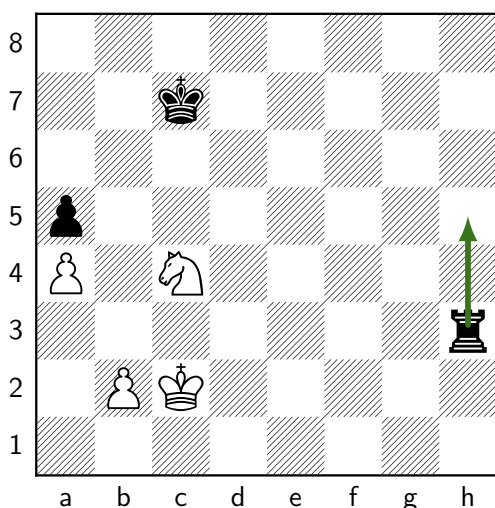
Figure 54: **Move 60, White: Kd2.** (a) Our position; Melanie plays Kd2 (c2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0003#2905` (ply 125) [structural distance  $d=20$ , QAP obj=1]: the master played Kb5; under  $\mu$  our *Kc2* plays the role of that game’s *Kc3*; cross-type roles: our *Ba7* as their *ra1*; transferred move c2d2 (**endorses this move**). (c) Retrieved from `ccr1_top-p0002#917` (ply 115) [structural distance  $d=17$ , QAP obj=70]: the master played Ke7; under  $\mu$  our *Kc2* plays the role of that game’s *Ke3*; transferred move c2c1. (d) Retrieved from `ccr1_top-p0000#3078` (ply 104) [structural distance  $d=22$ , QAP obj=66]: the master played Kc4; under  $\mu$  our *Kc2* plays the role of that game’s *Kb3*; cross-type roles: our *Ba7* as their *Pf3*, our *Ba4* as their *Pg3*; transferred move c2c3. Synthesis: 1 retrieved analog transferred to Kd2 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



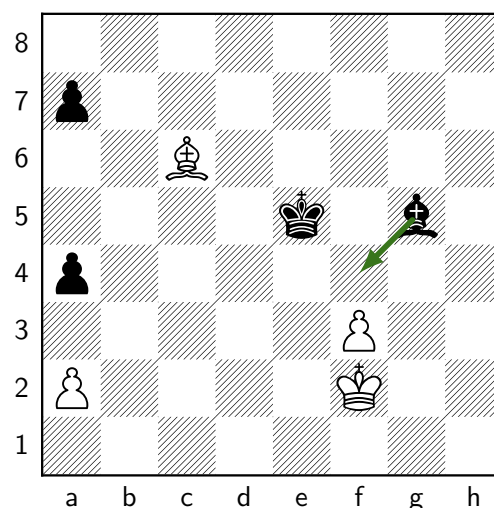
(a)



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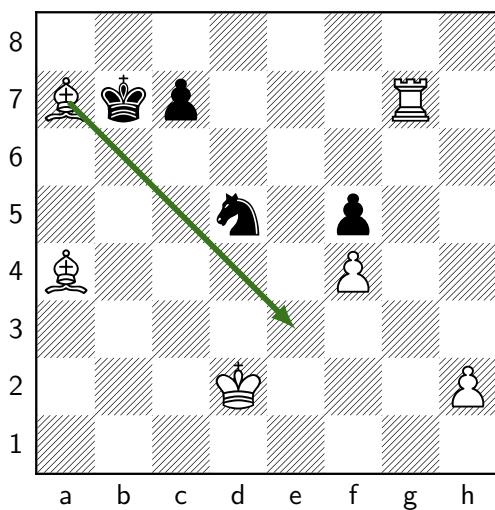


(c)

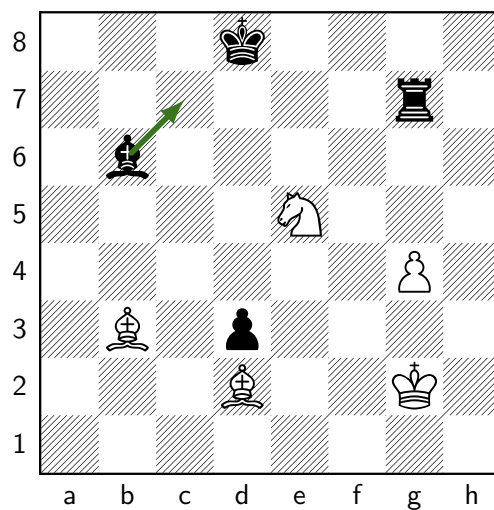


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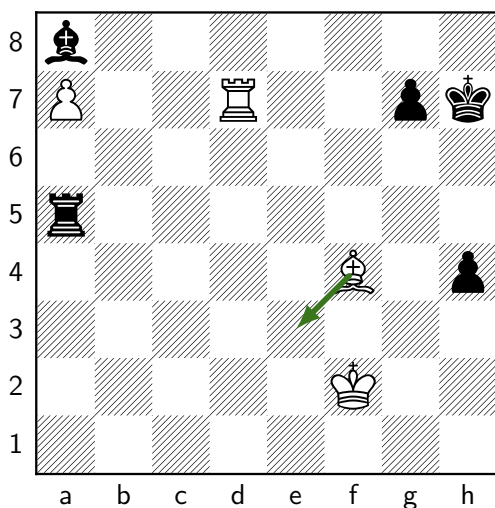
Figure 55: **Move 61, White: Rg7.** (a) Our position; Melanie plays Rg7 (g6g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top-p0002#2236 (ply 121) [structural distance  $d=18$ , QAP obj=24]: the master played Ra2+; under  $\mu$  our Rg6 plays the role of that game's Ra5; cross-type roles: our Ph2 as their bb6, our Ba7 as their Ph6; transferred move g6g7 (**endorses this move**). (c) Retrieved from ccr1\_top-p0003#807 (ply 123) [structural distance  $d=17$ , QAP obj=25]: the master played Rh5; under  $\mu$  our Rg6 plays the role of that game's Rh6; transferred move g6h6. (d) Retrieved from ccr1\_top-p0002#1381 (ply 109) [structural distance  $d=21$ , QAP obj=23]: the master played Bf4; under  $\mu$  our Ba7 plays the role of that game's Bg4; cross-type roles: our nd5 as their bc6; transferred move a7e3. Synthesis: 1 retrieved analog transferred to Rg7 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



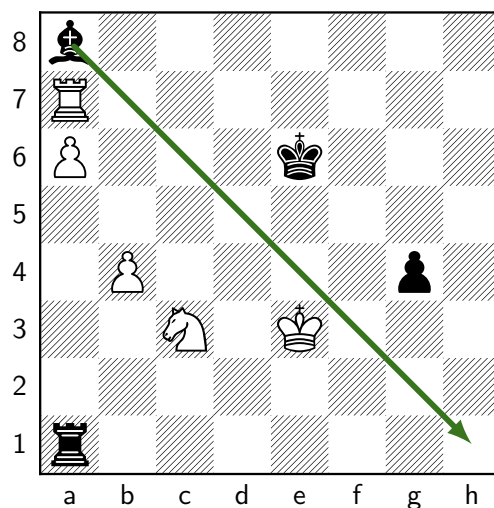
(a)



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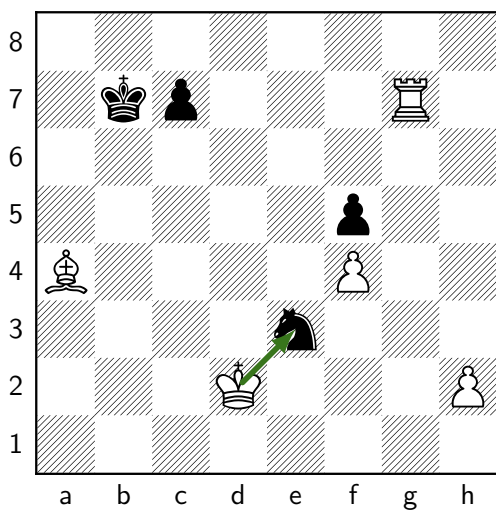


(c)

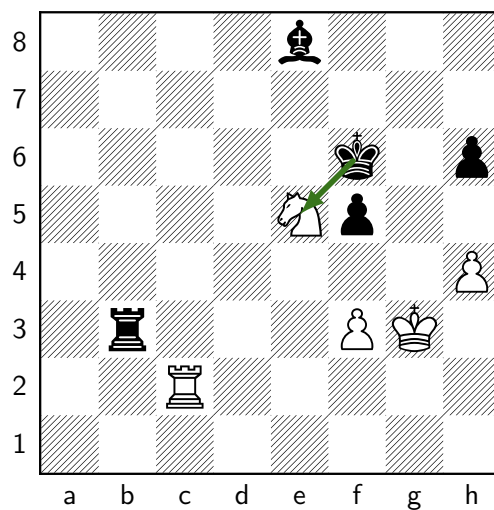


(d)

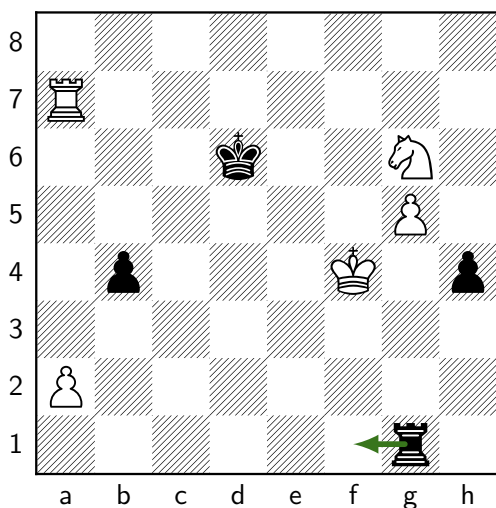
Figure 56: **Move 62, White: Be3**. (a) Our position; Melanie plays Be3 (a7e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2879` (ply 127) [structural distance  $d=22$ , QAP obj=21]: the master played Bc7; under  $\mu$  our Ba7 plays the role of that game’s Bb3; cross-type roles: our pf5 as their bb3, our Ph2 as their bd2; transferred move a7e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#991` (ply 120) [structural distance  $d=23$ , QAP obj=20]: the master played Be3; under  $\mu$  our Ba7 plays the role of that game’s Bf4; cross-type roles: our nd5 as their ph4, our Ph2 as their ra5; transferred move a7e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2918` (ply 111) [structural distance  $d=20$ , QAP obj=35]: the master played Bh1; under  $\mu$  our Ba7 plays the role of that game’s Ba1; cross-type roles: our kb7 as their ra7, our Ba4 as their Pg4; transferred move a7c5. Synthesis: 2 retrieved analogs transferred to Be3 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



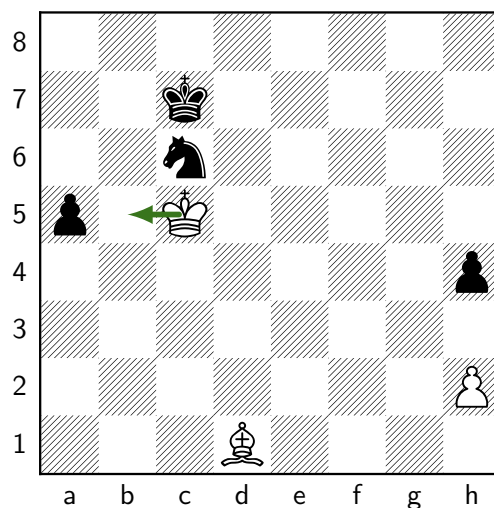
(a)



(b)

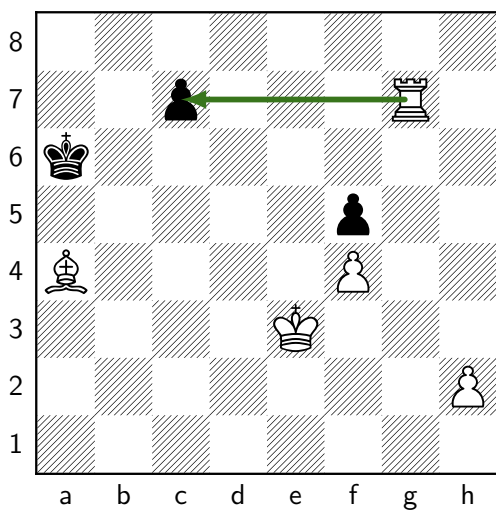


(c)

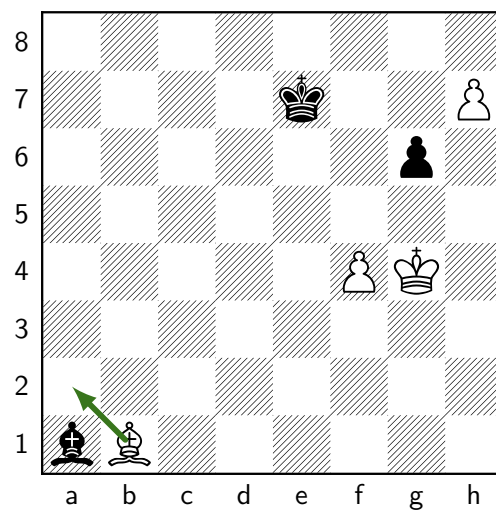


(d)

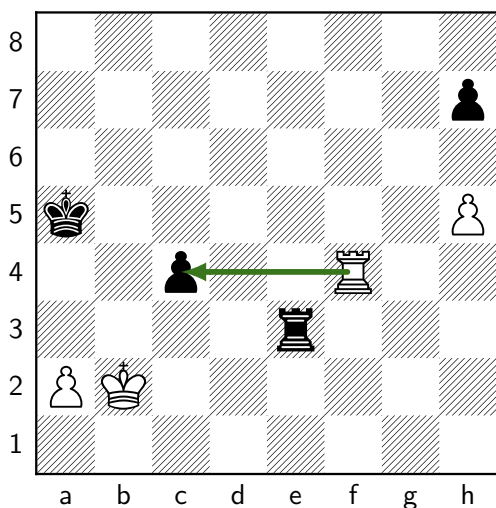
Figure 57: **Move 63, White: Kxe3**. (a) Our position; Melanie plays Kxe3 (d2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1548` (ply 117) [structural distance  $d=17$ , QAP obj=44]: the master played Kxe5; under  $\mu$  our Kd2 plays the role of that game’s Kf3; cross-type roles: our Pf4 as their Be8, our Ba4 as their Pf5; transferred move d2e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#210` (ply 113) [structural distance  $d=15$ , QAP obj=24]: the master played Rf1+; under  $\mu$  our Rg7 plays the role of that game’s Rg8; cross-type roles: our ne3 as their ra7, our Ba4 as their ng6; transferred move g7g8. (d) Retrieved from `ccr1_top_p0001#1838` (ply 120) [structural distance  $d=16$ , QAP obj=21]: the master played Kb5; under  $\mu$  our Kd2 plays the role of that game’s Kc5; cross-type roles: our Pf4 as their Bd1, our Ba4 as their Ph2; transferred move d2c1. Synthesis: 1 retrieved analog transferred to Kxe3 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



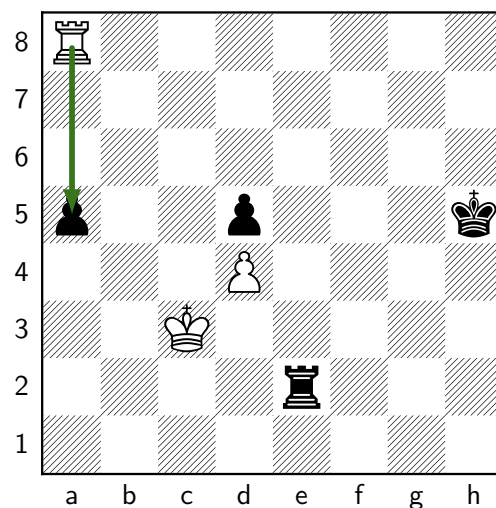
(a)



(b)

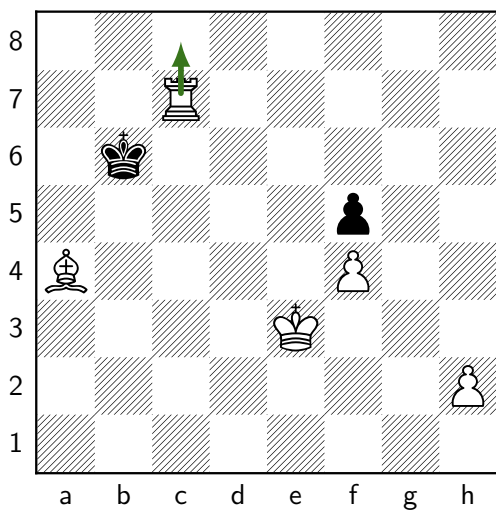


(c)

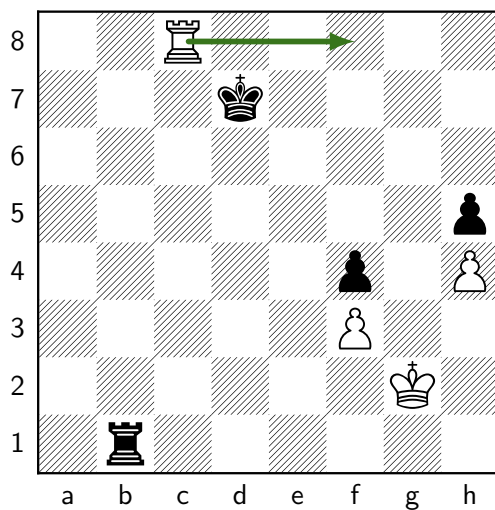


(d)

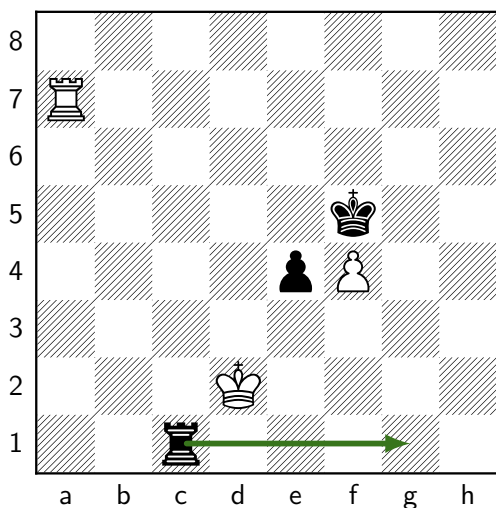
Figure 58: **Move 64, White: Rxc7 (g7c7).** (a) Our position; Melanie plays Rxc7 (g7c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3138` (ply 114) [structural distance  $d=8$ , QAP obj=16]: the master played Ba2; under  $\mu$  our Rg7 plays the role of that game's Bb1 (a **cross-type** correspondence); cross-type roles: our pf5 as their ba1; transferred move g7c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#423` (ply 116) [structural distance  $d=9$ , QAP obj=16]: the master played Rxc4; under  $\mu$  our Rg7 plays the role of that game's Rf4; cross-type roles: our pf5 as their re3, our Ba4 as their ph7; transferred move g7c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1638` (ply 122) [structural distance  $d=6$ , QAP obj=16]: the master played Rxa5; under  $\mu$  our Rg7 plays the role of that game's Ra8; cross-type roles: our Ph2 as their re2; transferred move g7c7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Rxc7 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



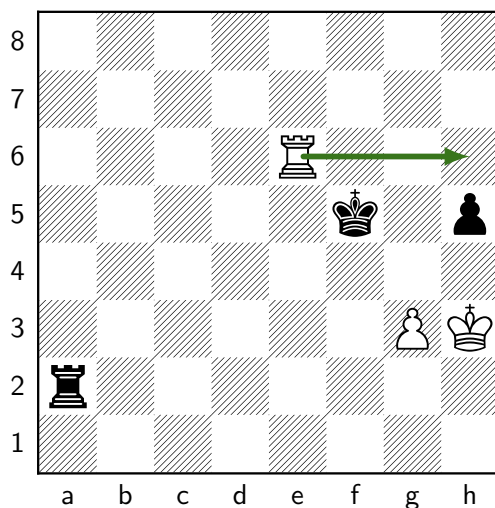
(a)



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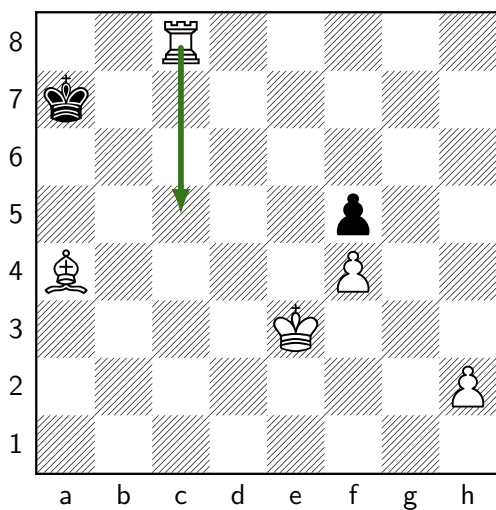


(c)

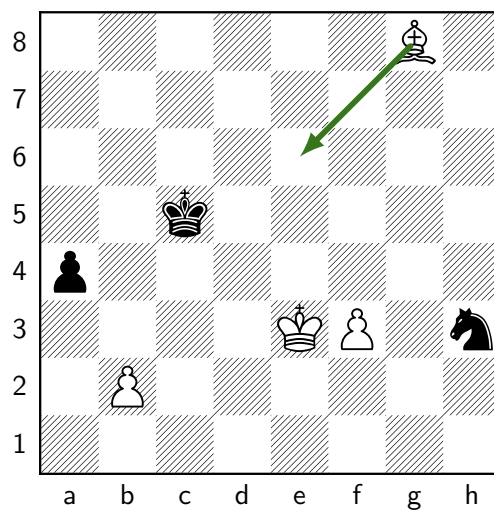


(d)

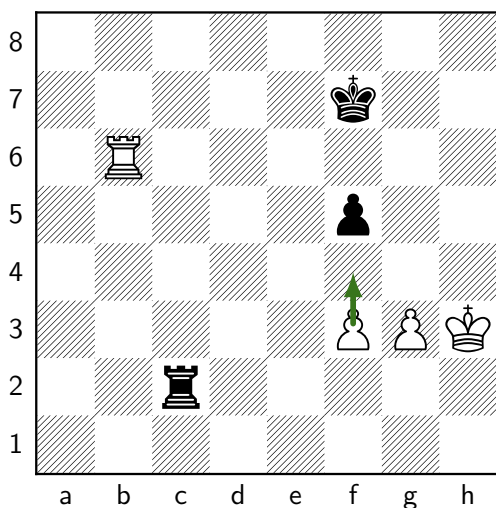
Figure 59: **Move 65, White: Rc8**. (a) Our position; Melanie plays Rc8 (c7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#471` (ply 140) [structural distance  $d=8$ , QAP obj=22]: the master played Rf8; under  $\mu$  our Rc7 plays the role of that game's Rc8; cross-type roles: our Ba4 as their rb1; transferred move c7c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#633` (ply 137) [structural distance  $d=8$ , QAP obj=21]: the master played Rg1; under  $\mu$  our Rc7 plays the role of that game's Rc8; cross-type roles: our Ph2 as their ra7; transferred move c7c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3965` (ply 114) [structural distance  $d=7$ , QAP obj=21]: the master played Rh6; under  $\mu$  our Rc7 plays the role of that game's Re6; cross-type roles: our Ph2 as their ra2; transferred move c7f7. Synthesis: 2 retrieved analogs transferred to Rc8 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



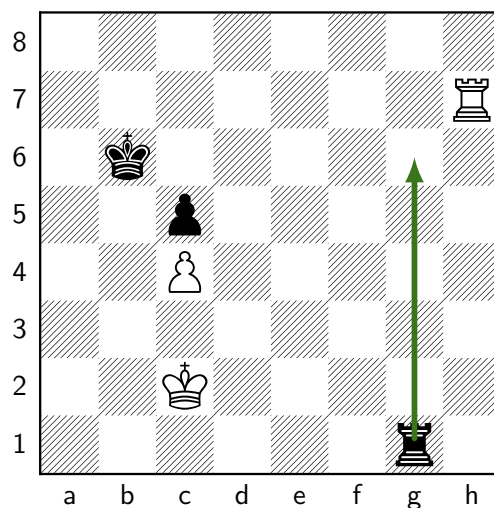
(a)



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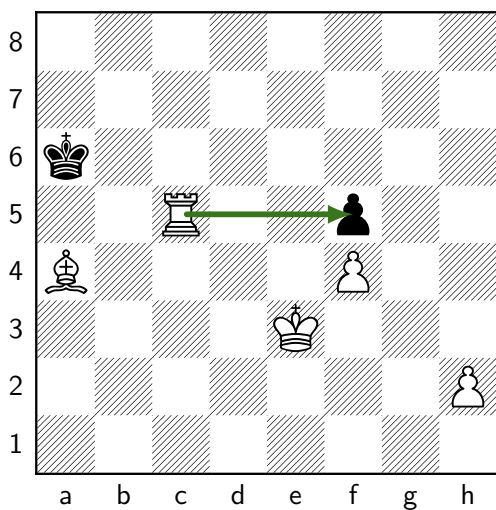
(c)



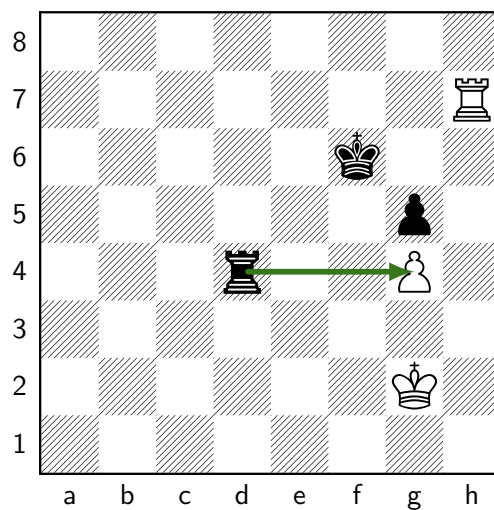
(d)

Figure 60: **Move 66, White: Rc5.** (a) Our position; Melanie plays Rc5 (c8c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1048` (ply 126) [structural distance  $d=4$ , QAP obj=3]: the master played Bc6; under  $\mu$  our Rc8 plays the role of that game’s Bg8 (a **cross-type** correspondence); cross-type roles: our pf5 as their nh3, our Ba4 as their pa4; transferred move c8c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1611` (ply 122) [structural distance  $d=4$ , QAP obj=3]: the master played f4; under  $\mu$  our Rc8 plays the role of that game’s Pf3 (a **cross-type** correspondence); cross-type roles: our Ba4 as their rc2, our Ph2 as their Rb6; transferred move c8c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#449` (ply 127) [structural distance  $d=5$ , QAP obj=3]: the master played Rg6; under  $\mu$  our Rc8 plays the role of that game’s Rg8; cross-type roles: our Ph2 as their rh7; transferred move c8c1. Synthesis: 2 retrieved analogs transferred to Rc5 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

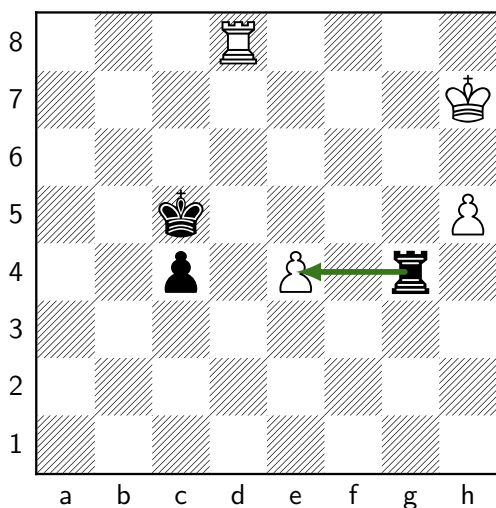




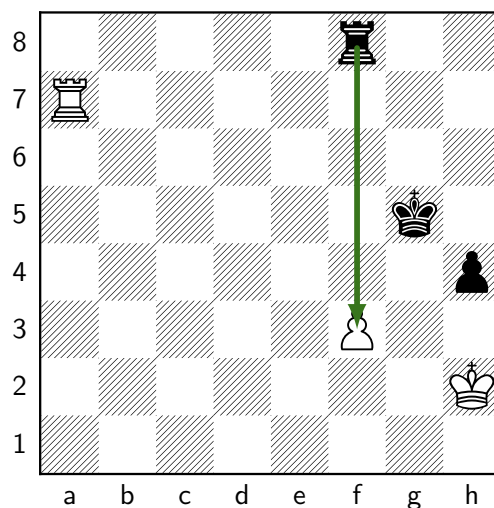
(a)



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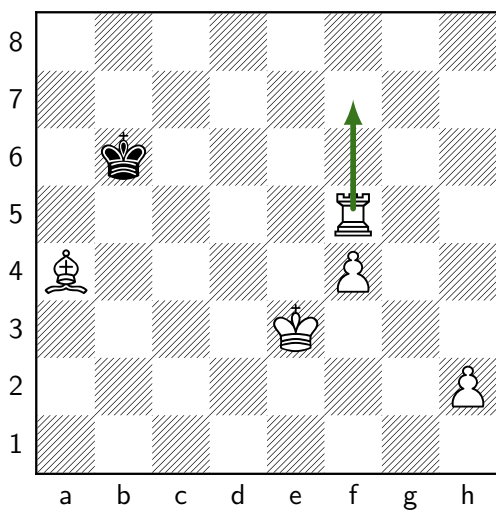


(c)

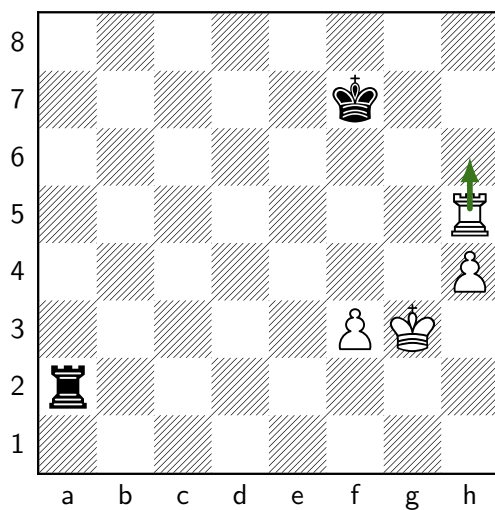


(d)

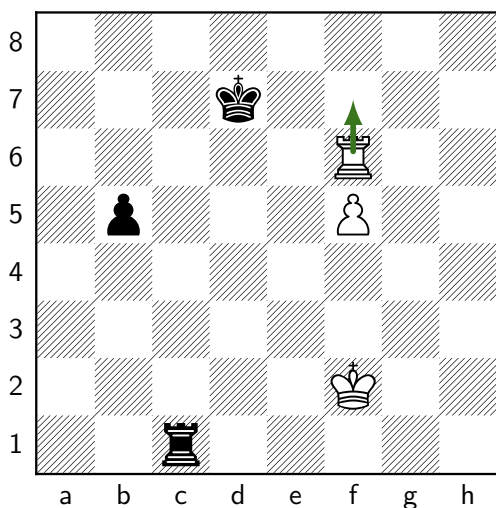
Figure 61: **Move 67, White: Rxf5**. (a) Our position; Melanie plays Rxf5 (c5f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1283` (ply 139) [structural distance  $d=6$ , QAP obj=22]: the master played Rxf4+; under  $\mu$  our Rc5 plays the role of that game's Rd5; cross-type roles: our Ph2 as their rh7; transferred move c5f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3961` (ply 131) [structural distance  $d=6$ , QAP obj=15]: the master played Rxe4; under  $\mu$  our Rc5 plays the role of that game's Rg5; cross-type roles: our Ph2 as their rd8, our Ba4 as their ph5; transferred move c5f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2202` (ply 147) [structural distance  $d=5$ , QAP obj=13]: the master played Rxf3; under  $\mu$  our Rc5 plays the role of that game's Rf1; cross-type roles: our Ph2 as their ra7; transferred move c5f5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rxf5 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



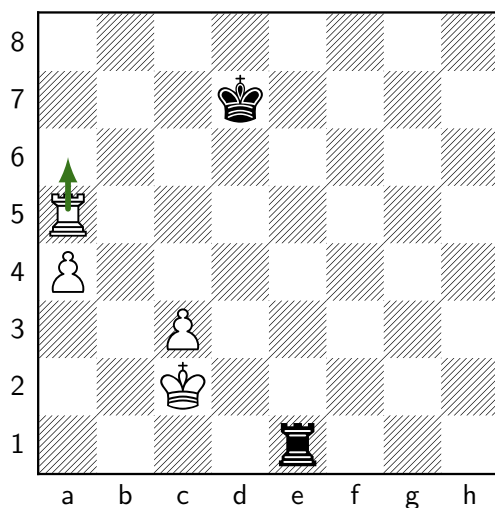
(a)



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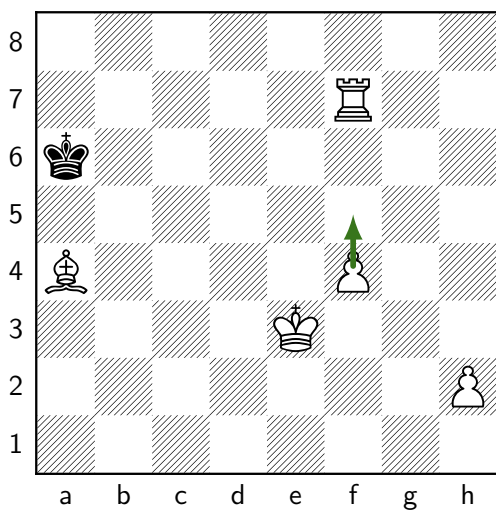


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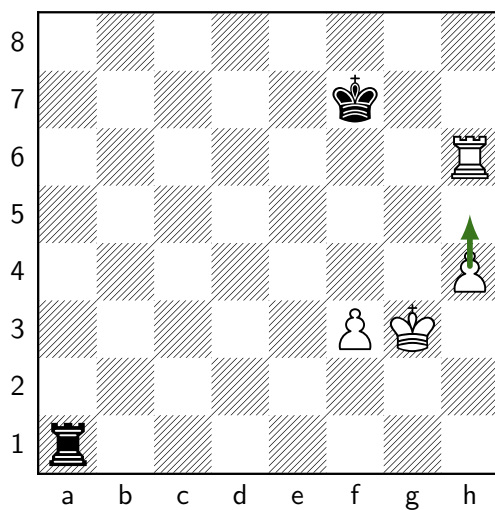


(d)

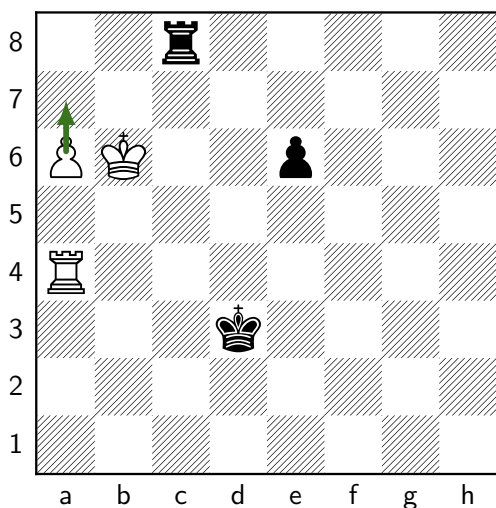
Figure 62: **Move 68, White: Rf7.** (a) Our position; Melanie plays Rf7 (f5f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1548` (ply 150) [structural distance  $d=8$ , QAP obj=3]: the master played Rh6; under  $\mu$  our Rf5 plays the role of that game’s Rh5; cross-type roles: our Ph2 as their ra2, our Ba4 as their Pf3; transferred move f5f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3310` (ply 122) [structural distance  $d=9$ , QAP obj=3]: the master played Rf7+; under  $\mu$  our Rf5 plays the role of that game’s Rf6; cross-type roles: our Ph2 as their rc1, our Ba4 as their pb5; transferred move f5f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2555` (ply 118) [structural distance  $d=3$ , QAP obj=2]: the master played Ra6; under  $\mu$  our Rf5 plays the role of that game’s Ra5; cross-type roles: our Ph2 as their re1, our Ba4 as their Pc3; transferred move f5f7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rf7 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



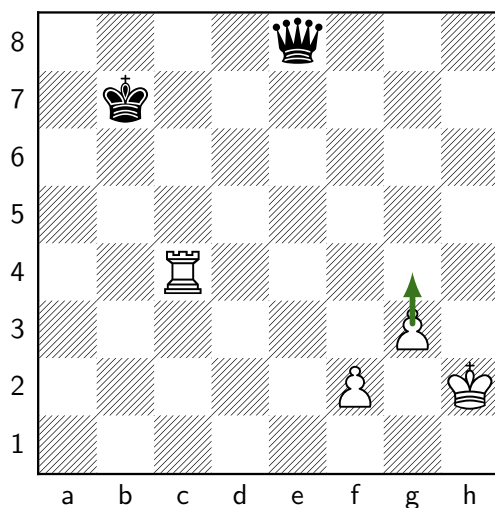
(a)



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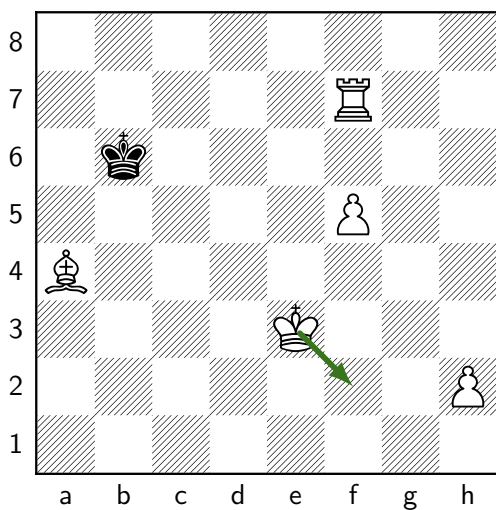


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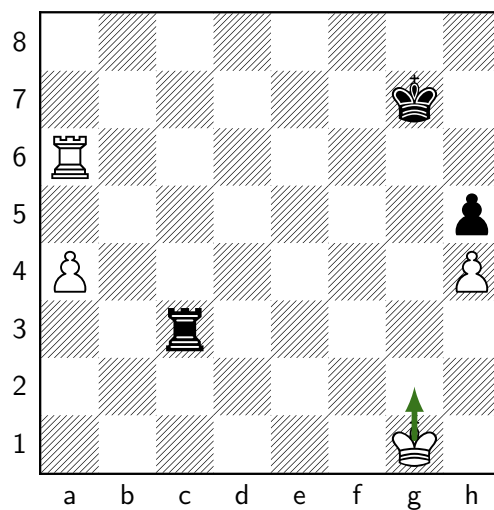


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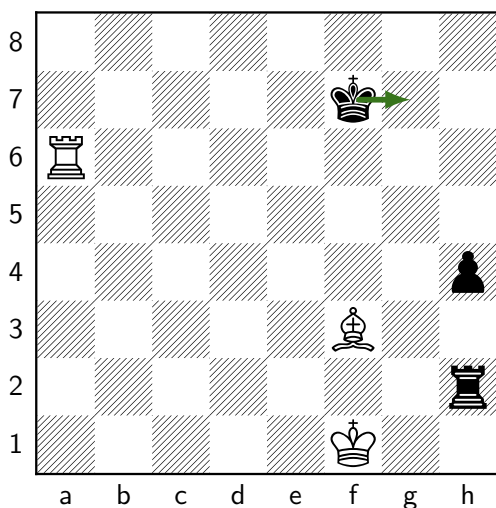
Figure 63: **Move 69, White: f5.** (a) Our position; Melanie plays f5 (f4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1548` (ply 152) [structural distance  $d=6$ , QAP obj=3]: the master played h5; under  $\mu$  our  $Pf_4$  plays the role of that game’s  $Ph_4$ ; cross-type roles: our  $Ph_2$  as their  $ra_1$ , our  $Ba_4$  as their  $Pf_3$ ; transferred move f4f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2730` (ply 150) [structural distance  $d=5$ , QAP obj=3]: the master played a7; under  $\mu$  our  $Pf_4$  plays the role of that game’s  $Pa_6$ ; cross-type roles: our  $Ba_4$  as their  $rc_8$ ; transferred move f4f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1696` (ply 134) [structural distance  $d=7$ , QAP obj=3]: the master played g4; under  $\mu$  our  $Pf_4$  plays the role of that game’s  $Pg_3$ ; cross-type roles: our  $Rf_7$  as their  $Pf_2$ , our  $Ba_4$  as their  $Rc_4$ ; transferred move f4f5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to f5 (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



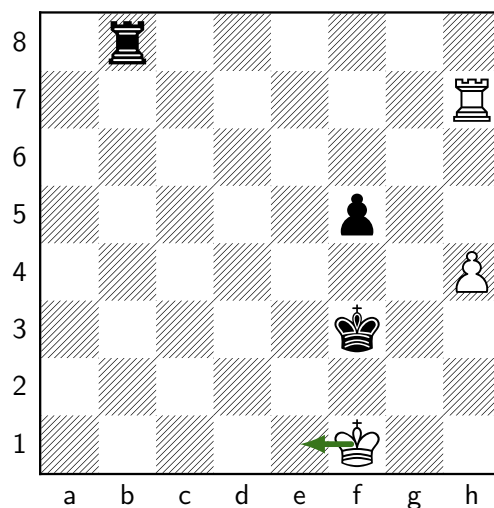
(a)



(b)



(c)



(d)

Figure 64: **Move 70, White: Kf2**. (a) Our position; Melanie plays Kf2 (e3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#106` (ply 122) [structural distance  $d=5$ , QAP obj=2]: the master played Kg2; under  $\mu$  our *Ke3* plays the role of that game’s *Kg1*; cross-type roles: our *Ba4* as their *rc3*; transferred move e3f2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3941` (ply 129) [structural distance  $d=5$ , QAP obj=2]: the master played Kg7; under  $\mu$  our *Ke3* plays the role of that game’s *Kf2*; cross-type roles: our *Ba4* as their *ra6*, our *Ph2* as their *bf3*; transferred move e3f2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1542` (ply 122) [structural distance  $d=5$ , QAP obj=1]: the master played Ke1; under  $\mu$  our *Ke3* plays the role of that game’s *Kf1*; cross-type roles: our *Ba4* as their *rb8*; transferred move e3f2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf2 (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

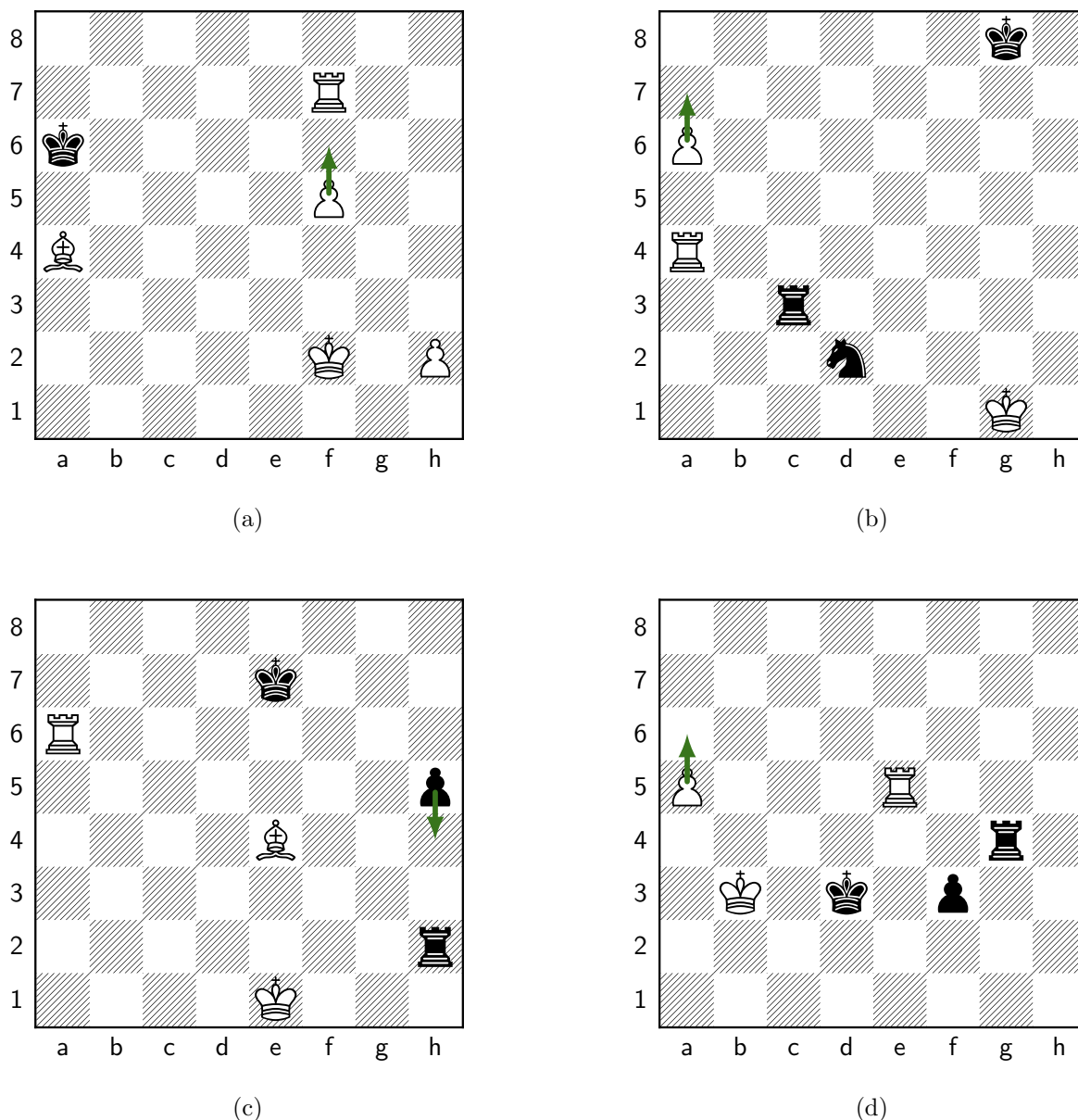
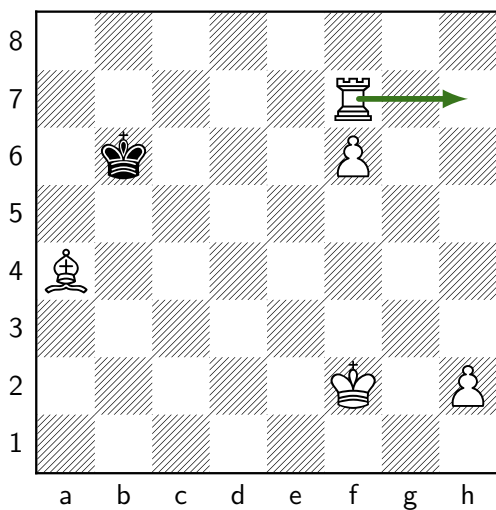
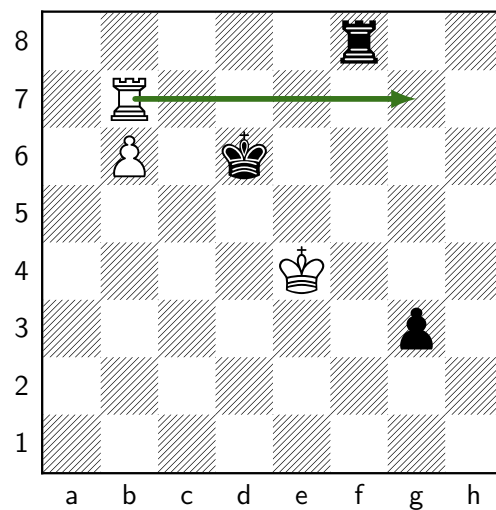


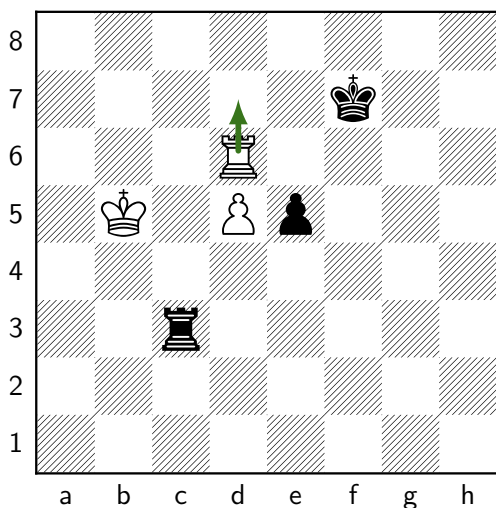
Figure 65: **Move 71, White: f6**. (a) Our position; Melanie plays f6 (f5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1329` (ply 128) [structural distance  $d=5$ , QAP obj=2]: the master played a7; under  $\mu$  our *Pf5* plays the role of that game’s *Pa6*; cross-type roles: our *Ph2* as their *nd2*, our *Ba4* as their *rc3*; transferred move f5f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3941` (ply 125) [structural distance  $d=5$ , QAP obj=2]: the master played h4; under  $\mu$  our *Pf5* plays the role of that game’s *Ph4*; cross-type roles: our *Ph2* as their *ra6*; transferred move f5f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2934` (ply 130) [structural distance  $d=6$ , QAP obj=2]: the master played a6; under  $\mu$  our *Pf5* plays the role of that game’s *Pa5*; cross-type roles: our *Ba4* as their *rg4*; transferred move f5f6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to f6 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



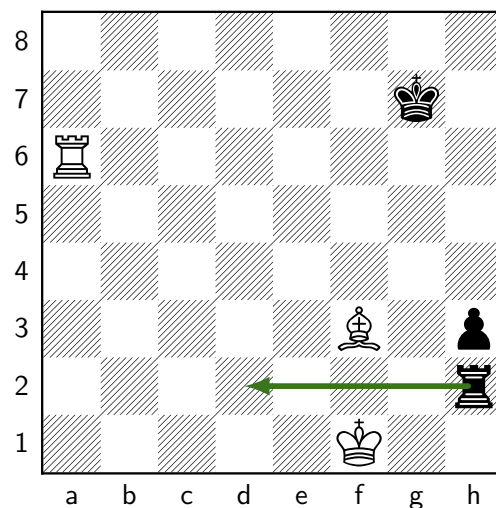
(a)



(b)

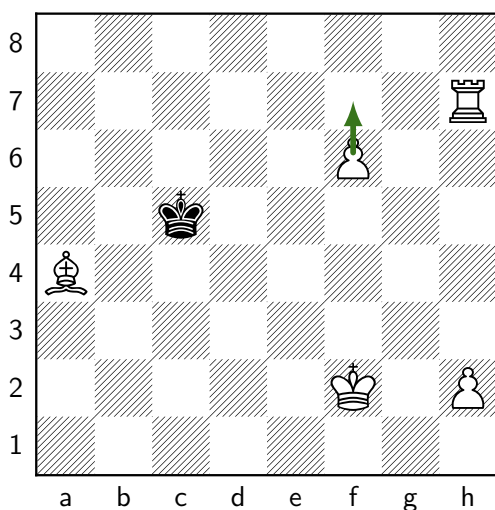


(c)

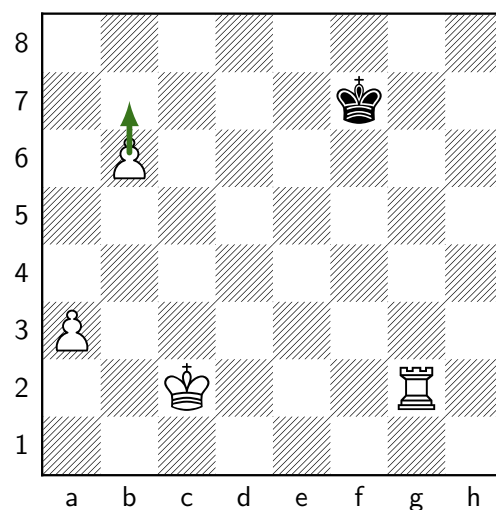


(d)

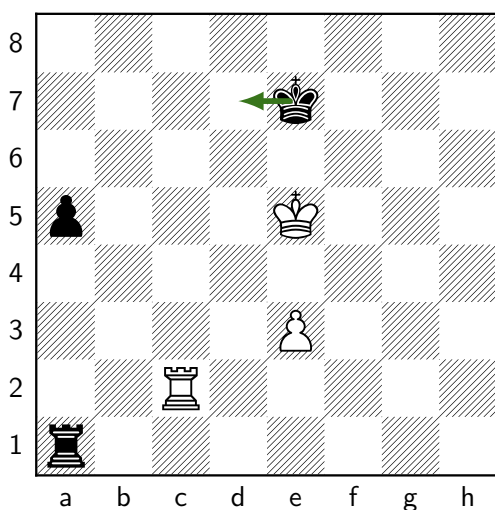
Figure 66: **Move 72, White: Rh7**. (a) Our position; Melanie plays Rh7 (f7h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#438` (ply 154) [structural distance  $d=5$ , QAP obj=3]: the master played Rg7; under  $\mu$  our  $Rf7$  plays the role of that game's  $Rb7$ ; cross-type roles: our  $Ba4$  as their  $rf8$ ; transferred move f7h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#790` (ply 128) [structural distance  $d=6$ , QAP obj=3]: the master played Rd7+; under  $\mu$  our  $Rf7$  plays the role of that game's  $Rd6$ ; cross-type roles: our  $Ba4$  as their  $rc3$ ; transferred move f7h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3941` (ply 133) [structural distance  $d=8$ , QAP obj=3]: the master played Rd2; under  $\mu$  our  $Rf7$  plays the role of that game's  $Rh7$ ; cross-type roles: our  $Ph2$  as their  $ra6$ ; transferred move f7h7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rh7 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



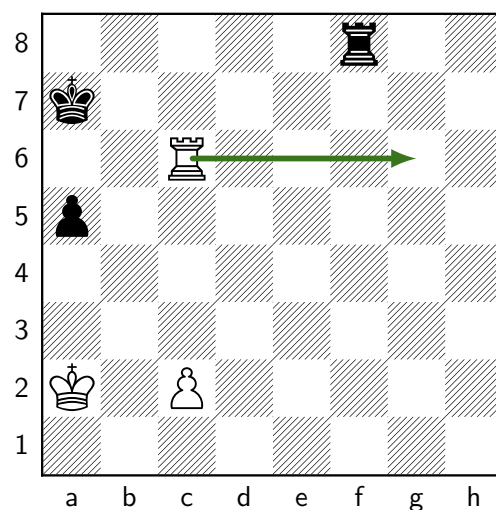
(a)



(b)

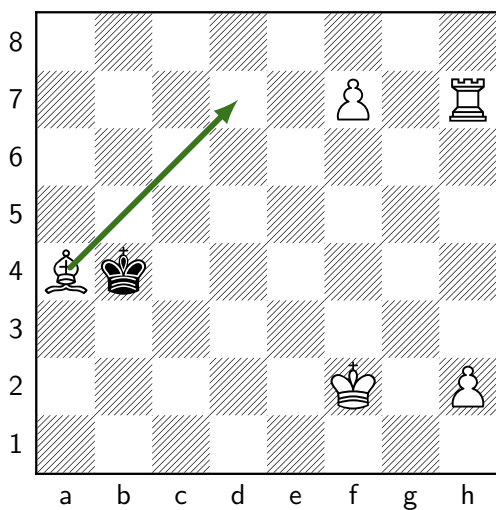


(c)

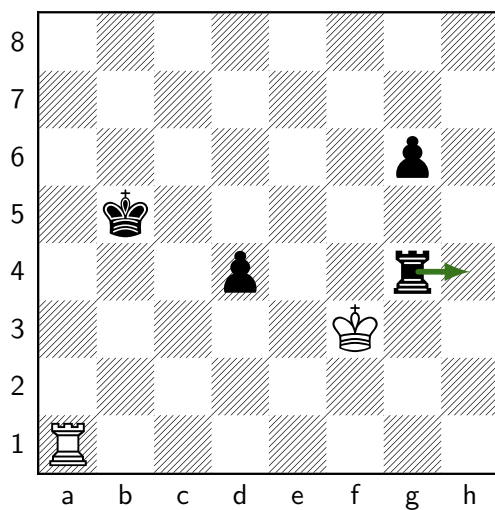


(d)

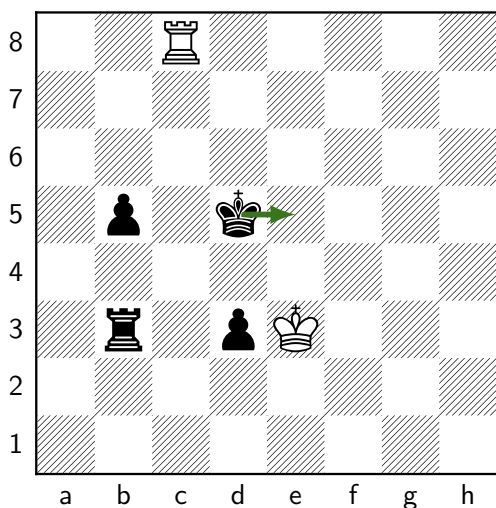
Figure 67: **Move 73, White: f7.** (a) Our position; Melanie plays f7 (f6f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0000#3589` (ply 160) [structural distance  $d=6$ , QAP obj=1]: the master played b7; under  $\mu$  our *Pf6* plays the role of that game’s *Pb6*; cross-type roles: our *Ba4* as their *Rg2*; transferred move f6f7 (**endorses this move**). (c) Retrieved from `ccr1_top.p0002#1972` (ply 141) [structural distance  $d=5$ , QAP obj=1]: the master played Kd7; under  $\mu$  our *Kf2* plays the role of that game’s *Ke2*; cross-type roles: our *Ba4* as their *rc2*; transferred move f2e2. (d) Retrieved from `ccr1_top.p0000#2511` (ply 142) [structural distance  $d=5$ , QAP obj=1]: the master played Rg6; under  $\mu$  our *Rh7* plays the role of that game’s *Rc6*; cross-type roles: our *Ba4* as their *pa5*, our *Pf6* as their *rf8*; transferred move h7b7. Synthesis: 1 retrieved analog transferred to f7 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



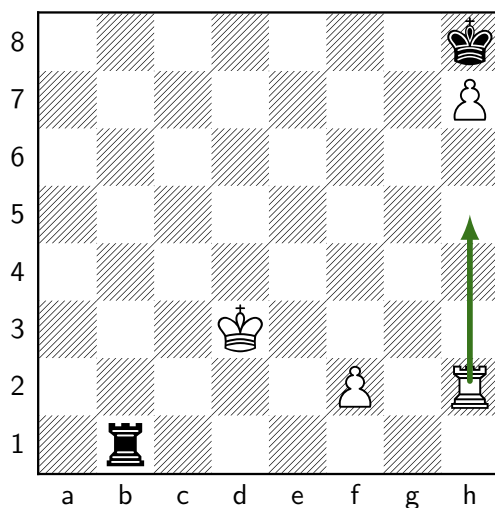
(a)



(b)



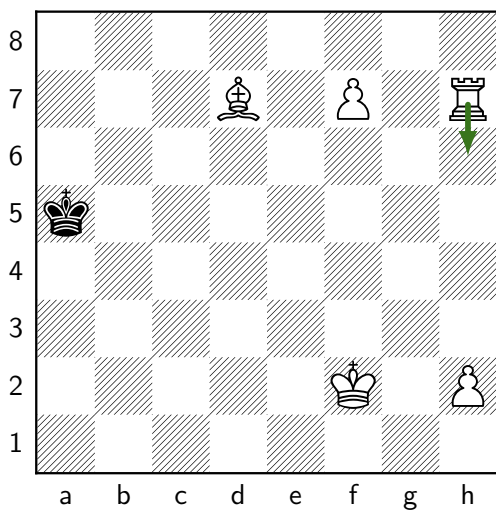
(c)



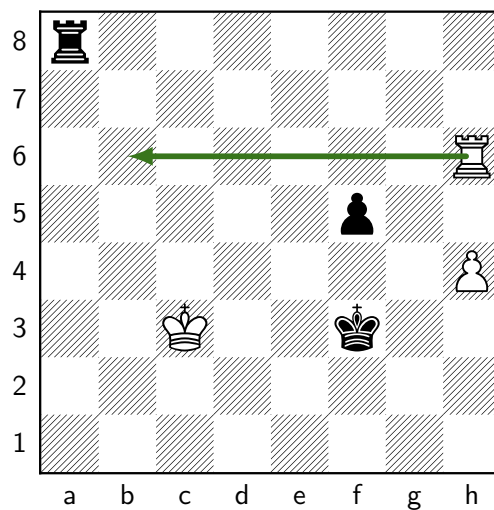
(d)

Figure 68: **Move 74, White: Bd7.** (a) Our position; Melanie plays Bd7 (a4d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1880` (ply 141) [structural distance  $d=6$ , QAP obj=11]: the master played Rh4; under  $\mu$  our *Ba4* plays the role of that game’s *Rg5* (a **cross-type** correspondence); cross-type roles: our *Rh7* as their *Pd4*, our *Pf7* as their *ra1*; transferred move **a4d7 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#2041` (ply 133) [structural distance  $d=9$ , QAP obj=8]: the master played Ke5; under  $\mu$  our *Kf2* plays the role of that game’s *Kd4*; cross-type roles: our *Ph2* as their *rc8*, our *Ba4* as their *Pd3*; transferred move **f2f3**. (d) Retrieved from `ccr1_top_p0003#2938` (ply 142) [structural distance  $d=7$ , QAP obj=8]: the master played Rh5; under  $\mu$  our *Rh7* plays the role of that game’s *Rh2*; cross-type roles: our *Ph2* as their *rb1*, our *Ba4* as their *Ph7*; transferred move **h7h5**. Synthesis: 1 retrieved analog transferred to Bd7 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

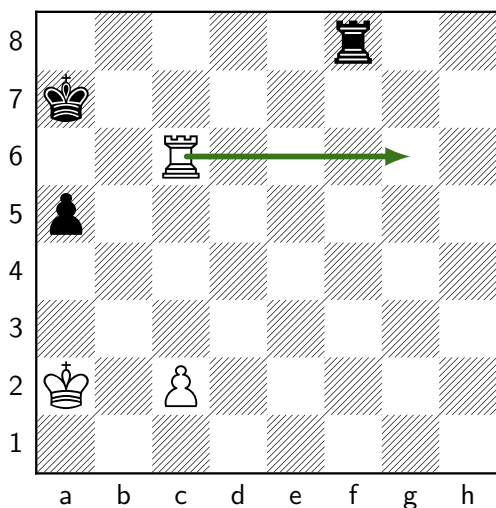




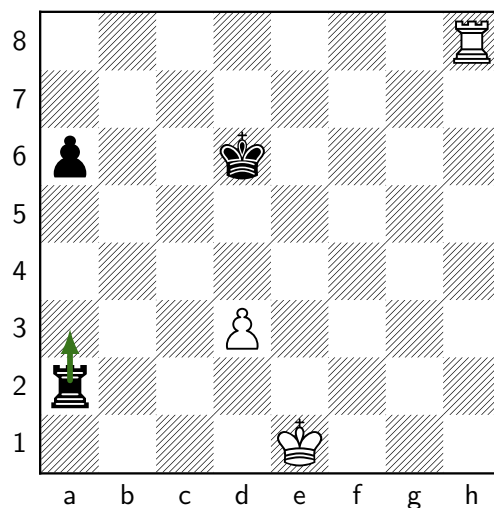
(a)



(b)



(c)



(d)

Figure 69: **Move 75, White: Rh6**. (a) Our position; Melanie plays Rh6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2292` (ply 142) [structural distance  $d=9$ , QAP obj=2]: the master played Rb6; under  $\mu$  our Rh7 plays the role of that game's Rh6; cross-type roles: our Bd7 as their ra8; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2511` (ply 142) [structural distance  $d=9$ , QAP obj=1]: the master played Rg6; under  $\mu$  our Rh7 plays the role of that game's Rc6; cross-type roles: our Bd7 as their rf8; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2257` (ply 133) [structural distance  $d=9$ , QAP obj=1]: the master played Ra3; under  $\mu$  our Rh7 plays the role of that game's Ra7; cross-type roles: our Bd7 as their rh8; transferred move h7h6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rh6 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

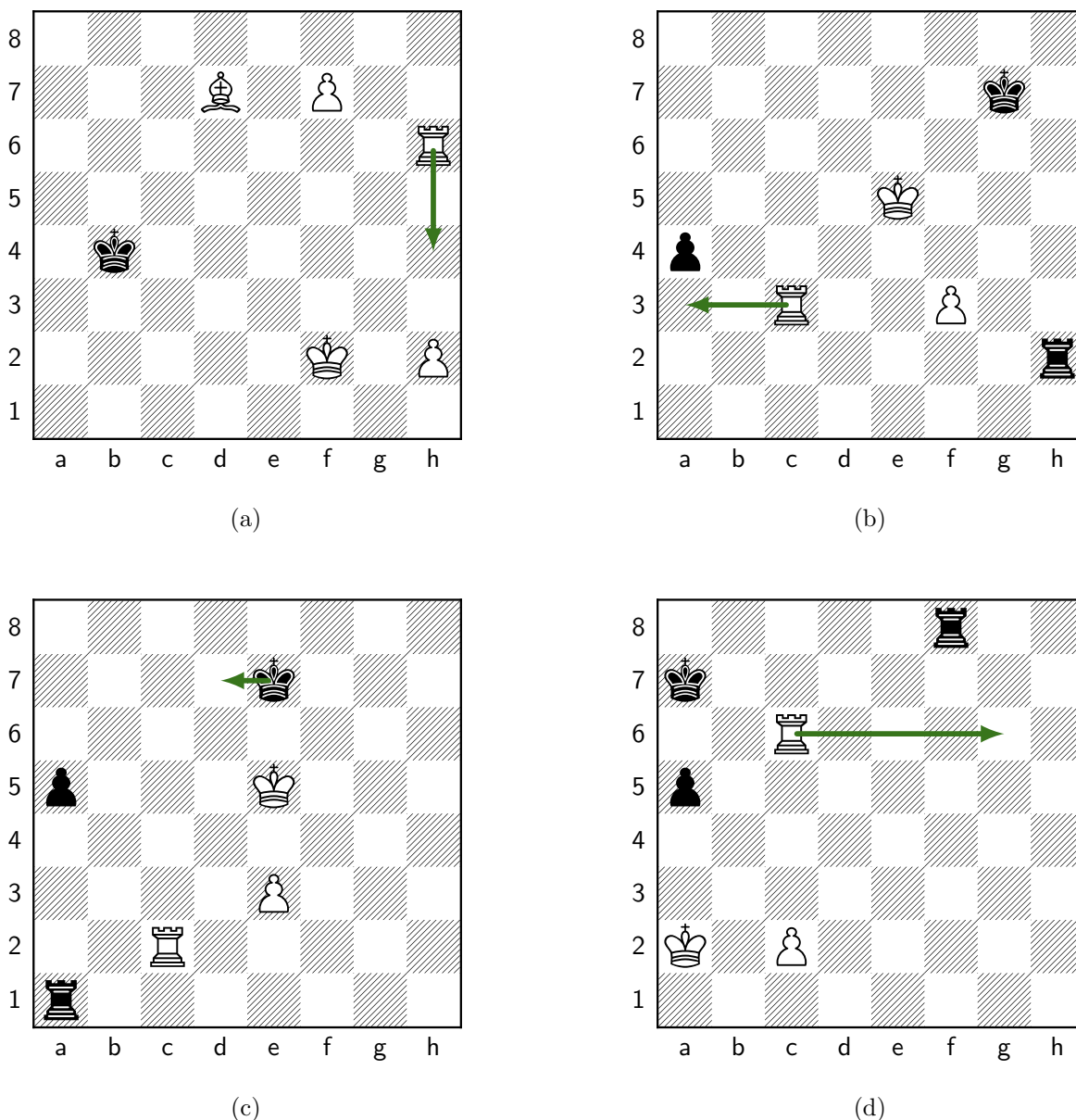
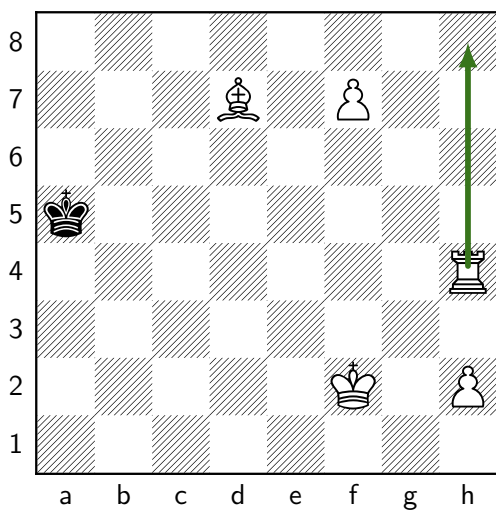
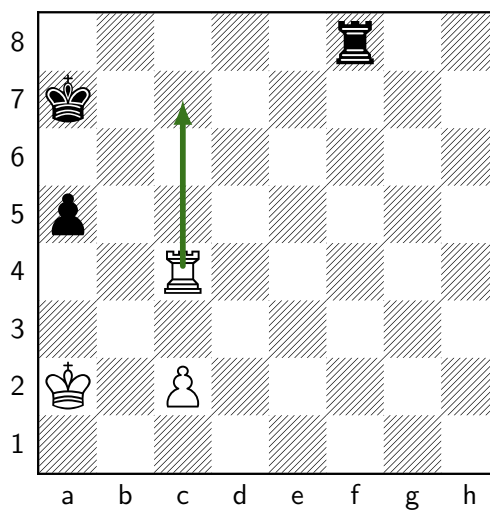


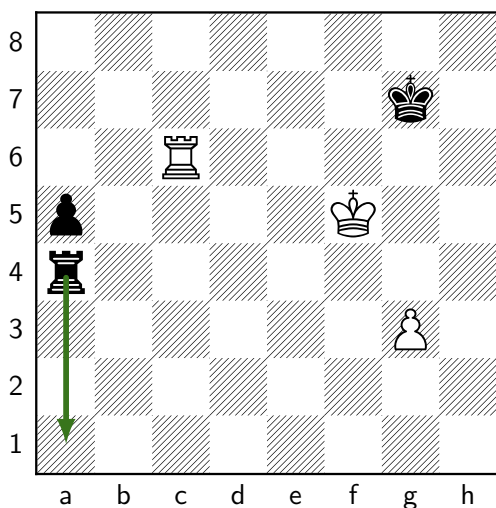
Figure 70: **Move 76, White: Rh4+.** (a) Our position; Melanie plays Rh4+ (h6h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2490` (ply 148) [structural distance  $d=6$ , QAP obj=1]: the master played Ra3; under  $\mu$  our Rh6 plays the role of that game’s Rc3; cross-type roles: our Bd7 as their rh2; transferred move h6h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1972` (ply 141) [structural distance  $d=5$ , QAP obj=2]: the master played Kd7; under  $\mu$  our Kf2 plays the role of that game’s Ke2; cross-type roles: our Bd7 as their pe3, our Pf7 as their rc2; transferred move f2e2. (d) Retrieved from `ccr1_top_p0000#2511` (ply 142) [structural distance  $d=5$ , QAP obj=2]: the master played Rg6; under  $\mu$  our Rh6 plays the role of that game’s Rc6; cross-type roles: our Bd7 as their pa5, our Pf7 as their rf8; transferred move h6a6. Synthesis: 2 retrieved analogs transferred to Rh4+ (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



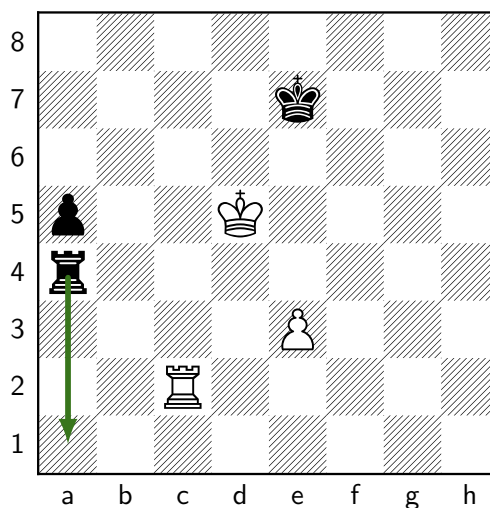
(a)



(b)

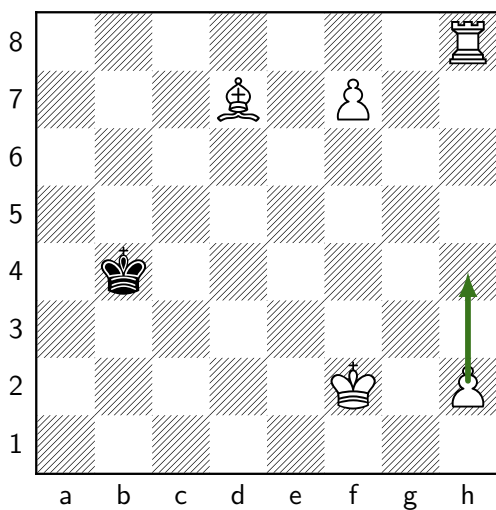


(c)

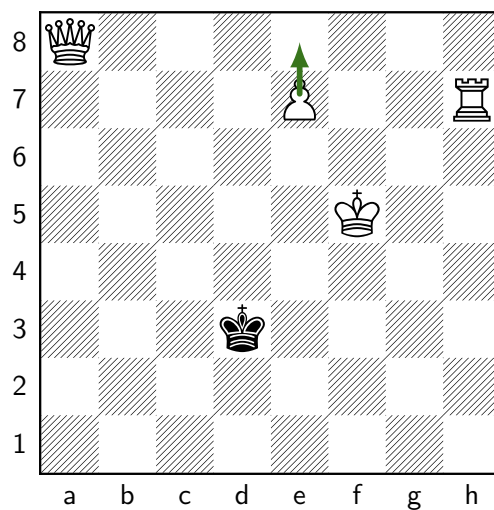


(d)

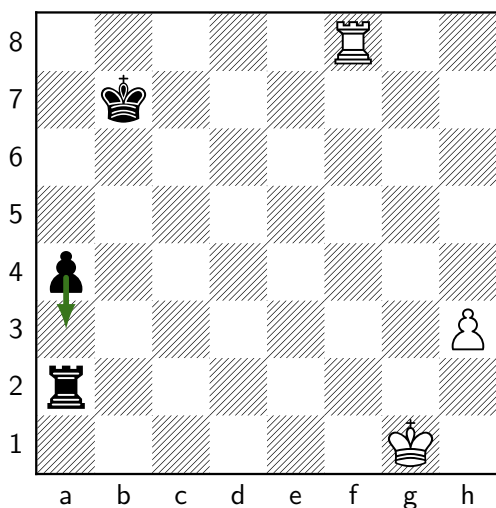
Figure 71: **Move 77, White: Rh8.** (a) Our position; Melanie plays Rh8 (**h4h8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2511` (ply 138) [structural distance  $d=7$ , QAP obj=2]: the master played Rc7+; under  $\mu$  our Rh4 plays the role of that game’s Rc4; cross-type roles: our Bd7 as their pa5, our Pf7 as their rf8; transferred move h4h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3362` (ply 139) [structural distance  $d=9$ , QAP obj=1]: the master played Ra1; under  $\mu$  our Rh4 plays the role of that game’s Ra5; cross-type roles: our Bd7 as their rc6; transferred move h4h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1972` (ply 139) [structural distance  $d=8$ , QAP obj=1]: the master played Ra1; under  $\mu$  our Rh4 plays the role of that game’s Ra5; cross-type roles: our Bd7 as their pe3, our Pf7 as their rc2; transferred move h4h8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rh8 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



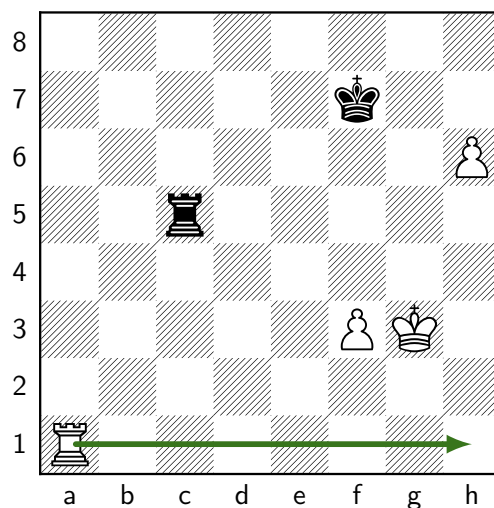
(a)



(b)

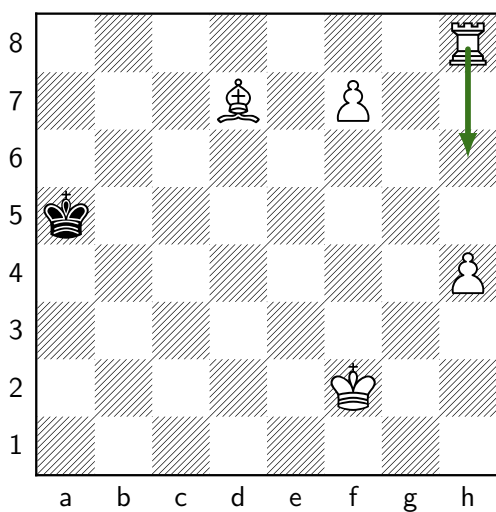


(c)

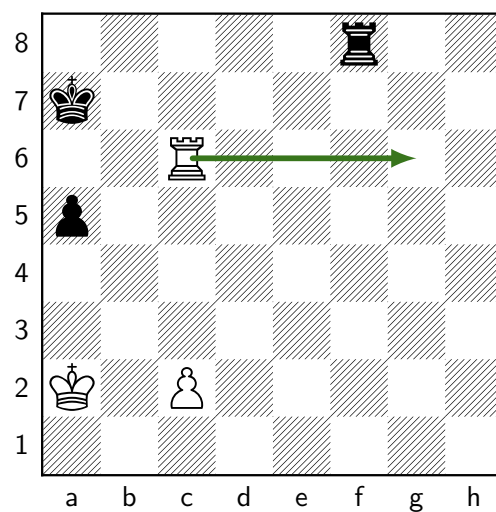


(d)

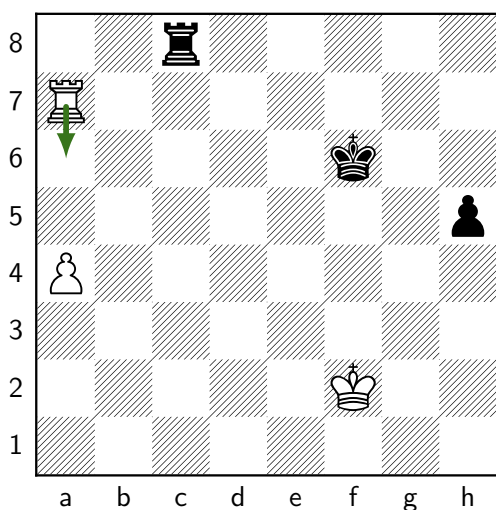
Figure 72: **Move 78, White: h4.** (a) Our position; Melanie plays **h4** (**h2h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#323` (ply 154) [structural distance  $d=4$ , QAP obj=1]: the master played **e8=Q**; under  $\mu$  our *Ph2* plays the role of that game’s *Pe7*; cross-type roles: our *Bd7* as their *Qa8*; transferred move **h2h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#9` (ply 169) [structural distance  $d=5$ , QAP obj=1]: the master played **a3**; under  $\mu$  our *Ph2* plays the role of that game’s *Pa5*; cross-type roles: our *Bd7* as their *rf8*; transferred move **h2h4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#851` (ply 142) [structural distance  $d=7$ , QAP obj=0]: the master played **Rh1**; under  $\mu$  our *Ph2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Bd7* as their *Ph6*; transferred move **h2h4** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **h4** (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



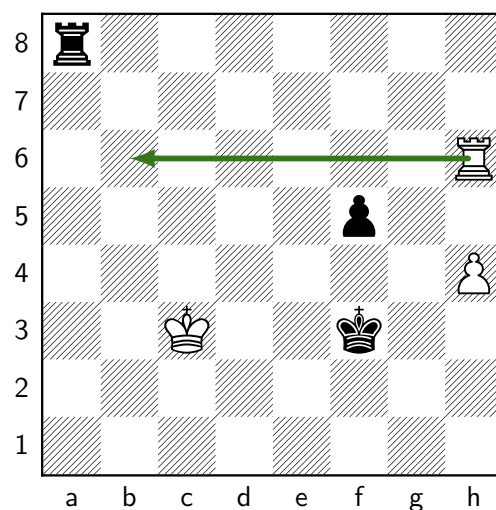
(a)



(b)

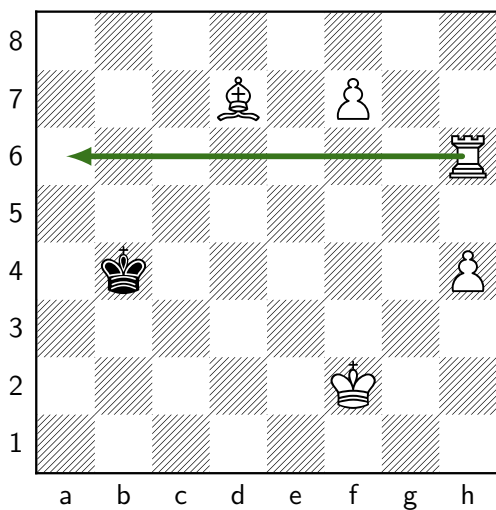


(c)

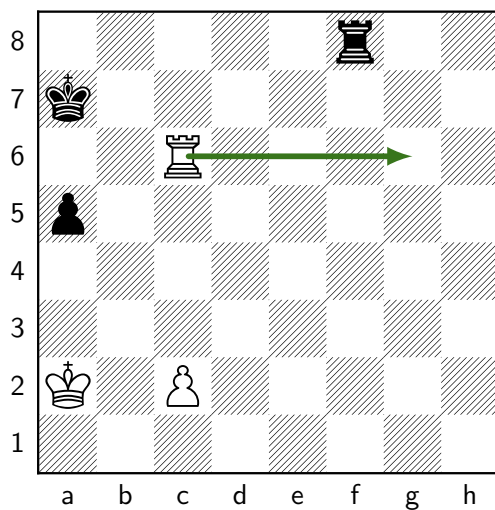


(d)

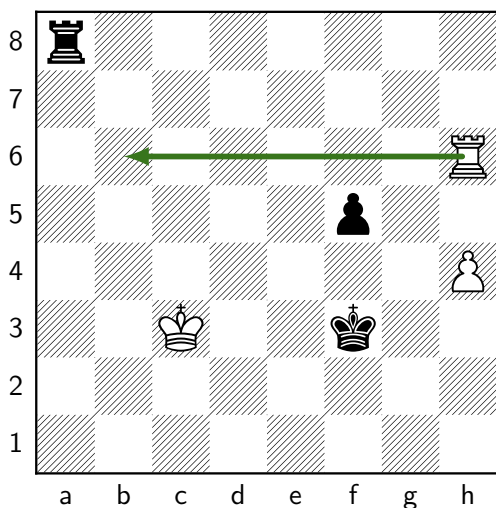
Figure 73: **Move 79, White: Rh6**. (a) Our position; Melanie plays Rh6 (h8h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2511` (ply 142) [structural distance  $d=5$ , QAP obj=2]: the master played Rg6; under  $\mu$  our Rh8 plays the role of that game's Rc6; cross-type roles: our Bd7 as their rf8; transferred move h8h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#106` (ply 146) [structural distance  $d=5$ , QAP obj=1]: the master played Ra6+; under  $\mu$  our Rh8 plays the role of that game's Ra7; cross-type roles: our Bd7 as their rc8; transferred move h8h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2292` (ply 142) [structural distance  $d=5$ , QAP obj=1]: the master played Rb6; under  $\mu$  our Rh8 plays the role of that game's Rh6; cross-type roles: our Bd7 as their ra8; transferred move h8h6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rh6 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



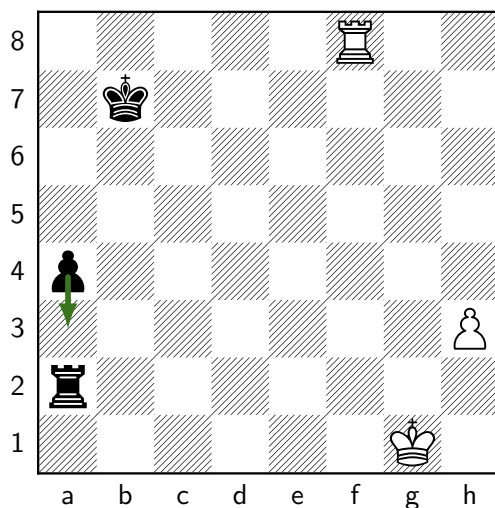
(a)



(b)

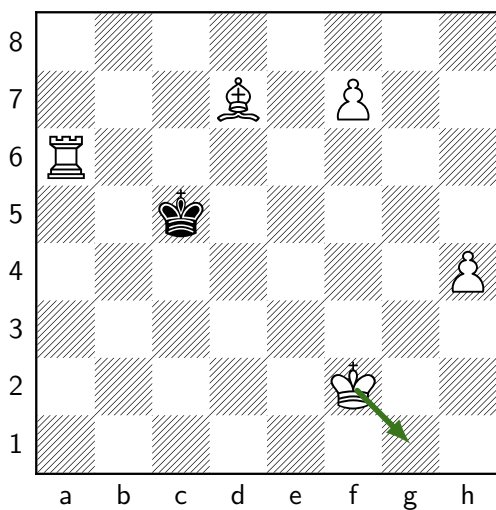


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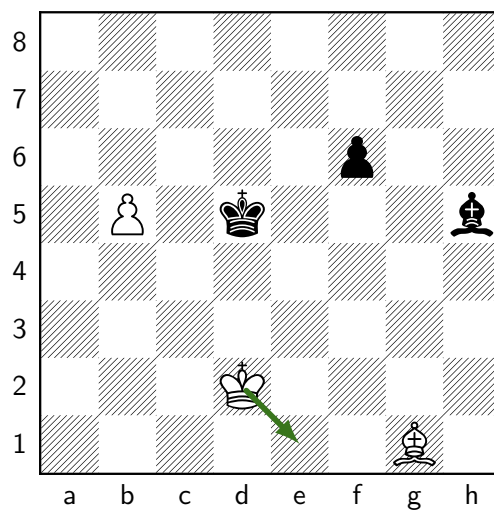


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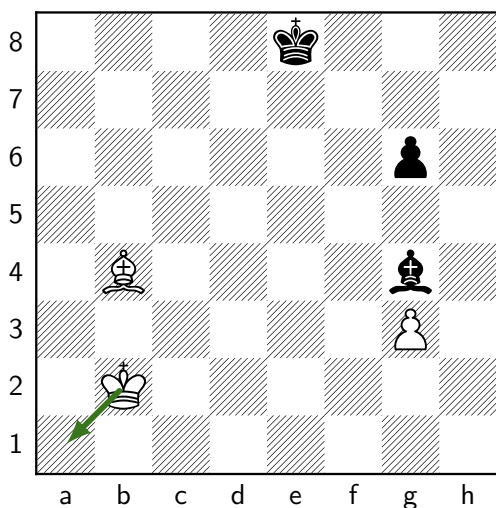
Figure 74: **Move 80, White: Ra6**. (a) Our position; Melanie plays Ra6 (h6a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2511` (ply 142) [structural distance  $d=5$ , QAP obj=1]: the master played Rg6; under  $\mu$  our Rh6 plays the role of that game’s Rc6; cross-type roles: our Bd7 as their pa5, our Pf7 as their rf8; transferred move h6a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2292` (ply 142) [structural distance  $d=4$ , QAP obj=2]: the master played Rb6; under  $\mu$  our Rh6 plays the role of that game’s Rh6; cross-type roles: our Bd7 as their pf5, our Pf7 as their ra8; transferred move h6g6. (d) Retrieved from `ccr1_top_p0000#9` (ply 169) [structural distance  $d=5$ , QAP obj=2]: the master played a3; under  $\mu$  our Ph4 plays the role of that game’s Pa5; cross-type roles: our Bd7 as their rf8; transferred move h4h5. Synthesis: 1 retrieved analog transferred to Ra6 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



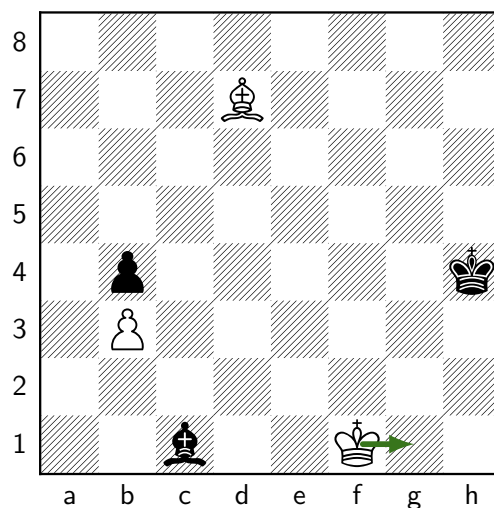
(a)



(b)

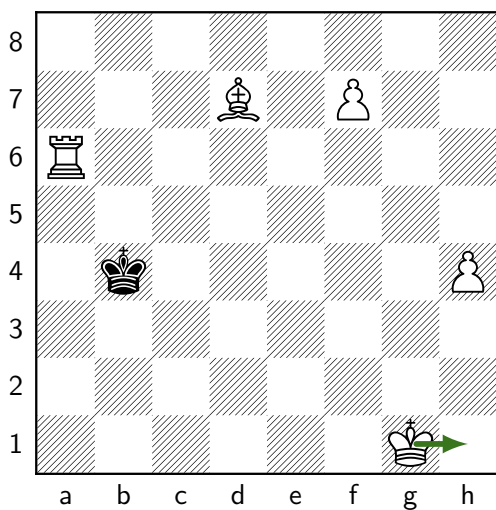


(c)

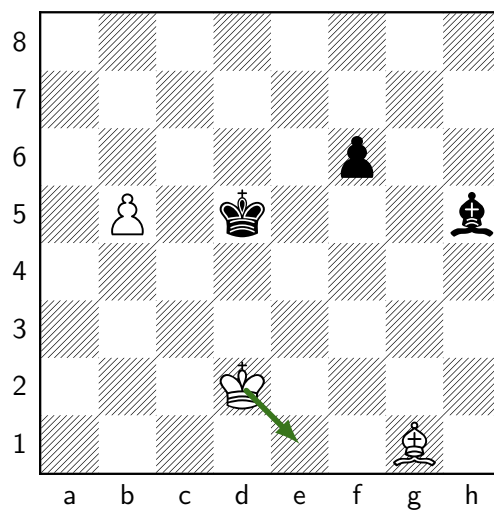


(d)

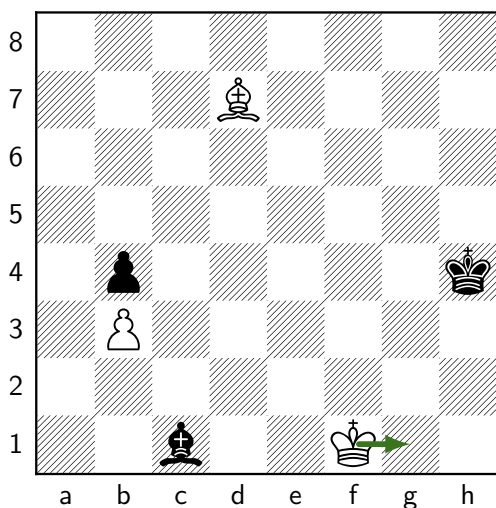
Figure 75: **Move 81, White: Kg1**. (a) Our position; Melanie plays Kg1 (f2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1488` (ply 156) [structural distance  $d=5$ , QAP obj=1]: the master played K e1; under  $\mu$  our Kf2 plays the role of that game’s Kd2; cross-type roles: our Ra6 as their Pb5, our Ph4 as their bh5; transferred move f2g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3122` (ply 144) [structural distance  $d=5$ , QAP obj=1]: the master played Ka1; under  $\mu$  our Kf2 plays the role of that game’s Kb2; cross-type roles: our Ra6 as their Bb4; transferred move f2g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2168` (ply 148) [structural distance  $d=5$ , QAP obj=1]: the master played Kg1; under  $\mu$  our Kf2 plays the role of that game’s Kf1; cross-type roles: our Ph4 as their bc1, our Bd7 as their Pb3; transferred move f2g1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg1 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



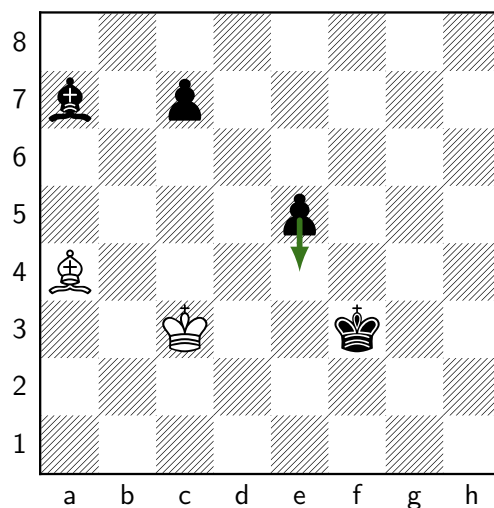
(a)



(b)



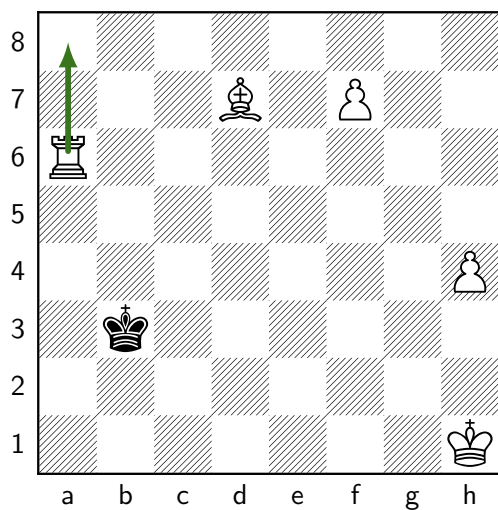
(c)



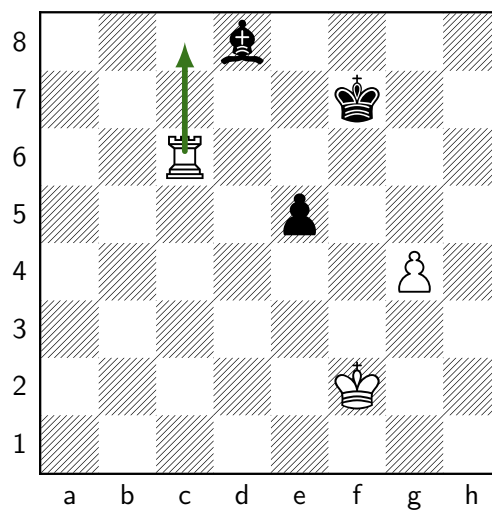
(d)

Figure 76: **Move 82, White: Kh1**. (a) Our position; Melanie plays Kh1 (g1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1488` (ply 156) [structural distance  $d=5$ , QAP obj=1]: the master played K $e1$ ; under  $\mu$  our K $g1$  plays the role of that game’s K $d2$ ; cross-type roles: our R $a6$  as their P $b5$ , our P $h4$  as their b $h5$ ; transferred move g1h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2168` (ply 148) [structural distance  $d=5$ , QAP obj=1]: the master played K $g1$ ; under  $\mu$  our K $g1$  plays the role of that game’s K $f1$ ; cross-type roles: our P $h4$  as their b $c1$ , our B $d7$  as their P $b3$ ; transferred move g1h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#312` (ply 147) [structural distance  $d=3$ , QAP obj=1]: the master played e $4$ ; under  $\mu$  our B $d7$  plays the role of that game’s P $e4$  (a **cross-type** correspondence); cross-type roles: our R $a6$  as their B $a7$ , our P $f7$  as their b $a4$ ; transferred move d7f5. Synthesis: 2 retrieved analogs transferred to Kh1 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

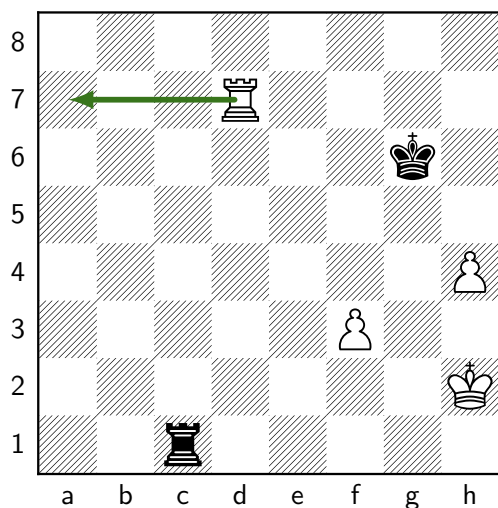




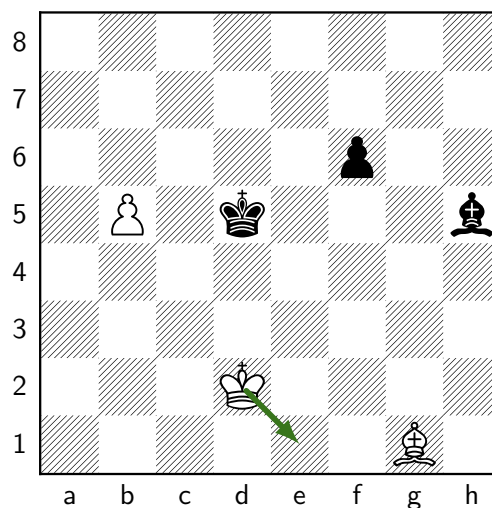
(a)



(b)



(c)



(d)

Figure 77: **Move 83, White: Ra8.** (a) Our position; Melanie plays Ra8 (a6a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3801` (ply 160) [structural distance  $d=5$ , QAP obj=1]: the master played Rc8; under  $\mu$  our Ra6 plays the role of that game's Rc6; cross-type roles: our Bd7 as their pe5, our Ph4 as their bd8; transferred move a6a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#179` (ply 150) [structural distance  $d=4$ , QAP obj=0]: the master played Ra7; under  $\mu$  our Ra6 plays the role of that game's Rd7; cross-type roles: our Ph4 as their rc1, our Bd7 as their Pf3; transferred move a6a8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1488` (ply 156) [structural distance  $d=5$ , QAP obj=1]: the master played Ke1; under  $\mu$  our Kh1 plays the role of that game's Kd2; cross-type roles: our Ra6 as their Bg1, our Bd7 as their Pb5; transferred move h1g1. Synthesis: 2 retrieved analogs transferred to Ra8 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

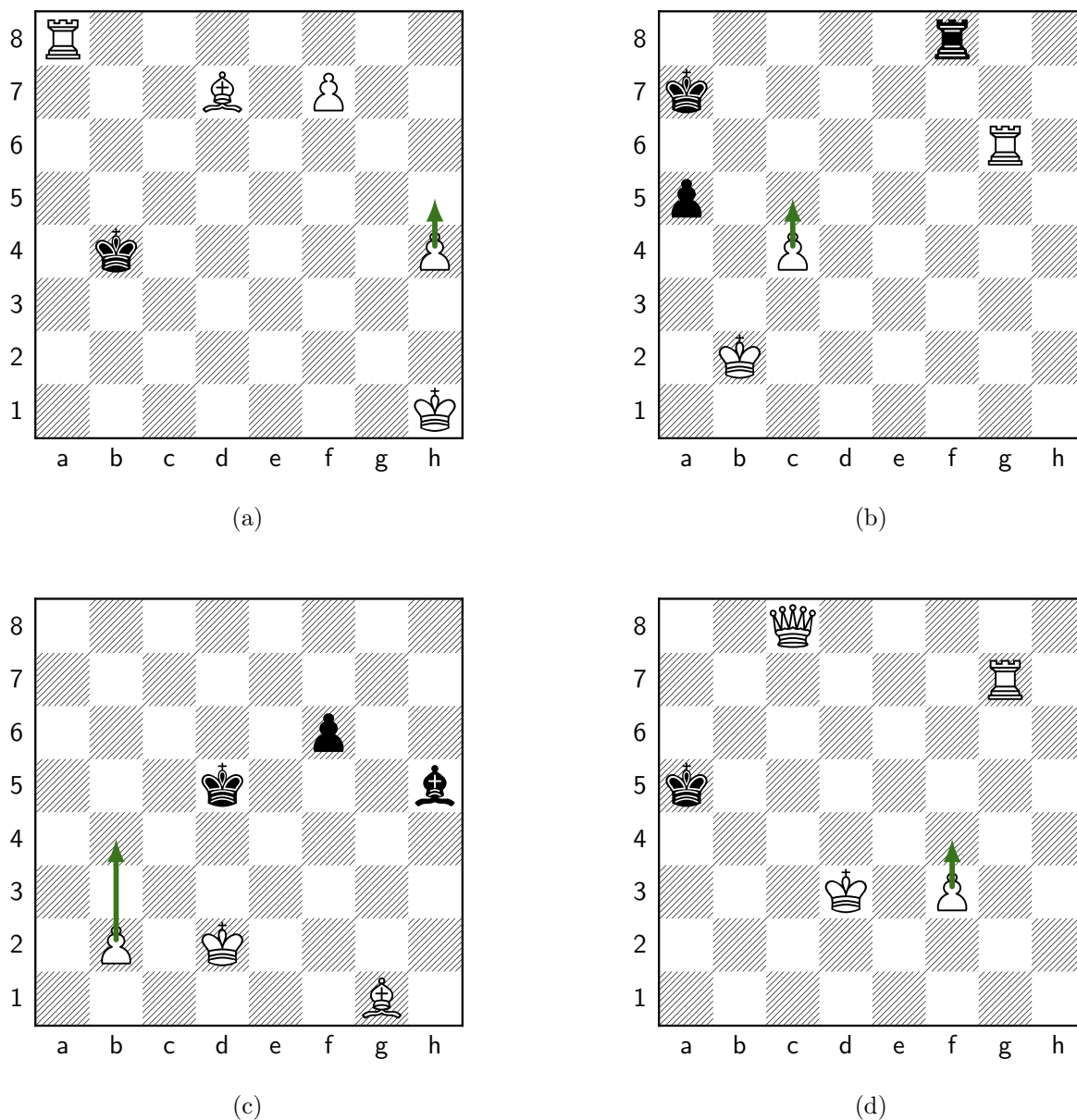
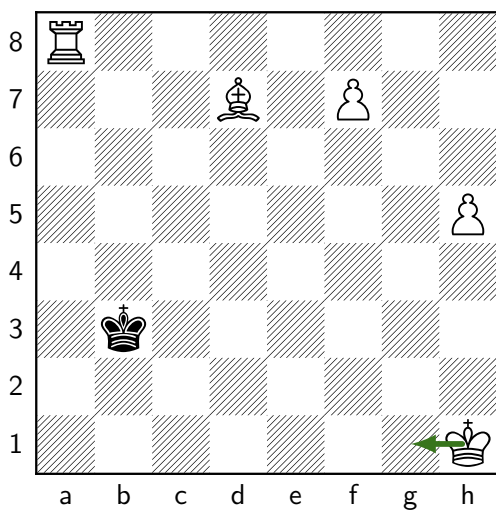
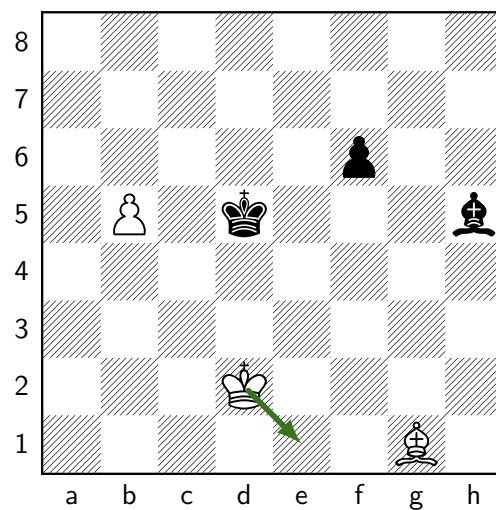


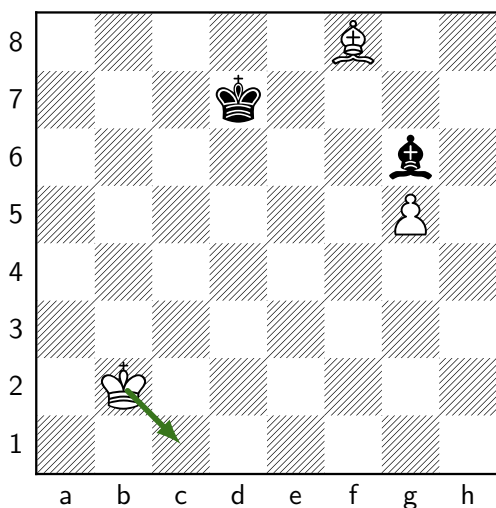
Figure 78: **Move 84, White: h5.** (a) Our position; Melanie plays h5 (h4h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2511` (ply 154) [structural distance  $d=4$ , QAP obj=1]: the master played c5; under  $\mu$  our  $Ph4$  plays the role of that game’s  $Pc4$ ; cross-type roles: our  $Bd7$  as their  $rf8$ ; transferred move h4h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1488` (ply 152) [structural distance  $d=5$ , QAP obj=1]: the master played b4; under  $\mu$  our  $Ph4$  plays the role of that game’s  $Pb2$ ; cross-type roles: our  $Ra8$  as their  $bh5$ ; transferred move h4h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#915` (ply 176) [structural distance  $d=5$ , QAP obj=0]: the master played f4; under  $\mu$  our  $Ph4$  plays the role of that game’s  $Pf3$ ; cross-type roles: our  $Bd7$  as their  $Rg7$ , our  $Ra8$  as their  $Qc8$ ; transferred move h4h5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to h5 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



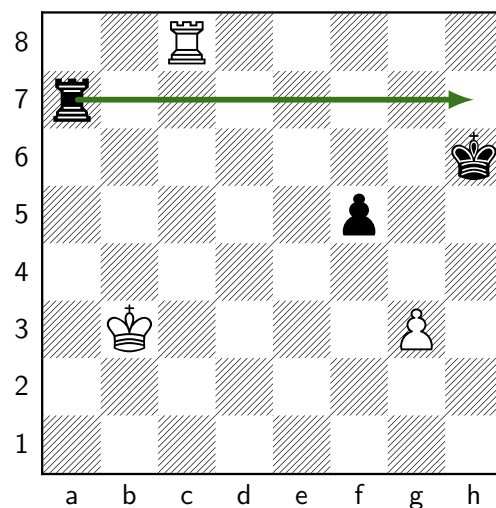
(a)



(b)

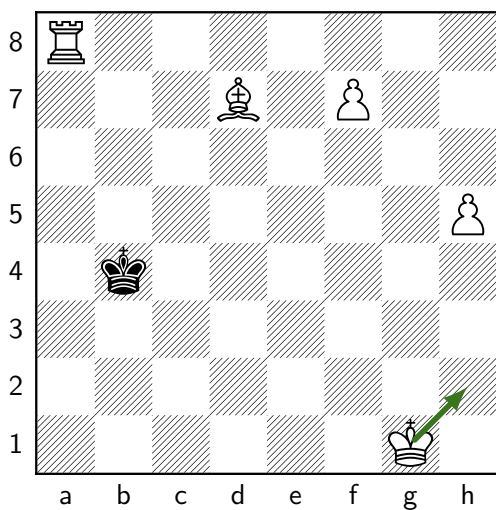


(c)

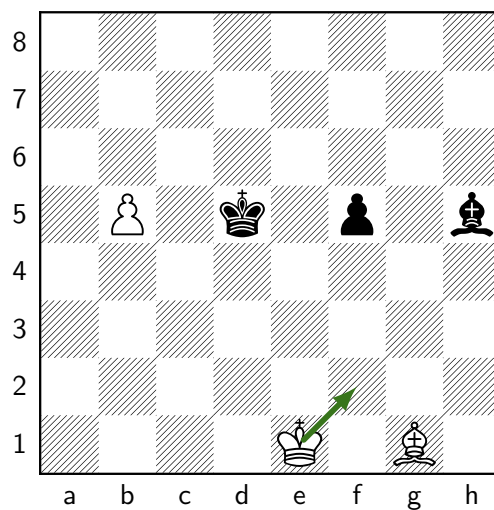


(d)

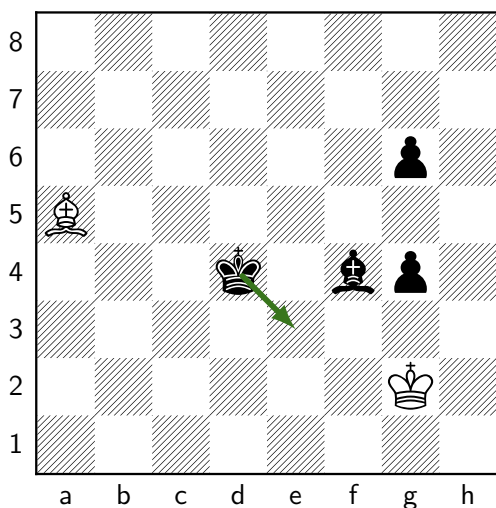
Figure 79: **Move 85, White: Kg1.** (a) Our position; Melanie plays Kg1 (**h1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1488` (ply 156) [structural distance  $d=5$ , QAP obj=1]: the master played K $e1$ ; under  $\mu$  our  $Kh1$  plays the role of that game’s  $Kd2$ ; cross-type roles: our  $Ra8$  as their  $Bg1$ , our  $Bd7$  as their  $Pb5$ ; transferred move **h1g1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#1609` (ply 170) [structural distance  $d=5$ , QAP obj=1]: the master played K $c1$ ; under  $\mu$  our  $Kh1$  plays the role of that game’s  $Kb2$ ; cross-type roles: our  $Ra8$  as their  $Bf8$ ; transferred move **h1g1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#3891` (ply 157) [structural distance  $d=5$ , QAP obj=1]: the master played R $h7$ ; under  $\mu$  our  $Ra8$  plays the role of that game’s  $Ra2$ ; cross-type roles: our  $Ph5$  as their  $rc8$ , our  $Bd7$  as their  $pg3$ ; transferred move **a8a1**. Synthesis: 2 retrieved analogs transferred to Kg1 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



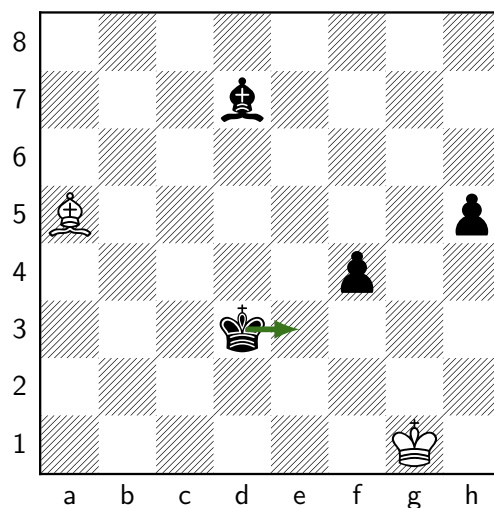
(a)



(b)

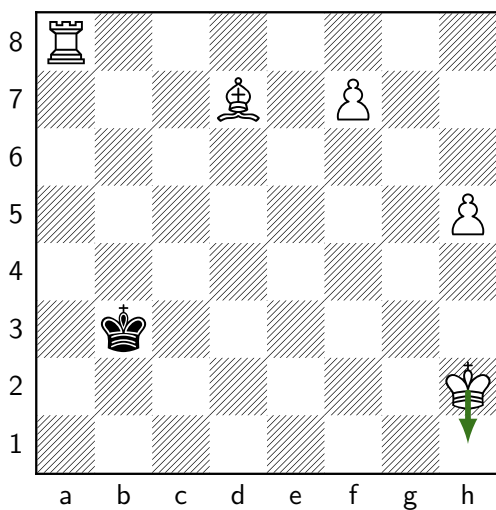


(c)

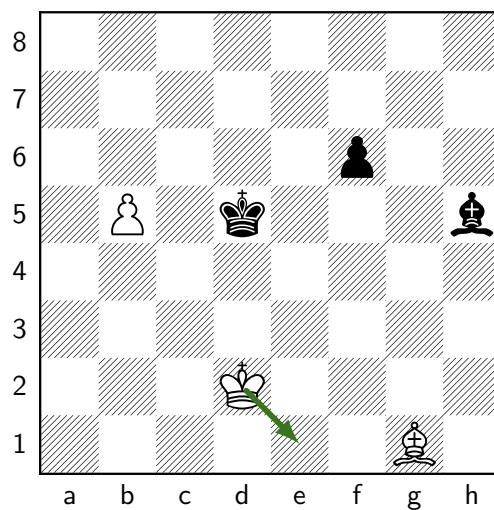


(d)

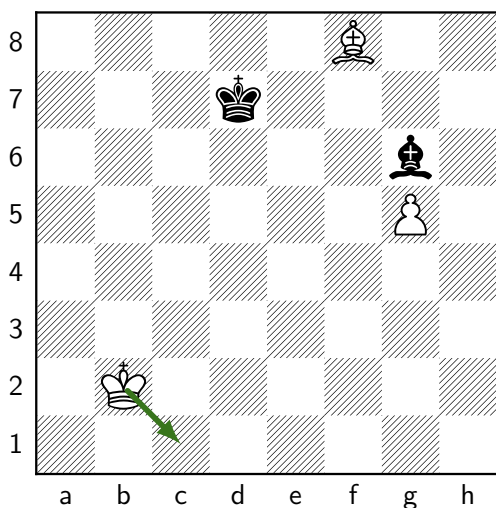
Figure 80: **Move 86, White: Kh2.** (a) Our position; Melanie plays Kh2 (**g1h2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1488` (ply 158) [structural distance  $d=5$ , QAP obj=1]: the master played Kf2; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Ra8* as their *Pb5*, our *Pf7* as their *bh5*; transferred move **g1h2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#471` (ply 187) [structural distance  $d=4$ , QAP obj=0]: the master played Ke3; under  $\mu$  our *Kg1* plays the role of that game’s *Kd5*; cross-type roles: our *Pf7* as their *ba5*, our *Ra8* as their *Bf4*; transferred move **g1h2 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#3944` (ply 175) [structural distance  $d=3$ , QAP obj=0]: the master played Ke3; under  $\mu$  our *Kg1* plays the role of that game’s *Kd6*; cross-type roles: our *Ph5* as their *Bd7*, our *Ra8* as their *Pf4*; transferred move **g1h2 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kh2 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



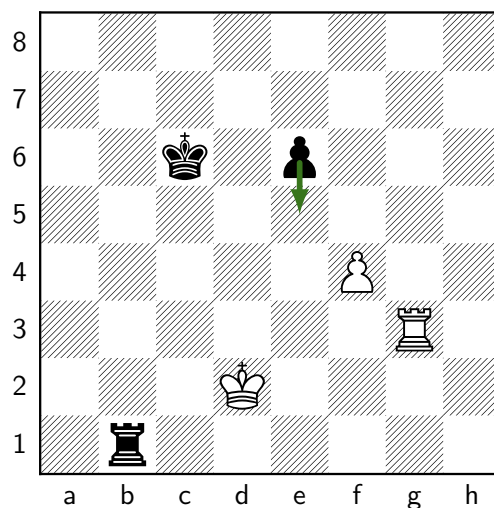
(a)



(b)

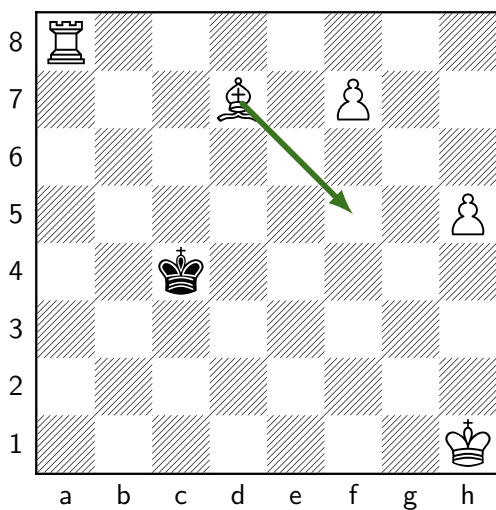


(c)

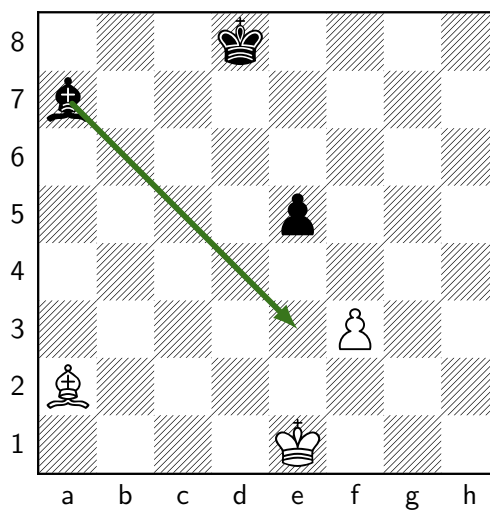


(d)

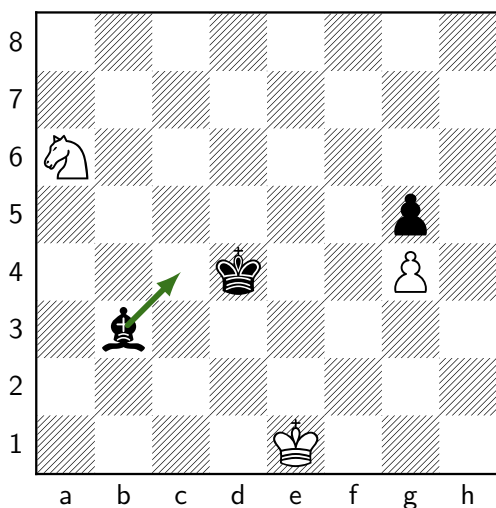
Figure 81: **Move 87, White: Kh1.** (a) Our position; Melanie plays Kh1 (**h2h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1488` (ply 156) [structural distance  $d=5$ , QAP obj=1]: the master played K $e1$ ; under  $\mu$  our K $h2$  plays the role of that game’s K $d2$ ; cross-type roles: our R $a8$  as their B $g1$ , our B $d7$  as their P $b5$ ; transferred move **h2h1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#1609` (ply 170) [structural distance  $d=5$ , QAP obj=1]: the master played K $c1$ ; under  $\mu$  our K $h2$  plays the role of that game’s K $b2$ ; cross-type roles: our R $a8$  as their B $f8$ ; transferred move **h2h1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#3912` (ply 187) [structural distance  $d=5$ , QAP obj=1]: the master played **e5**; under  $\mu$  our B $d7$  plays the role of that game’s P $e3$  (a **cross-type** correspondence); cross-type roles: our P $f7$  as their r $g3$ ; transferred move **d7f5**. Synthesis: 2 retrieved analogs transferred to Kh1 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



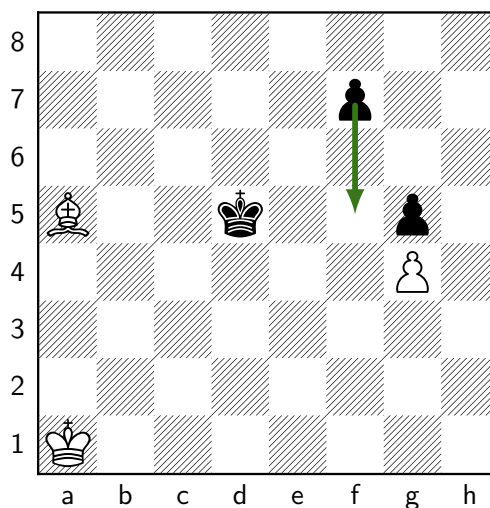
(a)



(b)

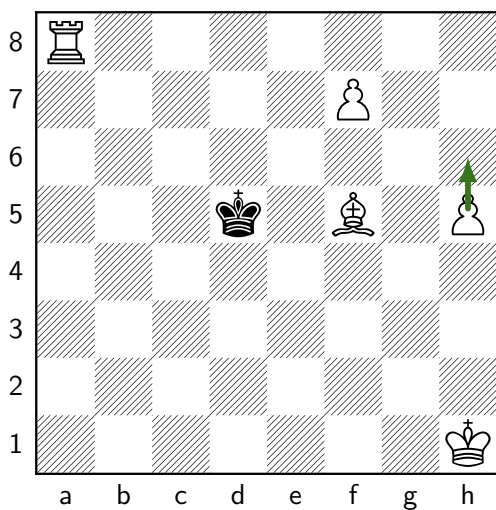


(c)

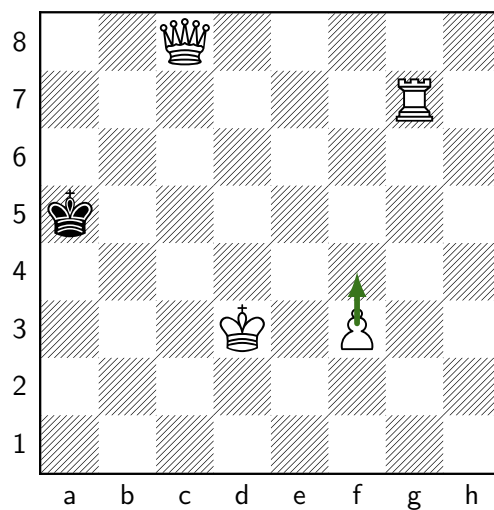


(d)

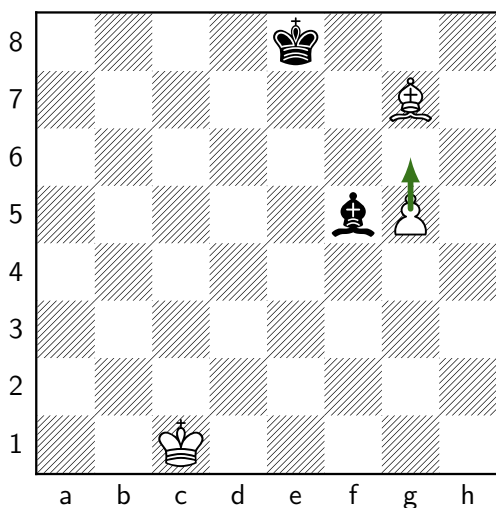
Figure 82: **Move 88, White: Bf5.** (a) Our position; Melanie plays Bf5 (d7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1495` (ply 185) [structural distance  $d=4$ , QAP obj=0]: the master played Be3; under  $\mu$  our Bd7 plays the role of that game’s Ba2; cross-type roles: our Ra8 as their pf3, our Ph5 as their ba2; transferred move d7f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1573` (ply 161) [structural distance  $d=5$ , QAP obj=0]: the master played Bc4; under  $\mu$  our Bd7 plays the role of that game’s Bb6; cross-type roles: our Ph5 as their na6, our Ra8 as their pg4; transferred move d7f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1531` (ply 161) [structural distance  $d=5$ , QAP obj=0]: the master played f5; under  $\mu$  our Bd7 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Pf7 as their ba5, our Ra8 as their Pg5; transferred move d7f5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bf5 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



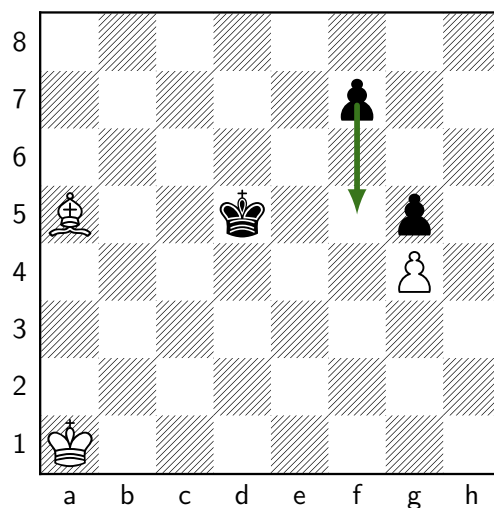
(a)



(b)

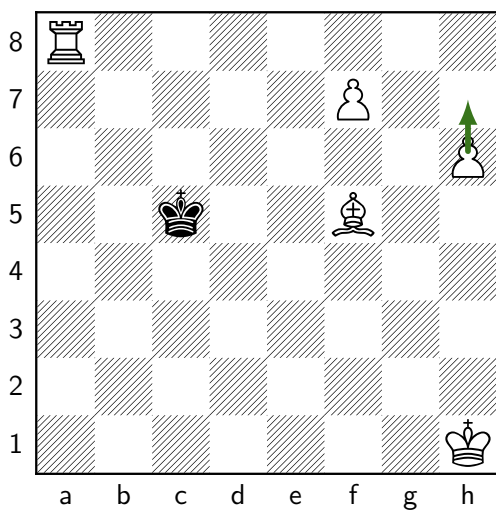


(c)

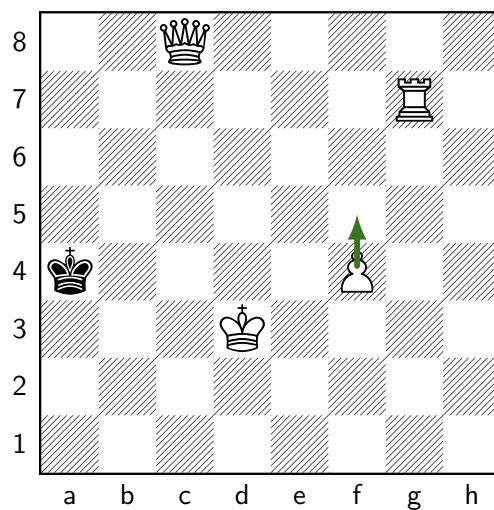


(d)

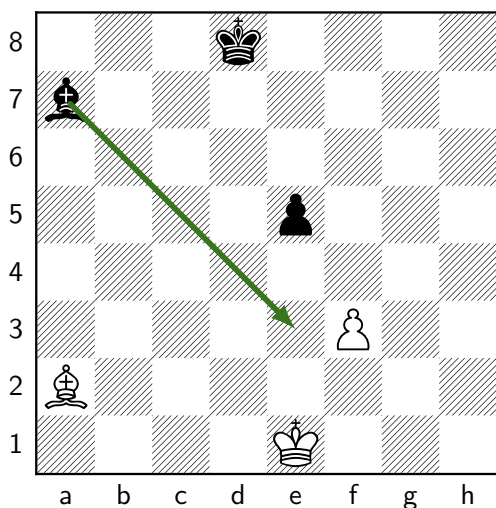
Figure 83: **Move 89, White: h6**. (a) Our position; Melanie plays h6 (h5h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#915` (ply 176) [structural distance  $d=4$ , QAP obj=0]: the master played f4; under  $\mu$  our *Ph5* plays the role of that game’s *Pf3*; cross-type roles: our *Pf7* as their *Rg7*, our *Bf5* as their *Qc8*; transferred move h5h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1609` (ply 174) [structural distance  $d=5$ , QAP obj=0]: the master played g6; under  $\mu$  our *Ph5* plays the role of that game’s *Pg5*; cross-type roles: our *Pf7* as their *bf5*; transferred move h5h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1531` (ply 161) [structural distance  $d=5$ , QAP obj=0]: the master played f5; under  $\mu$  our *Ph5* plays the role of that game’s *Pf2*; cross-type roles: our *Pf7* as their *ba5*, our *Ra8* as their *Pg5*; transferred move h5h6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to h6 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



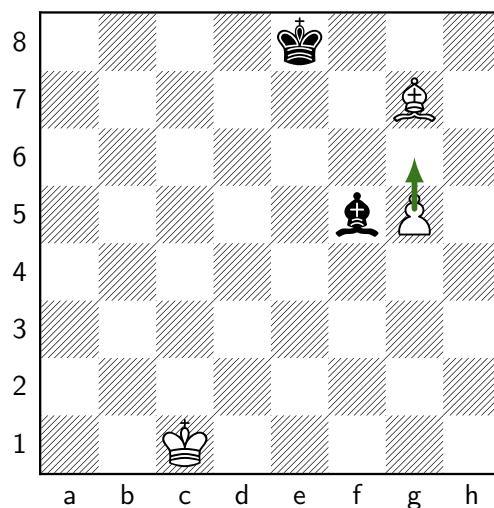
(a)



(b)



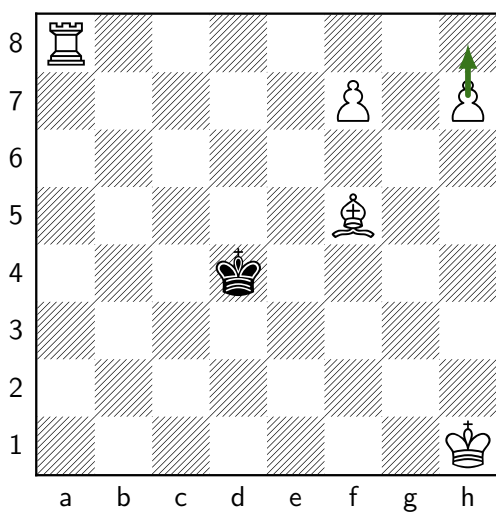
(c)



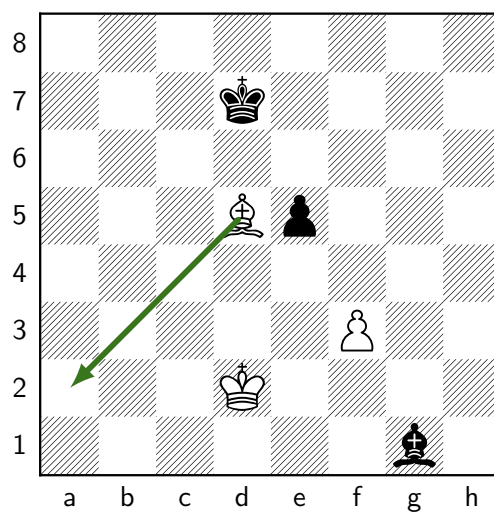
(d)

Figure 84: **Move 90, White: h7.** (a) Our position; Melanie plays h7 (h6h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#915` (ply 178) [structural distance  $d=4$ , QAP obj=0]: the master played f5; under  $\mu$  our *Ph6* plays the role of that game’s *Pf4*; cross-type roles: our *Pf7* as their *Rg7*, our *Bf5* as their *Qc8*; transferred move h6h7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1495` (ply 185) [structural distance  $d=5$ , QAP obj=0]: the master played Be3; under  $\mu$  our *Ph6* plays the role of that game’s *Ba2* (a **cross-type** correspondence); cross-type roles: our *Ra8* as their *pf3*; transferred move h6h7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1609` (ply 174) [structural distance  $d=5$ , QAP obj=0]: the master played g6; under  $\mu$  our *Ph6* plays the role of that game’s *Pg5*; cross-type roles: our *Pf7* as their *Bg7*; transferred move h6h7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to h7 (summed QAP objective 0, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

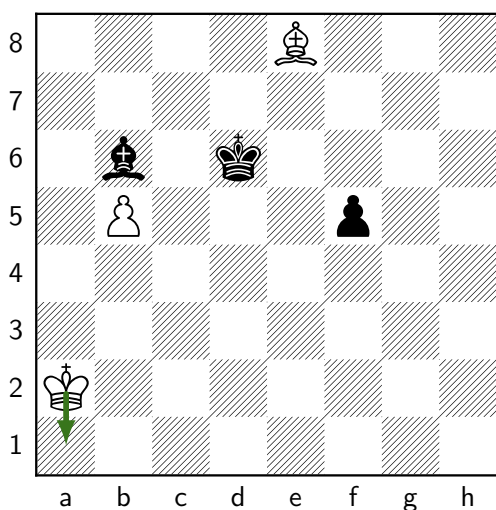




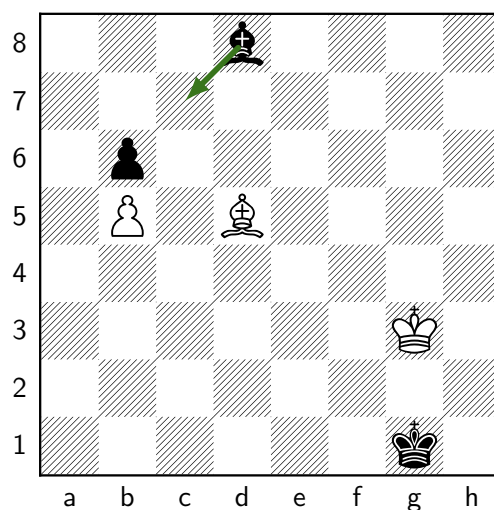
(a)



(b)

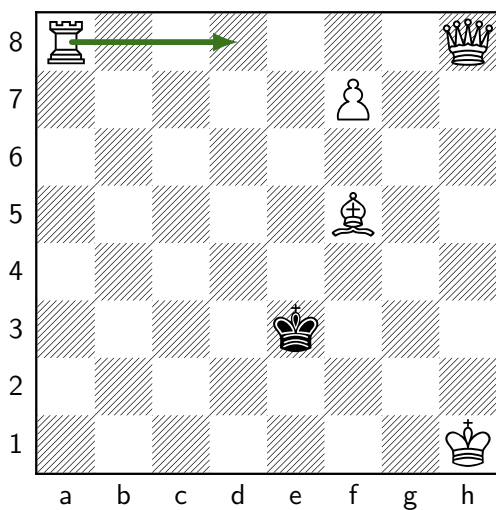


(c)

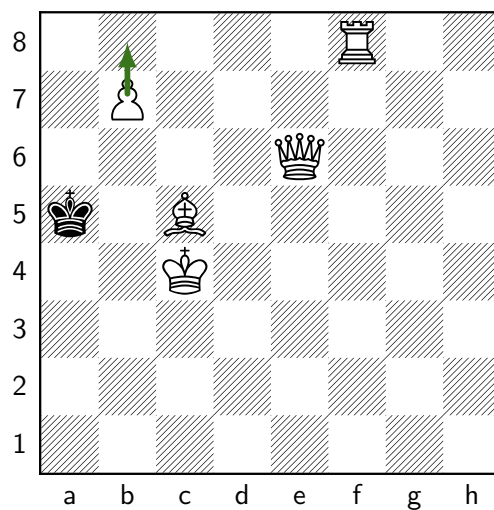


(d)

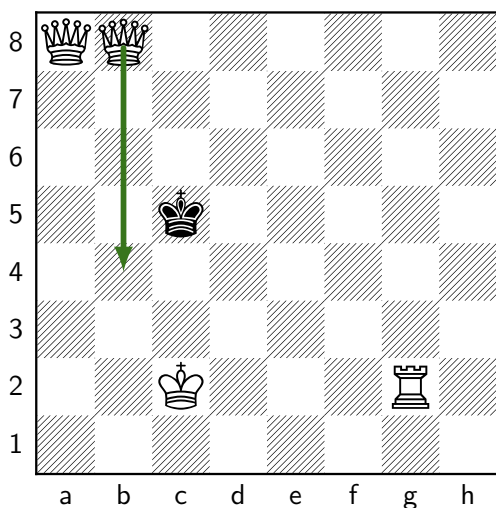
Figure 85: **Move 91, White: h8=Q+.** (a) Our position; Melanie plays h8=Q+ (h7h8q). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1495` (ply 196) [structural distance  $d=5$ , QAP obj=3]: the master played Ba2; under  $\mu$  our Bf5 plays the role of that game’s Bd5; cross-type roles: our Pf7 as their bg1, our Ra8 as their pe5; transferred move f5h3. (c) Retrieved from `ccr1_top_p0002#2862` (ply 164) [structural distance  $d=5$ , QAP obj=2]: the master played Ka1; under  $\mu$  our Kh1 plays the role of that game’s Ka2; cross-type roles: our Ra8 as their bb6; transferred move h1g1. (d) Retrieved from `ccr1_top_p0000#2889` (ply 197) [structural distance  $d=7$ , QAP obj=2]: the master played Bc7+; under  $\mu$  our Bf5 plays the role of that game’s Bd1; cross-type roles: our Ra8 as their bd5; transferred move f5c2. Synthesis: no retrieved analog transferred to h8=Q+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



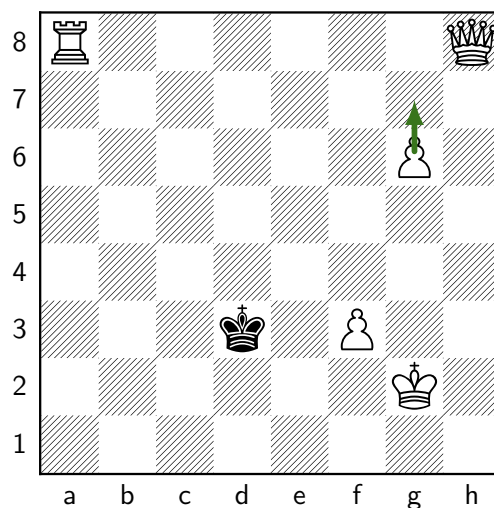
(a)



(b)

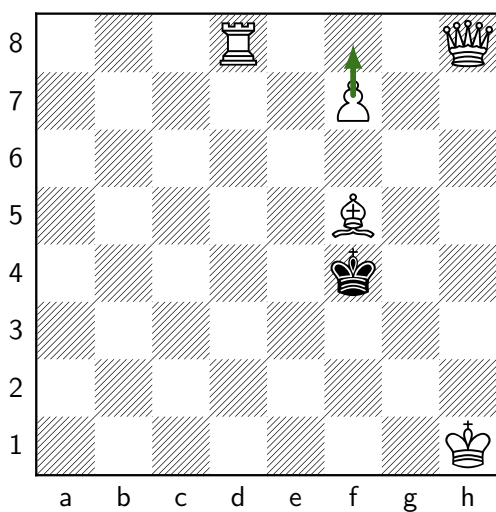


(c)

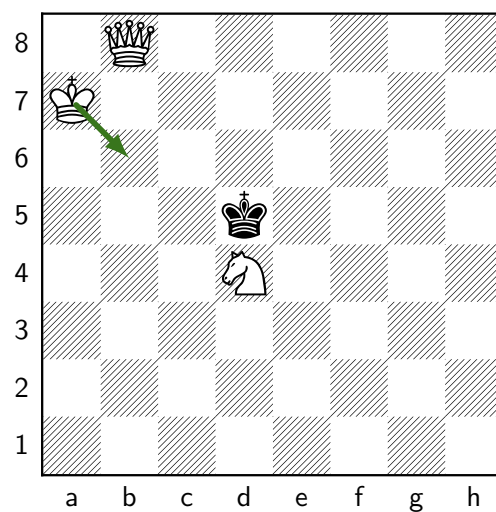


(d)

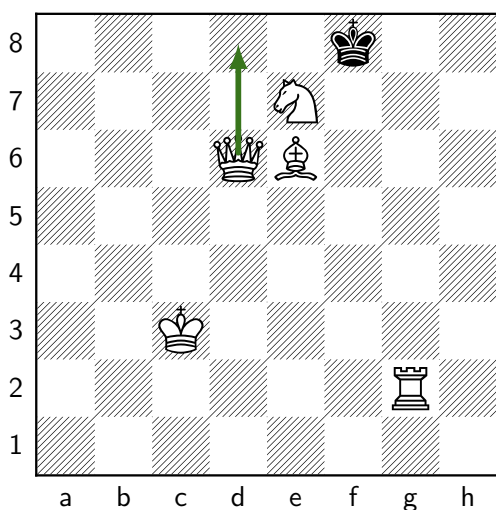
Figure 86: **Move 92, White: Rd8.** (a) Our position; Melanie plays Rd8 (a8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#934` (ply 178) [structural distance  $d=13$ , QAP obj=1]: the master played b8=Q; under  $\mu$  our *Ra8* plays the role of that game's *Pb7* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Rf8*; transferred move a8d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3589` (ply 174) [structural distance  $d=9$ , QAP obj=2]: the master played Qb4+; under  $\mu$  our *Bf5* plays the role of that game's *Qb8* (a **cross-type** correspondence); cross-type roles: our *Qh8* as their *Rg2*, our *Ra8* as their *Qa8*; transferred move f5d3. (d) Retrieved from `ccr1_top_p0003#377` (ply 174) [structural distance  $d=10$ , QAP obj=1]: the master played g7; under  $\mu$  our *Pf7* plays the role of that game's *Pg6*; cross-type roles: our *Bf5* as their *Pf3*; transferred move f7f8n. Synthesis: 2 retrieved analogs transferred to Rd8 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



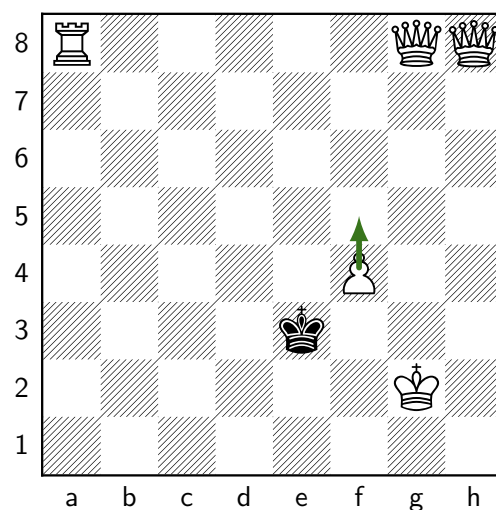
(a)



(b)

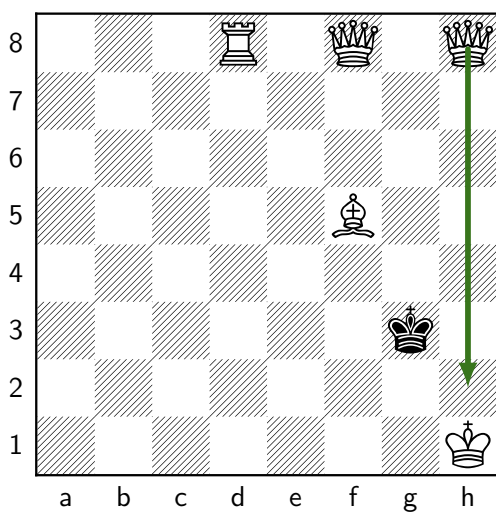


(c)

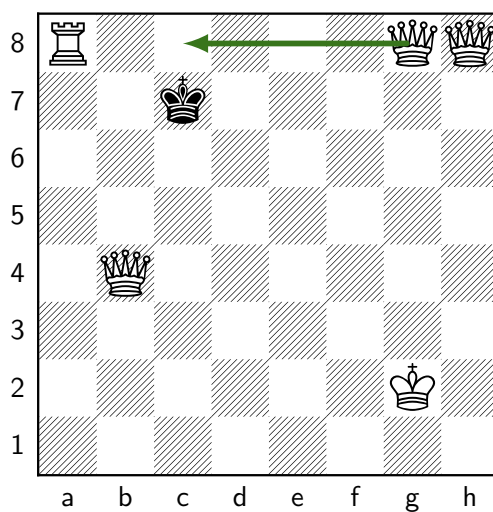


(d)

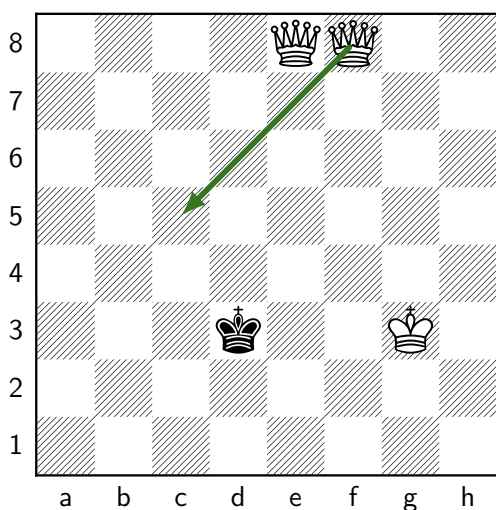
Figure 87: **Move 93, White: f8=Q.** (a) Our position; Melanie plays f8=Q (f7f8q). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#576` (ply 186) [structural distance  $d=19$ , QAP obj=19]: the master played Kb6; under  $\mu$  our *Kh1* plays the role of that game’s *Ka7*; cross-type roles: our *Bf5* as their *Nd4*; transferred move h1h2. (c) Retrieved from `ccr1_top_p0001#664` (ply 170) [structural distance  $d=17$ , QAP obj=19]: the master played Qd8#; under  $\mu$  our *Qh8* plays the role of that game’s *Qd6*; cross-type roles: our *Pf7* as their *Rg2*, our *Rd8* as their *Be6*; transferred move h8g7. (d) Retrieved from `ccr1_top_p0003#377` (ply 180) [structural distance  $d=20$ , QAP obj=8]: the master played f5; under  $\mu$  our *Bf5* plays the role of that game’s *Pf4* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Qh8*; transferred move f5e6. Synthesis: no retrieved analog transferred to f8=Q – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



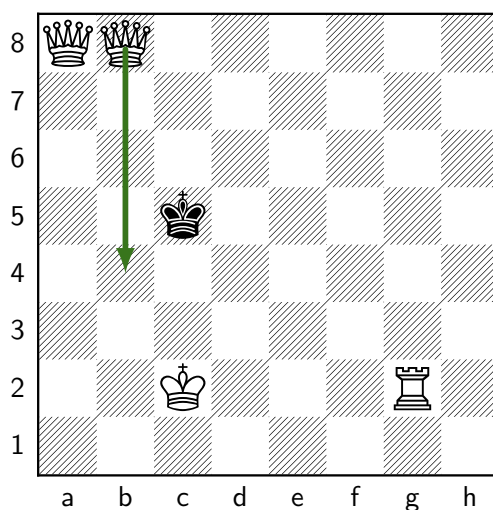
(a)



(b)

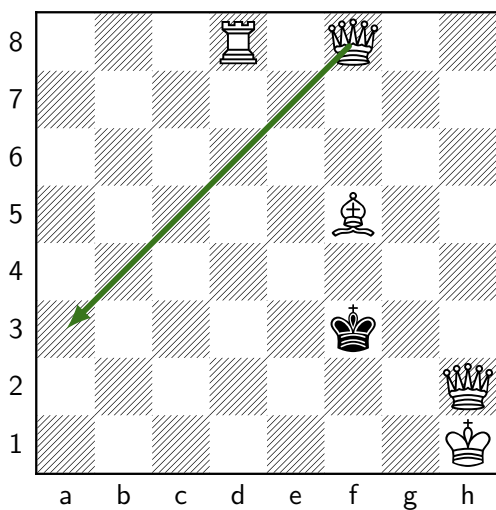


(c)

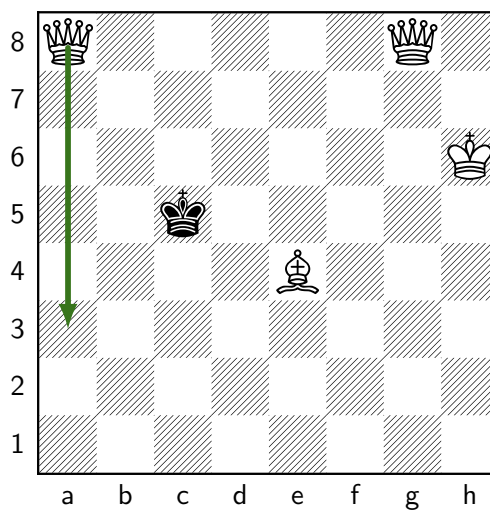


(d)

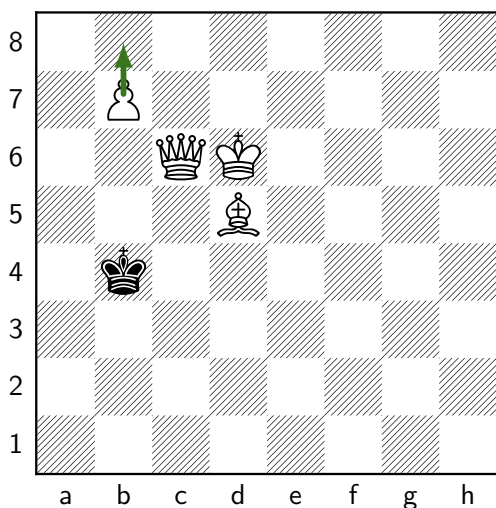
Figure 88: **Move 94, White: Qh2+.** (a) Our position; Melanie plays Qh2+ (h8h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#377` (ply 190) [structural distance  $d=7$ , QAP obj=4]: the master played Qc8#; under  $\mu$  our Qh8 plays the role of that game's Qg8; cross-type roles: our Bf5 as their Qb4; transferred move h8h4. (c) Retrieved from `ccrl_top_p0000#1578` (ply 192) [structural distance  $d=36$ , QAP obj=2]: the master played Qc5; under  $\mu$  our Qf8 plays the role of that game's Qf8; transferred move f8c5. (d) Retrieved from `ccrl_top_p0000#3589` (ply 174) [structural distance  $d=15$ , QAP obj=2]: the master played Qb4+; under  $\mu$  our Bf5 plays the role of that game's Qb8 (a **cross-type** correspondence); cross-type roles: our Qh8 as their Rg2; transferred move f5g4. Synthesis: no retrieved analog transferred to Qh2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



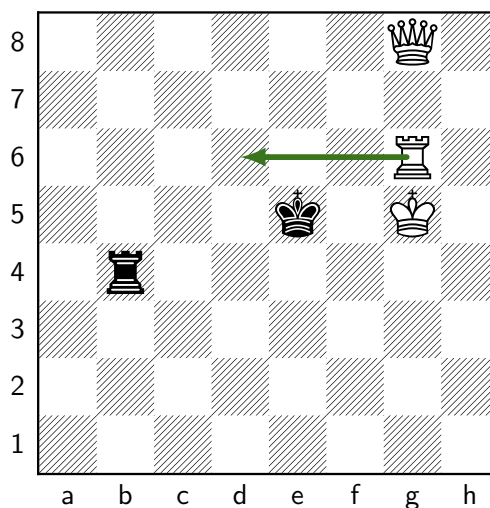
(a)



(b)



(c)



(d)

Figure 89: **Move 95, White: Qa3#.** (a) Our position; Melanie plays Qa3# (f8a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3574` (ply 178) [structural distance  $d=15$ , QAP obj=1]: the master played Qa3+; under  $\mu$  our Qf8 plays the role of that game’s Qa8; cross-type roles: our Qh2 as their Be4, our Bf5 as their Qg8; transferred move f8a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#589` (ply 198) [structural distance  $d=16$ , QAP obj=3]: the master played b8=Q+; under  $\mu$  our Rd8 plays the role of that game’s Pb7 (a **cross-type** correspondence); transferred move d8a8. (d) Retrieved from `ccr1_top_p0003#1484` (ply 180) [structural distance  $d=22$ , QAP obj=3]: the master played Rd6; under  $\mu$  our Qh2 plays the role of that game’s Rg6 (a **cross-type** correspondence); cross-type roles: our Bf5 as their Qg8; transferred move h2f4. Synthesis: 1 retrieved analog transferred to Qa3# (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.